

DETAILED e-TENDER DOCUMENT FOR

Design, survey, manufacture, supply, erection, testing, commissioning of 10 kWp & 5 kWp Solar Photovoltaic Power Plant (Hybrid) at Block Offices etc. including Warranty / Guarantee, Annual Maintenance Contract and insurance coverage for 5 (five) years on turn-key basis throughout the state of Tripura

**DNIe-T No.F.6 (542)/TREDA/NCES/2025-26/2110,
dated 12/06/2026**



TRIPURA RENEWABLE ENERGY DEVELOPMENT AGENCY

(Department of Power, Government of Tripura)

Vigyan Bhawan, Pandit Nehru Complex, Gorkhabasti, Agartala, Tripura

Tele-fax: 0381-2326139, email-tredatenders25@gmail.com

URL: www.treda.tripura.gov.in



TRIPURA RENEWABLE ENERGY DEVELOPMENT AGENCY

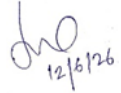
(Power Department, Government of Tripura)

Vigyan Bhawan, Pandit Nehru Complex, Gorkhabasti, Agartala, Tripura

Tele-fax: 0381-2326139, email- tredatenders25@gmail.com

NAME OF WORK: Design, survey, manufacture, supply, erection, testing, commissioning of 10 kWp & 5 kWp Solar Photovoltaic Power Plant (Hybrid) at Block Offices etc. including Warranty /Guarantee, Annual Maintenance Contract and insurance coverage for 5 (five) years on turn-key basis throughout the state of Tripura.

1. Certified that this DNIE-T contains 105 (one hundred five) pages numbered from 1 to 105.
2. Header of each page of DNIE-T bears the text “**e-TENDER FOR SPV POWER PLANTS AT BLOCK OFFICES IN TRIPURA**” under Special Assistance to States for Capital Investment.
3. The Bid Information Sheet of above e-Tender is shown in Page No. **3** to Page No. **5**.


12/6/26

(D S DAS)
Joint Director

SIGNATURE OF THE BIDDER WITH SEAL & DATE


12/6/26

TRIPURA RENEWABLE ENERGY DEVELOPMENT AGENCY

(Department of Power, Government of Tripura)

Vigyan Bhawan, Pandit Nehru Complex, Gorkhabasti, Agartala, Tripura

Tele-fax: 0381-2326139, e-mail: tredatenders25@gmail.comURL: www.treda.tripura.gov.in**BID INFORMATION SHEET**

1.	Name of work	Design, survey, manufacture, supply, erection, testing, commissioning of 10 kWp & 5 kWp Solar Photovoltaic Power Plant (Hybrid) at Block Offices etc. including Warranty / Guarantee, Annual Maintenance Contract and insurance coverage for 5 (five) years on turn-key basis throughout the state of Tripura
2.	Estimated cost	Rs.12,66,67,376/- (Rupees twelve crore sixty six lakh sixty seven thousand three hundred seventy six only)
3.	Tender Fee	Rs. 2,000/- (Rupees two thousand) only are to be paid electronically using on-line facility provided in the e-procurement portal (Non-refundable)
4.	Bid Security / EMD	Rs. 31,66,684/- (Rupees thirty one lakh sixty six thousand six hundred eighty four) only are to be paid in the shape of Bank Guarantee on any Nationalized Bank / Scheduled Bank guaranteed by RBI
5.	Completion period for the work	12 (twelve) months from the date of issue of Letter of Award (LoA)
6.	Date of Publishing of Tender	12/06/2026
7.	Bidding portal	https://tripuratenders.gov.in
8.	Document download Start date & Time from e-procurement portal	12/06/2026 at 06:00 PM
9.	Date of start of seeking clarification	12/06/2026 at 06:00 PM
10.	Closing date for seeking clarification	19/06/2026 at 12:30 PM
11.	Pre-bid meeting	19/06/2026 at 03:30 PM at the office of the Director General, Tripura Renewable Energy Development Agency, Vigyan Bhawan, Gorkhabasti, Agartala, West Tripura District, Pin: 799006.
12.	Bid Submission End Date & time	03/07/2026 at 03:00 PM

SIGNATURE OF THE BIDDER WITH SEAL & DATE



13.	Online Bid Opening Date & time of Technical bid	03/07/2026 at 03:30 PM
14.	Last date of submission of The Original Bank Guarantee for BID SECURITY / EARNEST MONEY DEPOSIT to Tendering Authority	Within 10 (ten) days (During office hours) from the date of opening of online technical bid.
15.	Online Bid Opening Date & time of Price bid	Date, time and venue for Price bid opening shall be communicated to Technically Qualified Bidders through e-tender portal.
16.	Place of Opening of Bids	In the office of the Director General, Tripura Renewable Energy Development Agency (A constituent organization of Department of Power, Govt. of Tripura), Vigyan Bhawan, 2nd Floor, Pandit Nehru Complex, Gorkhabasti, Agartala, West Tripura District, Pin: 799006.
17.	Bid Validity	180 (One hundred eighty) days from the date of opening of the tender.
18.	Officer Inviting Bids	Sri Debabrata Sukladas, Joint Director, TREDA

NOTE: All the above-mentioned time are as per clock time of e-procurement website <https://tripuratenders.gov.in>.



(D S DAS)
Joint Director

IMPORTANT NOTE:

- 1) Tender documents may be downloaded from Central Public Procurement Portal <https://tripuratenders.gov.in>. Aspiring Bidders who have not enrolled/registered in e-procurement should enroll/register before participating through the website <https://tripuratenders.gov.in>. The enrollment for the bidder is free of cost. Bidders need to go through the tender document where instructions are given.
- 2) Bidders can access tender documents on the website, fill them with all relevant information and submit the completed tender document into electronic tender on the website <https://tripuratenders.gov.in>.
- 3) Tender and supporting documents as per DNIE-T should be uploaded through website <https://tripuratenders.gov.in>. Hard copy of the tender documents will not be accepted. The successful bidder(s) may be required to produce original documents on getting communications from TREDA before issue of formal work



order.

- 4) Prospective bidders are requested to remain updated for any Notices/Amendments/Corrigendum etc. to the DNle-T document through the website **<https://tripuratenders.gov.in>**. No separate Notices would be issued for such Notices/Amendments/Clarifications etc. in the print media or individually. All the information related to this DNle-T shall only be uploaded in the website **<https://tripuratenders.gov.in>**.

ds
12/6/26

(D S DAS)

Joint Director

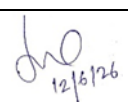
SIGNATURE OF THE BIDDER WITH SEAL & DATE

ds
12/6/26

2. TABLE OF CONTENTS

TOPIC		
PART - A		
PART - 1		PAGE NO.
e-tender Notice		8
Checklist of Annexure		9
Particulars of e-tender		11
PART - 2	INSTRUCTION TO BIDDERS	
SECTION - 1	The e-tender document	12
SECTION - 2	Bidder to inform fully	16
SECTION - 3	Eligibility conditions	22
SECTION - 4	Preparation of e-tender	26
SECTION - 5	Uploading of e-tender	29
SECTION - 6	e-tender opening and evaluation	30
SECTION - 7	Procedure for Finalisation of Bid	32
PART 3	GENERAL CONDITION OF THE CONTRACT	35
PART 4	SCOPE OF WORK & TECHNICAL SPECIFICATIONS	49
PART 5	DETAILS OF SCOPE OF WARRANTY/GUARANTEE AND ANNUAL MAINTENANCE CONTRACT (AMC)	72
PART 6	TECHNICAL INFORMATION	78
PART 7	QUALITY CERTIFICATION, STANDARDS AND TESTING FOR SOLAR PV POWER PLANTS	79
FORMAT 1	Format for Particulars of Bidder	81
FORMAT 2	Format for Schedule of experience / credentials	83
FORMAT 3	Format For Annual Turnover Data and Information on Profitability	85
FORMAT 4	Format of Bank Guarantee for Performance Bank Guarantee (PBG)	86
FORMAT 5	Format for Power of Attorney	87
FORMAT 6	Format for authorization letter	89
FORMAT 7	Format for Affidavit – No Black Listing	90
FORMAT 8	Format for Declaration	91
FORMAT 9	Format for Covering letter	93
FORMAT 10	Format for queries of bidder on tender for pre-bid meeting	94

FORMAT 11	Format for Letter of Intent (LoI)	95
FORMAT 12	Format for Declaration to submit PART 6 and Test Certificates / BIS Certificate by Bidder	96
FORMAT 13	Format for Site Visit Declaration	97
FORMAT 14	Format of Bank Guarantee for Earnest Money Deposit (EMD/BID Security)	98
FORMAT 15	Format for information of BID CAPACITY CALCULATION	100
PART – B: PRICE BID		102


 (D S DAS)
Joint Director

SIGNATURE OF THE BIDDER WITH SEAL & DATE


 12/6/26

TRIPURA RENEWABLE ENERGY DEVELOPMENT AGENCY

(Department of Power, Government of Tripura)

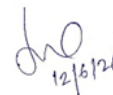
Vigyan Bhawan, Pandit Nehru Complex, Gorkhabasti, Agartala, Tripura

 Tele-fax: 0381-2326139, email-tredatenders25@gmail.com , URL: www.treda.tripura.gov.in
DNie-T No.F.6(542)/TREDA/NCES/2025-26/2110
Date 12/06/2026
NOTICE INVITING e-TENDER

On behalf of Tripura Renewable Energy Development Agency (TREDA), Joint Director, TREDA invites e-tenders from prospective bidders engaged in the business of Solar PV applications for the following work:

Sl. No.	Description	Quantity
1.	ITEM I: Design, survey, manufacture, supply, erection, testing, commissioning of 10 kWp Solar Photovoltaic Power Plant (Hybrid) at Block Offices etc. including Warranty / Guarantee, Annual Maintenance Contract and insurance coverage for 5 (five) years on turn-key basis throughout the state of Tripura	58 nos.
2.	ITEM II: Design, survey, manufacture, supply, erection, testing, commissioning of 5 kWp Solar Photovoltaic Power Plant (Hybrid) at Block Offices etc. including Warranty / Guarantee, Annual Maintenance Contract and insurance coverage for 5 (five) years on turn-key basis throughout the state of Tripura	29 nos.
TENDER FEE		Rs. 2,000/- (Rupees two thousand) only
BID SECURITY / EMD		Rs. 31,66,684/- (Rupees thirty one lakh sixty six thousand six hundred eighty four) only are to be paid in the shape of Bank Guarantee on any Nationalized Bank / Scheduled Bank guaranteed by RBI
PERIOD OF COMPLETION		12 (twelve) months from the date of issuance of Letter of Award
NOTICE INVITING e-TENDER		Nie-T No.F.6(542)/TREDA/NCES/2025-26/2109, dated 12/06/2026

- The other details related to this e-tender can be seen and downloaded from the website <https://www.tripuratenders.gov.in>.
- Amendment / Notification / Corrigendum / Addendum, if any, will be published only on the above website.



 (D S DAS)
Joint Director

3. **CHECKLIST OF ANNEXURE/DOCUMENT TO BE UPLOADED:** The following information/documents are to be annexed, flagged and uploaded by the Bidders along with the Technical BID. The list is indicative, any other document as required / specified in this DNIe-T are to be uploaded.

Sl. No.	Annexure No.	Particulars	Folder Number	Yes / No.
3.1	Annexure I	a) Blank copy of the e-tender document [Along with subsequent Amendment, Corrigendum, Notification, if any] signed and stamped on each page b) Scanned copy of BID SECURITY / EMD in the shape of BANK GUARANTEE	FOLDER 1	
3.2	Annexure II	Documents in support of General Eligibility of BIDDER as per Clause No. 3.1 of SECTION 3	FOLDER 2	
3.3	Annexure – III	Documents in support of Technical Eligibility of BIDDER as per Clause No. 3.2 of SECTION 3	FOLDER 3	
3.4	Annexure – IV	Documents in support of Financial Eligibility of BIDDER as per Clause No. 3.3 of SECTION 3	FOLDER 4	
3.5	Annexure V	Documents in support of BID CAPACITY of BIDDER as per Clause No. 3.4 of SECTION 3	FOLDER 5	
3.6	Annexure VI	a) Covering Letter as per FORMAT 9 b) DECLARATION by the bidder as per FORMAT 8 c) DECLARATION by the BIDDER to submit PART 6: Technical Information along with relevant documents as per FORMAT 12. d) SITE VISIT DECLARATION as per FORMAT 13	FOLDER 6	
3.7	Annexure VII	a) Power of Attorney of the BIDDER as per FORMAT 5 b) Authorization Letter of the BIDDER as per FORMAT 6 c) Affidavit by BIDDER to confirm that bidder has not been	FOLDER 7	

		Debarred or Blacklisted by any Government Department or Undertaking as per FORMAT 7 .		
		d) Any other document, as per DNle-T (Pl. specify)		

NOTE:

a) All the documents mentioned above as per DNle-T are mandatory qualifying requirements.

b) The bidder shall ensure that all the required documents, as mentioned in this bidding document, are uploaded along with the bid and in the PRESCRIBED FORMATS ONLY. Non - submission of the required documents or submission of the documents in a DIFFERENT FORMAT/ CONTENTS MAY LEAD TO THE REJECTIONS OF THE BID.

c) Please mark the Annexure number in right upper corner in all pages of the relevant uploaded documents and write Folder Name in the box.

d) Bids received without supporting documents as specified and mentioned in DNle-T shall be rejected.

SIGNATURE OF BIDDER WITH SEAL & DATE



Page 10 of 105

4. PARTICULARS OF e-TENDER:

4.1	e-tender No.	Nle-T No.F.6(542)/TREDA/NCES/2025-26/2109, dated 12/06/2026
4.2	Detailed e-tender No.	DNle-T No.F.6(542)/TREDA/NCES/2025-26/2110, dated 12/06/2026
4.3	Particulars of work	4.3.1 ITEM I: Design, survey, manufacture, supply, erection, testing, commissioning of 10 kWp Solar Photovoltaic Power Plant (Hybrid) at Block Offices etc. including Warranty /Guarantee, Annual Maintenance Contract and insurance coverage for 5 (five) years on turn-key basis throughout the state of Tripura. 4.3.2 ITEM II: Design, survey, manufacture, supply, erection, testing, commissioning of 5 kWp Solar Photovoltaic Power Plant (Hybrid) at Block Offices etc. including Warranty /Guarantee, Annual Maintenance Contract and insurance coverage for 5 (five) years on turn-key basis throughout the state of Tripura.
4.4	Period of completion	12 (twelve) months from the date of issuance of Letter of Award (LoA).
4.5	Period of validity of rates for acceptance	180 (One hundred eighty) days from the date of opening of tender.
4.6	Place of opening of e-tender	Office of the Director General, Tripura Renewable Energy Development Agency, Department of Power, Govt. of Tripura, Vigyan Bhawan, 2 nd Floor, Pandit Nehru Complex, Gorkhabasti, Agartala, West Tripura District, Pin: 799006.

- 4.7 The bidders can view the DNle-T and the time schedule (Key Dates) for all the tenders floated through the single portal e-procurement system on the Home Page at <https://www.tripuratenders.gov.in>.
- 4.8 Bidders are advised to study the tender Document carefully. Submission of e-Bid against this tender shall be deemed to have been done after careful study and examination of the procedures, terms and conditions of the tender Document with full understanding of its implications.
- 4.9 The bid should be submitted through e-Procurement website <https://www.tripuratenders.gov.in>.
- 4.10 The e-Bids will be electronically opened and bidders may view the bid opening through e-procurement portal. Physical presence of bidder at the venue will not be allowed.
- 4.11 In the event of date specified for e-Bids opening being declared a holiday for TREDA's office then the due date for opening of e-Bids shall be the following working day at the appointed time and place.
- 4.12 All the required documents including Price Schedule/BOQ should be uploaded by the Bidder electronically in the PDF/XLS format. The required electronic documents for each document label of Technical (Qualification details, e-Bid Form and Technical Specification details) schedules can be clubbed together to make single different files for each label. All the enclosures should be scanned and uploaded with bid as per requirement of e-procurement portal <https://tripuratenders.gov.in>

PART -2: INSTRUCTION TO BIDDERS (ITB)

SECTION -1: THE e-TENDER DOCUMENT

1.1 Scope of e-tender:

- 1.1.1** In connection with the Notice Inviting e-Tender (NIe-T), TREDA issues this Detailed e-tender document (DNIe-T) containing all the terms & conditions mentioned herein.
- 1.1.2** The Detailed e-tender along with the NIe-T, Amendment(s), Corrigendum(s), Notification(s), Clarification(s) and Addendum(s), if any, to be issued from time to time, shall be collectively termed as the bid document. Such a bid document shall be published on e-procurement website <https://tripuratenders.gov.in>.

1.2 Integrity violation:

- 1.2.1** The bidders observe the highest standards of ethics all the time.
- 1.2.2** TREDA defines, for the purposes of this provision, the terms set forth below as follows:
- 1.2.2.1** “Corrupt Practice” means the offering, giving, receiving or soliciting, directly or indirectly, anything of value to influence improperly the actions of another party;
- 1.2.2.2** “Fraudulent Practice” means any act or omission including a misrepresentation that knowingly or recklessly misleads or attempts to misled a party to obtain a financial or other benefit or to avoid an obligation;
- 1.2.2.3** “Coercive Practice” means impairing or harming or threatening to impair or harm, directly or indirectly, any Party or the property of a Party to influence improperly the actions of the other Party;
- 1.2.2.4** “Collusive Practice” means an arrangement between two (2) or more Parties designed to achieve an improper purpose, including influencing improperly the actions of the other Party;
- 1.2.2.5** “Obstructive Practice” means:
- 1.2.2.5.1** Deliberately destroying, falsifying, altering or concealing of evidence material to TREDA’s investigation;
- 1.2.2.5.2** Making false statements to investigators in order to materially impeded TREDA’s investigation;
- 1.2.2.5.3** Failing to comply with requests to provide information, documents or records in connection with TREDA’s Investigation;
- 1.2.2.5.4** Threatening, harassing or intimidating any Party to prevent it from disclosing its knowledge of matters relevant to the investigation or from perusing the investigation; or
- 1.2.2.5.5** Materially impeding TREDA’s Contractual Right of Audit or access to information.
- 1.2.2.6** “Integrity Violation” is an act which violates TREDA’s Policies, including Sub-clause No. 1.2.2.1 to Sub-clause No. 1.2.2.5 given above under ITB Clause No. 1.2.2 and the following abuse, conflict of interest, retaliation against whistleblowers or witnesses and other violations of TREDA’s Policies, including failure to adhere to the highest ethical standard.

- 1.2.3** TREDA will reject a bid, if it determines that the bidder has, directly or indirectly through an agent, engaged in Integrity Violation including but not limited to any Corrupt Practice, Fraudulent Practice, Coercive Practice, Collusive Practice and Obstructive Practice.
- 1.2.4** TREDA will impose punitive actions on any bidder or an individual, at any point time, in accordance with the bid document including blacklisting / debarment for a period of five (5) years to participate in TREDA managed, administrated or supported, activities or to benefit from TREDA managed, administrated or supported financially or otherwise if it at any time determines that the bidder or individual has directly or indirectly through an agent, engaged in Corrupt Practice, Fraudulent Practice, Coercive Practice, Collusive Practice, Obstructive Practice or Integrity Violation; and
- 1.2.5** TREDA will have the right to inspect the accounts, records, other documents, etc. of the bidders and relating to the bid submission and to have them audited at any point of time.

1.3 Contents of e-tender document: The e-tender procedure and contract terms are prescribed in the e-tender Documents. In addition to the e-tender Notice the Bidding documents include.

1.3.1 PART – A

1.3.1.1 PART – 1

1.3.1.1.1 e-tender Notice

1.3.1.1.2 Checklist of Annexure

1.3.1.1.3 Particulars of e-tender

1.3.1.2 PART 2: Instruction to bidder

1.3.1.2.1 SECTION -1 Contents of e-tender document

1.3.1.2.2 SECTION - 2 Bidder to inform fully

1.3.1.2.3 SECTION - 3 Eligibility condition

1.3.1.2.4 SECTION - 4 Preparation of e-tender

1.3.1.2.5 SECTION - 5 Uploading of e-tender

1.3.1.2.6 SECTION - 6 e-tender opening and evaluation

1.3.1.2.7 SECTION - 7 Procedure for Finalisation of Bid

1.3.1.3 PART 3: GENERAL CONDITION OF THE CONTRACT

1.3.1.4 PART 4: SCOPE OF WORK & TECHNICAL SPECIFICATIONS

1.3.1.5 PART 5: DETAILS OF WARRANTY/GUARANTEE AND ANNUAL MAINTENANCE CONTRACT OBLIGATIONS

1.3.1.6 PART 6: TECHNICAL INFORMATION

1.3.1.7 PART 7: QUALITY CERTIFICATION, STANDARDS AND TESTING FOR SOLAR PV POWER PLANTS AND OTHER ITEMS

1.3.2 PART – B: PRICE BID

[NOTE: The Bidder is expected to examine all instructions, forms, terms and specifications as mentioned in the e-tender document. Failure to furnish all information required by the e-tender documents or submission of a bid not substantially responsive to the Bid Document in every respect will be at the Bidder's risk and is likely to result in out-right rejection of the e-tender.]

1.4 Local Conditions: It shall be imperative on each bidder to fully inform himself of all local conditions and factors, which may have any effect on the execution of the works covered under these documents and specifications. TREDA shall not entertain any request for clarifications from the Bidder, regarding such local conditions.

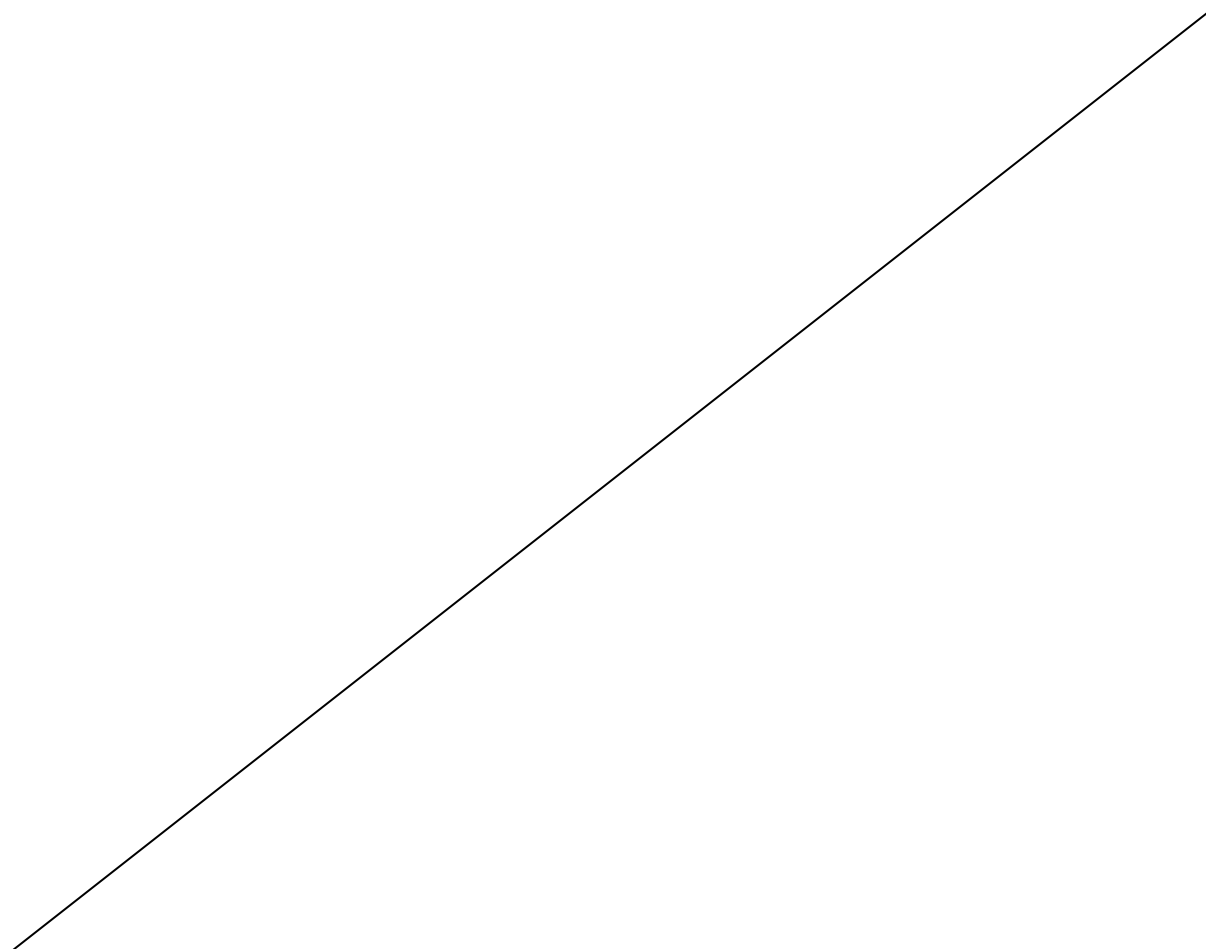
1.5 Clarification:

- 1.5.1** A prospective Bidder requiring any clarification of the e-tender Documents may contact TREDA in writing in writing through **<https://tripuratenders.gov.in>** in **FORMAT 10** within the timeline specified at **BID INFORMATION SHEET**.
- 1.5.2** TREDA will not be responsible for non-receipt of above communications.
- 1.5.3** Verbal clarifications and information given by the TREDA or its employees or its representatives shall not be in any way entertained.
- 1.5.4** The bidders have to remain updated with the e-procurement portal **<https://tripuratenders.gov.in>**.
- 1.5.5** The TREDA is not under any obligation to entertain or respond to suggestions made or to incorporate modifications sought for.

1.6 Amendment of e-Tender Document:

- 1.6.1** At any time prior to the due date for submission of the e-tender, TREDA may for any reason, whether at its own initiative or as a result of a request for clarification / suggestion by a prospective bidder, amend the tender document by issuing a notice.
- 1.6.2** The amendments will be notified on the **e-tender website <https://tripuratenders.gov.in>** at least **5 (five) days** before the last date of submission of the tender. TREDA will bear no responsibility or liability arising out of non-receipt of the information in time or otherwise. Bidders must check the website for any such amendment before submitting their bid.
- 1.6.3** If any Notification is to be issued within 5 (five) days from the last date of submission of tender, suitable time extension for bid submission would be given.
- 1.6.4** All the notices related to this bid which are required to be publicized shall be uploaded on website **<https://tripuratenders.gov.in>**.
- 1.7** The Bidder shall make independent enquiry and satisfy itself with respect to all the required information, inputs, conditions, including site conditions and circumstances and factors that may have any effect on its Bid. Once the Bidder has submitted the Bids, the Bidder shall be deemed to have examined the laws and regulations in force in India, the grid conditions, and prepare the Price Bid and other sections taking into account all such relevant conditions and also the risks, contingencies and other circumstances which may influence or affect the implementation of power plants. Accordingly, the Bidder acknowledges that, on being selected as the Selected Bidder, it shall not be relieved from any of its obligations foreseen under this Document nor shall be entitled to any extension of time for commissioning of the plants or financial compensation for any reason whatsoever.

- 1.8** The Bidders should particularly acquaint themselves with the technical requirements of integrating the power plant with the distribution system of the respective distribution licensee of the State, the regulations specified by Central Electricity Authority, grid operation as per the regulations / Codes as specified for the state of Tripura. The Tripura State Grid Code and the Tripura State Distribution Code should also be followed.
- 1.9** In their own interest, the Bidders are requested to familiarize themselves with the Electricity Act, 2003, scheme announced by Ministry of New and Renewable Energy for promotion of grid connected solar photovoltaic power plants, Regulatory Framework specified by the Tripura State Electricity Regulatory Commission (TERC) and any other local laws affecting the implementation of grid connected solar photovoltaic power plants and all other related acts, laws, rules and regulations prevalent in India, as amended from time to time. The TREDA shall not entertain any request for clarifications from the Bidders regarding the same. Non-awareness of these laws or such information shall not be a reason for the Bidder to request for extension in the Bid Deadline. The Bidder undertakes and agrees that, before submission of its Bid; all such factors as generally stated above, have been fully investigated and considered while submitting the Bid.



SECTION-2: BIDDERS TO INFORM FULLY

2.1 INTRODUCTION:

- 2.1.1. Government of Tripura under Part – XI (Implementation of SNA SPARSH Model) of the Scheme for Special Assistance to States for Capital Investment for 2024-25 has accorded Administrative & Financial Sanction for installation of Hybrid Solar Power Plant at Block Offices throughout the state of Tripura.
- 2.1.2. TREDA invited e-tender for selection of vendor for the following work:
 - 2.1.2.1. ITEM I: Design, survey, manufacture, supply, erection, testing, commissioning of 10 kWp Solar Photovoltaic Power Plant (Hybrid) at Block Offices etc. including Warranty /Guarantee, Annual Maintenance Contract and insurance coverage for 5 (five) years on turn-key basis throughout the state of Tripura
 - 2.1.2.2. ITEM II: Design, survey, manufacture, supply, erection, testing, commissioning of 5 kWp Solar Photovoltaic Power Plant (Hybrid) at Block Offices etc. including Warranty /Guarantee, Annual Maintenance Contract and insurance coverage for 5 (five) years on turn-key basis throughout the state of Tripura
- 2.1.3. The Solar Power Plants & Internal Electrification Works will be under the responsibility of the successful bidder for satisfactory & trouble-free operation for 5 (five) years Warranty / Guarantee and Annual Maintenance Contract (AMC).

- 2.2 LOCATION OF INSTALLATION:** Block / Panchayet Samity / Block Advisory Committee Offices etc. throughout the State of Tripura. Location of installation would be handed over during issuance of Letter of Award or despatch clearance.

2.3 MODE OF EXECUTION OF PROGRAMME:

- 2.3.1 The basis of evaluation of the bids shall be the cost/rate quoted in the Price Schedule. To further clarify, installation and commissioning cost and taxes etc. shall be inclusive to the cost of supply of complete system including 5 (FIVE) years comprehensive maintenance (25 years Performance warranty for PV Modules) for comparison and evaluation. The bidders are required to quote rate / cost on firm basis and no price variation on any account shall be considered.
- 2.3.2 The successful bidder shall provide maintenance services for FIVE (5) years with free replacement warranty on spare parts against manufacturing defects, normal wear & tear for five years. They shall also be required to set up their operation, repair and maintenance centres in Tripura with sufficient spares and technical manpower for providing effective repair/maintenance services to the beneficiaries / users.
- 2.3.3 The Programme shall be carried out as given here under: -
 - 2.3.3.1 The contractor shall be allowed to install the systems & other components conforming to the specifications mentioned in this document only after clearance from TREDA.
 - 2.3.3.2 The work covers design, manufacture, supply, erection, testing, commissioning and comprehensive maintenance including insurance coverage for FIVE (5) years of the Power Plant & IE works and 25 years Performance warranty for PV Modules.

2.3.3.3 TREDA reserves the right to change the site of installation before commencement of supply of materials.

2.4 TENDER FEE: - Rs. 2,000/- (Rupees two thousand) only to be paid electronically using the Online Payment Facility provided by the e-procurement Portal (Non-refundable).

2.5 BID SECURITY / EARNEST MONEY DEPOSIT (EMD): Rs. 31,66,684/- (Thirty one lakh sixty six thousand six hundred eighty four) only Bank Guarantee by RBI as per **FORMAT 14** of the DNIe-T.

2.5.1 The Bank Guarantee is to be necessary drawn-up before closing of bid.

2.5.2 The Bank Guarantee should be valid for a minimum period of 6 (six) months + 60 (sixty) days claim period.

2.5.3 Scanned copy of Bank Guarantee shall be uploaded in the <https://tripuratenders.gov.in> along with Technical bid.

2.5.4 The Bank Guarantee in physical form has to reach to Tendering Authority in a Sealed Envelope superscripting name of bidder, tender Reference Number and Tender ID in person or by post **within the stipulated time mentioned in the BID INFORMATION SHEET**. TREDA in no way will responsible for any postal delay or non-receipt of the Physical Bank Guarantee within the time specified.

2.5.5 TREDA reserve the right to take any punitive action including cancellation of bid against the bidder if anything found wrong in the BG submitted by the bidder.

2.5.6 Account Details of TREDA for preparation of Bank Guarantee towards BID SECURITY / EARNEST MONEY DEPOSIT (EMD) is furnished below:

2.5.6.1	Account Holder	:	Tripura Renewable Energy Development Agency
2.5.6.2	Bank	:	State Bank of India
2.5.6.3	Branch	:	T.L.A. House
2.5.6.4	Address	:	Colonel Chowmuhani, Agartala, West Tripura District, Pin: 799001
2.5.6.5	Account No.	:	10320318107
2.5.6.6	Account Type	:	Savings
2.5.6.7	IFSC	:	SBIN0005559
2.5.6.8	MICR	:	799002009

2.6 PROCEDURE FOR DEPOSITING TENDER FEE (TF):

2.6.1 Tender Fee is to be paid electronically using the online payment Facility provided in the e-procurement Portal.

2.6.2 After initiating the Bid Submission Process from “My Tender” option, an “Online Payment” page will appear which will display the total TF.

2.6.3 On submission of TF payment option, System will redirect to the SBI Bank MOPS window.

2.6.4 SBI MOPS will have two options for Net Banking – “SBI” & “Other Bank”. Bidder can choose any of the options as desired and can complete the Online Payment process.

2.7 The facilities may be extended as per norms of Tripura Industrial Investment Promotion Incentive Scheme – 2022 of Department of Industries &

Commerce, Govt. of Tripura to SSI Units registered with NSIC under its single point registration scheme having manufacturing unit (For one of the solar components specified at SECTION 3 of the DNle-T) at Tripura. The certificate issued by District Industries Office for having SSI units setup & other necessary documents which is mandatory to avail facilities should be uploaded along with technical bid.

2.8 RELEASE OF EARNEST MONEY DEPOSIT (EMD):

- 2.8.1 BID SECURITY / EMD in the form of BANK GUARANTEE shall be released to all the bidders including successful bidder after the Letter of Award (LoA) is uploaded in the e-procurement Portal <https://tripuratenders.gov.in>, on receipt of Performance Security (Performance Bank Guarantee) from the successful bidder.
- 2.8.2 The EMD of the successful bidder will be forfeited, if the successful bidder fails to submit "PERFORMANCE SECURITY [Performance Bank Guarantee/ Additional Performance Bank Guarantee (if required)]" within the stipulated period after issuance of Letter of Intent (LoI).
- 2.8.3 If any bidder withdraws his offer after opening Technical Bid before opening of Price bid or makes any modification in the terms and conditions of the tender which are not acceptable to the Tendering Authority, deposited BID SECURITY / EMD may be forfeited.
- 2.8.4 If any bidder withdraws his offer after opening of Price Bid within the period of bid validity or makes any modification in the terms and conditions of the tender which are not acceptable to the Tendering Authority, the deposited BID SECURITY / EMD may be forfeited.
- 2.8.5 If document(s) / certificate(s) submitted by the bidder(s) is/are found fake/false/fabricated, 100% BID SECURITY / EMD of the bidder(s) will be forfeited and the bidder will be suspended / debarred / blacklisted for 5(five) years from being eligible to submit Bids / proposals for contracts with TREDA along with other punitive actions deemed found suitable by TREDA will also be initiated.
- 2.8.6 The BID SECURITY / EMD by the bidders will not carry any interest.

2.9 PERFORMANCE SECURITY:

- 2.9.1 Successful bidder within 21 (twenty one) days of the issue of Letter of Intent (LoI) shall provide "PERFORMANCE BANK GUARANTEE (PBG)" and "ADDITIONAL PERFORMANCE BANK GUARANTEE" (Only for tenders less than 10% of the estimated cost put to the tender, the difference between the tender amount & 90% of the estimated cost put to the tender) as PERFORMANCE SECURITY for an amount equal to the 5% of the total contract value with a validity period including Warranty/Guarantee and Annual Maintenance Contract obligations + 60 (sixty) days claim period.
- 2.9.2 PERFORMANCE BANK GUARANTEE (PBG) and ADDITIONAL PERFORMANCE BANK GUARANTEE (if required) may be issued with a validity period initially for a period of 5 (five) years + 60 (sixty) days claim period which may be extended thereafter until completion of Warranty/Guarantee and Annual Maintenance Contract obligations.
- 2.9.3 The successful bidder, 1 (one) month prior to the date of expiry of the submitted Bank Guarantee as PBG & Additional PBG (If

- required), shall submit revalidated Bank Guarantee (PBG / Additional PBG / Performance Security) to TREDA.
- 2.9.4 Performance Bank Guarantee/ Additional Performance Guarantee (if required) should be submitted in favour of Director General, TREDA in the **FORMAT 4** (Performance Security Bond Form).
- 2.9.5 The Performance Security denominated in Indian Rupees shall be in the form of Bank Guarantee (BG) on Nationalized / Scheduled Bank guaranteed by the Reserve Bank of India.
- 2.9.6 Any request for time extension by the bidder for submission of “Performance Bank Guarantee/Additional Performance Bank Guarantee (if required)” will not be accepted.
- 2.9.7 On receipt of “Performance Bank Guarantee/Additional Performance Bank Guarantee (if required)” from the selected bidder, TREDA shall scrutinize the received instrument for its authenticity and validity for the Amount & Period.
- 2.9.8 The Performance Security shall be forfeited as follows without prejudice to the Bidder being liable for any further consequential loss or damage incurred to TREDA.
- 2.9.8.1 Non-execution of Contract Agreement after issuance of Letter of Award (LoA), hundred percent (100%) Performance Security amount submitted by the success bidder would be forfeited.
- 2.9.8.2 If the Successful Bidder is not able to commission the entire projects or part thereof to the satisfaction of TREDA, hundred percent (100%) Performance Security amount would be forfeited.
- 2.9.8.3 In the event of breach / violation or contravention of any terms and conditions contained herein by the agency i.e. if the selected bidder fails to execute the contract including compliance to Warranty/Guarantee, Annual Maintenance Contract and insurance coverage for satisfactory performance of the system, the Performance Security(s), part / whole as per decision of Tendering Authority, will be forfeited from the guarantor and the bidder will be suspended/ debarred / blacklisted for 5(five) years from being eligible to submit Bids / proposals for contracts with TREDA along with other punitive actions deemed found suitable by TREDA will also be initiated.
- 2.9.8.4 In all the above cases corresponding non-commissioned quantity shall stand cancelled.
- 2.9.8.5 If submitted Performance Bank Guarantee/ Additional Performance Bank Guarantee (if required) (Performance Security) is not revalidated within 1 (one) month prior to the date of expiry, hundred percent (100%) Performance Security amount would be forfeited.
- 2.9.8.6 If document(s) / certificate(s) submitted by the bidder(s) is/are found fake/false/fabricated after issuance of Letter of Award (LoA) / execution of Contract Agreement & even during the execution of works, Performance Bank Guarantee / Additional Performance Bank Guarantee (if required) (Performance Security) of the bidder will be forfeited and the bidder will be suspended / debarred / blacklisted for 5(five) years from being eligible to submit Bids / proposals for contracts with TREDA along with other punitive actions deemed found suitable by TREDA will also be initiated.

2.9.9 The Performance security shall be released after successful operation of Solar Power Plant including IE works and with the compliance of entire obligations in the contract from the date of commissioning.

2.10 BID INFORMATION:

2.10.1	Document Description	2.10.1.1 The bidding process under this is installation & commissioning of 87 nos. (10 kWp X 58 nos. & 5 kWp X 29 nos.) Hybrid SPV Power Plant including Internal Electrification Works at Block Offices. 2.10.1.2 Bidder must have to offer rates for all items of the PRICE BID SCHEDULE.
2.10.2	Broad Scope	2.10.2.1 ITEM I: Design, survey, manufacture, supply, erection, testing, commissioning of 10 kWp Solar Photovoltaic Power Plant (Hybrid) including Internal Electrification Works at Block Offices etc. as per the scope of work & technical specifications mentioned in the PART 4 of the DNle-T including Warranty /Guarantee, Annual Maintenance Contract and insurance coverage for 5 (five) years on turn-key basis throughout the state of Tripura. 2.10.2.2 ITEM II: Design, survey, manufacture, supply, erection, testing, commissioning of 5 kWp Solar Photovoltaic Power Plant (Hybrid) including Internal Electrification Works at Block Offices etc. as per the scope of work & technical specifications mentioned in the PART 4 of the DNle-T including Warranty /Guarantee, Annual Maintenance Contract and insurance coverage for 5 (five) years on turn-key basis throughout the state of Tripura. 2.10.2.3 Total timeline up to Commissioning for the above Scope of Work is 12 (twelve) months from the date of issue of Letter of Award (LoA).
2.10.3	Tender Fee	Rs. 2,000/- (Rupees two thousand) only are to be paid electronically using on-line facility provided in the e-procurement portal (Non-refundable).
2.10.4	Bid Security/ EMD	Rs. 31,66,684/- (Rupees thirty one lakh sixty six thousand six hundred eighty four) only are to be paid in the shape of Bank Guarantee on any Nationalized Bank / Scheduled Bank guaranteed by RBI
2.10.5	Performance Security	“Performance Bank Guarantee” for an amount equal to the 5% of the total contract value and “Additional Bank Guarantee” (if required) with a validity period including Warranty/Guarantee and Annual Maintenance Contract obligations + 60 (sixty) days claim period.

2.11 The bidder shall be responsible for Annual Maintenance including insurance coverage of the installed Solar PV system & IE Works for a period of 5 (FIVE) years, during which TREDA will monitor the project for effective & satisfactory performance in line with conditions specified elsewhere in the bid document.

- 2.12** The SPV power plant must obtain electrical safety inspection certificate from concerned authority for the system more than 10 KW capacities.
- 2.13** Tender without **TENDER FEE** and **BID SECURITY / EMD** shall not be accepted.
- 2.14** Any damage of the existing installations made by the successful bidder during execution of this work shall be made good as it was at his own cost & risk.
- 2.15 COST OF BIDDING:** The bidder shall bear all the costs associated with the preparation and submission of his offer, and the TREDA will in no case be responsible or liable for those costs, under any conditions. The Bidder shall not be entitled to claim any costs, charges and expenses of and incidental to or incurred by him through or in connection with submission of bid even though TREDA may elect to modify / withdraw the invitation of Bid.
- 2.16 ZERO DEVIATION:** This is a ZERO Deviation Bidding Process. Bidder is to ensure compliance of all provisions of the Bid Document and submit their Bid accordingly. Tenders with any deviation to the bid conditions shall be liable for rejection.
- 2.17 RIGHT TO ACCEPT / REJECT THE BID:** TREDA reserves the right to accept or reject any Bid or annul the bidding process and reject all such bids at any time prior to award of contract, without thereby incurring any liability to the affected applicant(s) or any obligation to inform the affected applicant(s) of the ground for such decision.

SIGNATURE OF BIDDER WITH SEAL & DATE



SECTION – 3: ELIGIBILITY CONDITION

3.1 GENERAL ELIGIBILITY	
CRITERIA	DOCUMENTS TO BE UPLOADED ALONG WITH TECHNICAL BID
3.1.1 The bidder must have any of the following LEGAL STATUS:	a) Duly filled FORMAT - 1
3.1.1.1 Body incorporated in India under the Companies Act, 1956 or Companies Act, 2013 including any amendment there to OR	b) Copy of Certificate of Incorporation from Registrar of Companies for Body incorporated in India under the Companies Act, 1956 or Companies Act, 2013 including any amendment there to OR
3.1.1.2 Government of India / State Government Undertakings.	c) Documents to satisfy the legal status as Government of India / State Government Undertakings.
3.1.2 The bidder must have to fulfil any one of the following TECHNICAL STATUS	
3.1.2.1 <u>SPV SYSTEM MANUFACTURER:</u> Manufacturer of either: 3.1.2.1.1 SPV Module [Enlisted under ALMM of MNRE published by MNRE time to time] AND / OR 3.1.2.1.2 Power Conditioning Unit (PCU) of Off-grid /Hybrid SPV Power Plant AND / OR 3.1.2.1.3 Tubular GEL Battery	a) The any of the following documents to satisfy the eligibility as SPV SYSTEM MANUFACTURER: i. PV Module: BIS Certificate and latest ALMM List of MNRE ii. Power Conditioning Unit (PCU) of SPV Power Plants: BIS Certificate / Test Report iii. Tubular GEL Battery: BIS Certificate / Test Report [NOTE: For Test Reports should be in in compliance with the Clause No. 8 under PART 3 of the DNIE-T]
3.1.2.2 <u>SPV SYSTEM INTEGRATOR:</u> The bidder may be System Integrator i.e. the components of Off-grid/Hybrid SPV Power Plants / Grid connected SPV Power Plant (RTS / Ground Mounted) procured by the bidder through outsourcing & installed by themselves.	Work execution in compliance with the DNIE-T to satisfy eligibility <u>SPV SYSTEM INTEGRATOR</u>

3.2	TECHNICAL ELIGIBILITY
CRITERIA	DOCUMENTS TO BE UPLOADED ALONG WITH TECHNICAL BID
3.2.1 The bidder should have successfully completed successful supply, installation & commissioning of Hybrid/Off-grid Solar Power Plant AND / OR Grid connected Solar Power Plants (RTS / Ground Mounted) in India during last seven (7) years ending last day of the month previous to the month of the floating of the tender and should have either of the followings: 3.2.1.1 THREE (3) Letter of Awards (LoAs) & corresponding completed works of Hybrid/Off-grid Solar Power Plant AND / OR Grid connected Solar Power Plants (RTS / Ground Mounted) not less than the aggregated capacity of 290 kWp each. 3.2.1.2 TWO (2) Letter of Awards (LoAs) & corresponding completed works of Hybrid/Off-grid Solar Power Plant AND / OR Grid connected Solar Power Plants (RTS / Ground Mounted) not less than the aggregated capacity of 363 kWp each. 3.2.1.3 ONE (1) Letter of Award (LoA) & corresponding completed works of Hybrid /Off-grid Solar Power Plant AND / OR Grid connected Solar Power Plants (RTS / Ground Mounted) not less than the aggregated capacity of 580 kWp each. 3.2.2 Letter of Award / Work Order, corresponding Works completion / Commissioning Certificate / Grid Synchronization Report issued in favour of the bidder will only be accepted. 3.2.3 Letter of Award / Work Order, corresponding Works completion / Commissioning Certificates/ Grid Synchronization Report should be issued by the authorised representative of State Nodal Agency for RE / DISCOM (Government / Private) / Govt. Departments / Government Organization of the concerned State. 3.2.4 Letter of Award / Work Order, Works	a) Duly filled FORMAT - 2 b) Copies of the Letter of Awards / Work Orders, corresponding Works completion / Commissioning Certificates / Grid Synchronization Report as Experience of executing of works. c) In case of Power Purchase Agreement (PPA), PPA signed between Government Department and the bidder for Grid connected SPV Power Plants, Installation / Commissioning Certificate / Grid Synchronization Report from authorised representative of concerned Govt. Departments / DISCOM (Government / Private) shall be submitted along with PPA.

completion / Commissioning Certificates/ Grid Synchronization Report of any Collaborator / Sister Concern / Principal Manufacturer / Lead Member of Joint Venture of the bidder submitted as experience proof of the bidder will not be accepted.	
3.3 FINANCIAL ELIGIBILITY	
CRITERIA	DOCUMENTS TO BE UPLOADED ALONG WITH TECHNICAL BID
3.3.1 The financial capability of the bidder to take up the proposed work to be supported by AUDITED BALANCE SHEET for any four (4) Financial Years out of last five (5) Financial years [i.e. 2021-22, 2022-23, 2023-24, 2024-25 and 2025-26 (In case of 2025-26, provisional balance sheet along with CA certificate)]. 3.3.2 Minimum Average Annual Turnover (MAAT) of the bidder in the any four (4) financial years out of last five (5) financial years (i.e. 2021-22, 2022-23, 2023-24, 2024-25 and 2025-26) should be minimum Rs. 6.34 crore (Rupees six crore thirty four lakh) only. 3.3.3 The bidder should have positive NET WORTH in last four (4) Financial Years out of five (5) Financial Years (i.e. 2021-22, 2022-23, 2023-24, 2024-25 and 2025-26) 3.3.4 The Minimum Average Annual Turnover (MAAT), NET WORTH & Profitability in FORMAT 3 should be certified by practicing CA of the bidder. 3.3.5 The MAAT, NET WORTH & PROFITABILITY submitted along with the bid shall be of bidder, MAAT, NET WORTH & PROFITABILITY of group of Companies will not be considered as financial eligibility of the bidder. 3.3.6 The bidder must have the followings: 3.3.6.1 PAN Card 3.3.6.2 GST Registration Certificate 3.3.6.3 Income Tax Return for 4 Assessment Years out of last 5 Assessment Years (i.e. 2020-21, 2021-22, 2022-23, 2023-24 & 2024-25).	a) AUDITED BALANCE SHEET for any four (4) financial years out of last five (5) Financial Years [i.e. 2021-22, 2022-23, 2023-24, 2024-25 and 2025-26 (In case of 2025-26, provisional balance sheet)] duly certified by Chartered Account in original on their letterhead with Membership No., FRN, UDIN, contact details (e-mail and Mobile Number). b) The Minimum Average Annual Turnover (MAAT), Net Worth & Profitability in FORMAT 3 duly certified by Chartered Account in original on their letterhead with Membership No., FRN, UDIN, Contact details (e-mail and Mobile Number). c) Documents specified at Clause No. 3.3.6.

3.4 BID CAPACITY

CRITERIA			DOCUMENTS TO BE UPLOADED ALONG WITH TECHNICAL BID
3.4.1 The bidder(s) must have AVAILABLE BID CAPACITY equal or more than the estimated cost of work put to the tender. 3.4.2 The BID CAPACITY shall be calculated using the following formula: BID CAPACITY = (A*N*M)-B Where: A= Maximum value of engineering works in respect to the SOLAR PV Projects executed in any one year during the last five (5) years taking into account the completed as well as work in progress. The value of works shall be brought to current costing level by enhancing at a simple rate of 7% per annum. N= Number of years prescribed for completion of work for which bids has been invited (N should be taken as 0.5 for the period up to six months and more than six months up to twelve months as 1 and so on) i.e. 1. M = Multiplying factor i.e. 1.5 B= Value of existing commitments and ongoing works during the period of “N” years for which bids has been invited.			FORMAT 15 (PART A) and FORMAT 15 (PART B) duly certified by Chartered Account in original on their letterhead with Membership No., FRN, UDIN, Contact details (e-mail and Mobile Number).
3.5	TENDER FEE	Rs. 2,000/- (Rupees two thousand) only	Payment through e-procurement portal
3.6	BID SECURITY / EMD	Rs. 31,66,684/- (Rupees thirty one lakh sixty six thousand six hundred eighty four) only	In the shape of Bank Guarantee on any Nationalized Bank / Scheduled Bank guaranteed by RBI

- 3.7** Bidders shall maintain strict adherence while filling the formats as specified in this document during submission of bids. Wherever, information has been sought in specified formats, the Bidder shall refrain from referring to brochures / pamphlets. Non-adherence to formats and / or submission of incomplete information may be a ground for declaring the Bid as non-responsive. Each format has to be duly filled in, signed and stamped by the authorized signatory of the Bidder then scanned and uploaded in the Technical Bid.

SECTION-4: PREPARATION OF e-TENDER

- 4.1 LANGUAGE OF BID AND MEASURE:** The e-tender prepared by the Bidder and all correspondence and documents relating to the bid exchanged by the Bidder and TREDA shall be written in the **English** provided that any printed literature furnished by the Bidder may be written in **another language** so long as accompanied by an English translation of its pertinent passages for purpose of interpretation, the English translation should be attested by **NOTARY PUBLIC**. Units of measurement shall be MKS system.
- 4.2 DOCUMENTS COMPRISING THE BID:** The e-tender prepared by the Bidder shall comprise the following components and should be uploaded in the e-procurement portal in the respective folders as specified at Checklist Of Annexure / Documents:
- 4.2.1 Blank copy of the e-tender document [Along with subsequent Amendment, Corrigendum and Notification, if any] signed and stamped on each page, as a confirmation by the Bidder to accept all technical specifications / commercial conditions.
 - 4.2.2 Scanned copy of BID SECURITY / EMD in the shape of BANK GUARANTEE in the format as provided in e-tender document
 - 4.2.3 Documents in support of GENERAL ELIGIBILITY of BIDDER as per Clause No. 3.1 of SECTION 3
 - 4.2.4 Documents in support of TECHNICAL ELIGIBILITY of BIDDER as per Clause No. 3.2 of SECTION 3
 - 4.2.5 Documents in support of FINANCIAL ELIGIBILITY of BIDDER as per Clause No. 3.3 of SECTION 3
 - 4.2.6 Documents in support of BID CAPACITY of BIDDER as per Clause No. 3.4 of SECTION 3
 - 4.2.7 COVERING LETTER by BIDDER in the format as provided in e-tender document
 - 4.2.8 DECLARATION by the BIDDER in the format as provided in e-tender document.
 - 4.2.9 DECLARATION by BIDDER to submit “PART 6: Technical Information along with relevant documents” in the format as provided in e-tender document.
 - 4.2.10 SITE VISIT DECLARATION by BIDDER in the format as provided in e-tender document.
 - 4.2.11 POWER OF ATTORNEY along with Board Resolution of BIDDER for the person representing bidder in the format as provided in e-tender document
 - 4.2.12 AUTHORIZATION LETTER of BIDDER for the person representing bidder in the format as provided in e-tender document
 - 4.2.13 AFFIDAVIT by BIDDER to confirm that BIDDER has not been Debarred or Blacklisted by any Government Department or Undertaking in the format as provided in e-tender document
 - 4.2.14 Any other document, as per DNIe-T (Pl. specify).
- 4.3** The bidder shall ensure that all the required documents, as mentioned in this bidding document, are uploaded along with the bid and in the **PRESCRIBED FORMATS ONLY**. Non - submission of the required documents or submission of the documents in a **DIFFERENT FORMAT/ CONTENTS** may lead to the rejections of the bid.

4.4 BID PRICE:

- 4.4.1 The Bidder shall indicate prices on the appropriate price bid schedule.
- 4.4.2 Bidder shall quote rate for all components of Bill of Quantity (BOQ).
- 4.4.3 The rate quoted in the Bill of Quantity (BOQ) which is in MS-Excel (Macro enabled) should be in Indian Rupees and be firm, inclusive all the charges including GST etc. as required.
- 4.4.4 DUTIES AND TAXES:
 - 4.4.4.1 The price quoted should include all taxes as applicable. Except as otherwise, specifically provided in the contract, the bidder shall bear and pay all taxes, duties, levies and charges assessed on the bidder by all Municipal, State or National Govt. Authorities. All taxes, duties & levies on the works Contract, if any, shall be to the bidders account and no separate claim in this regard will be entertained by TREDA.
 - 4.4.4.2 TDS on Income as applicable will be deducted at source as per the prevailing laws.
 - 4.4.4.3 GST as applicable will be deducted at source as per the prevailing laws and rules of Government of India and Government of Tripura.
 - 4.4.4.4 LABOUR CESS as applicable will be deducted at source as per the prevailing laws and rules of Government of India / Government of Tripura.
 - 4.4.4.5 In case of variation in applicable GST (+ or -) during the Contract Period, the same will be adjusted accordingly.

- 4.5 **BANK ACCOUNT LINKED WITH GOODS AND SERVICES TAX IDENTIFICATION NUMBER (GSTIN):** Every participating bidder to mandatorily disclose the **BANK ACCOUNT NUMBER** which is linked with their **GSTIN** at the time of bid submission. No payments shall be released by the Government Department / Agency to any other Bank Account except the one linked with the GST Registered Number of the successful bidder.

- 4.6 **BID CURRENCIES:** Prices shall be quoted in Indian Rupees (INR) only.

4.7 PERIOD OF VALIDITY OF e-TENDER:

- 4.7.1 Validity of the offer should be 180 (one hundred eighty) days from the date of opening of tender.
- 4.7.2 In exceptional circumstances; the TREDA will solicit the Bidder's consent to an extension of the period of validity. The request and the response there of, shall be made in writing.

4.8 BID SECURITY / EARNEST MONEY DEPOSIT (EMD):

- 4.8.1 The bidder shall submit, as part of its bid, BID SECURITY / EMD in the shape of BANK GUARANTEE as per **FORMAT 14**.
- 4.8.2 Any bid not secured with the TENDER FEE and EARNEST MONEY DEPOSIT will be rejected by the TREDA as non-responsive.
- 4.8.3 BID SECURITY / EMD shall be released to all the bidders including success bidder after the Letter of Award (LoA) is issued through e-procurement Portal <https://tripuratenders.gov.in> on receipt of PERFORMANCE SECURITY from the successful bidder within the stipulated time period stipulated in DNIE-T.

**4.9 FORMAT AND SIGNING OF e-TENDER:**

- 4.9.1 The bid must contain the name, residence and places of business of the persons making the e-tender and must be signed and sealed by the Bidder with his usual signature. The name and designations of all persons signing should be typed or printed below the signature.
- 4.9.2 e-tender by Corporation/ Company must be signed with the legal name of the Corporation/ Company/Firm by the "President", "Managing Director" or by the "Secretary" or other designation or a person duly authorized.
- 4.9.3 The downloaded copy of the e-tender shall be signed by the Bidder or a person duly authorized. A Letter of Authorization shall be submitted along with Power-Of-Attorney. All the pages of the bid shall be initialled by the person or persons signing the e-tender.
- 4.9.4 The bid shall contain no interlineations, erasures or overwriting except as necessary to correct errors made by the Bidder in which case such corrections shall be initialled by the person or persons signing the e-tender.

SIGNATURE OF BIDDER WITH SEAL & DATE

SECTION-5: UPLOADING OF e-TENDER

- 5.1 Bid should be uploaded online in e-procurement portal **<https://tripuratenders.gov.in>**.
- 5.2 The tender must be complete in all technical and commercial respect and should contain requisite certificate, drawings, informative literature etc. required as per the DNIE-T.
- 5.3 First part **(PART-A)** should contain technical bid comprising of all technical documents as per DNIE-T. All parts of tender documents except PRICE BID should be uploaded as per e-procurement mode within due date and time. Scanned copy of documents as specified in DNIE-T should be uploaded.
- 5.4 The Bidder should submit PRICE BID in Second part. Second part **(PART-B)** should contain price bid only and should be uploaded as per e-procurement mode within due date and time. Anything in regard of financial condition, payment terms, rebate etc. mentioned in price bid may make the tender invalid. Therefore, it is in the interest of the Bidder not to write anything extra except price.
- 5.5 The original copy of uploaded document i.e. First part **(PART-A)** is not required to be submitted to TREDA. However, if it is required so, may be submitted on getting letter from TREDA.
- 5.6 **DEADLINE FOR SUBMISSION OF BIDS:** Bids must be uploaded by the bidder in e-procurement portal **<https://tripuratenders.gov.in>** on or before **03/07/2026 up to 03:00 PM.**

SECTION - 6: e-TENDER OPENING AND EVALUATION

6.1 OPENING OF e-TENDER: The procedure of opening of the e-tender shall be as under:

6.1.1 FIRST PART (PART-A) i.e. Technical bid of the tender would be opened online by TREDA on **03/07/2026 at 03:30 PM** in the office of the Director General, TREDA, Vigyan Bhawan, Gorkhabasti, Agartala, West Tripura District, Pin – 799006. Bidders can view the opening of tender through e-procurement portal **<https://tripuratenders.gov.in>**. **Physical presence of the bidder or authorised representative of the bidder is not permitted.**

6.1.2 RESPONSIVENESS CHECK OF TECHNICAL BID: The Technical Bid (PART A) submitted by Bidders shall be scrutinized to establish responsiveness to the requirements laid down in the SECTION 3 and CLAUSE NO. 4.2 (DOCUMENTS COMPRISING THE BID) under SECTION-4 of the DNIE-T. Any of the following may cause the Bid to be considered “Informal / Non-responsive”, at the sole discretion of TREDA:

- 6.1.2.1 Bids that are incomplete, i.e. not accompanied by any of the applicable formats inter alia covering letter, power of attorney supported by a board resolution, other documents specified at Clause No. 4.2 under SECTION 4 of the DNIE-T etc.
- 6.1.2.2 Bid not signed by authorized signatory and /or stamped in the manner indicated in this DNIE-T;
- 6.1.2.3 Material inconsistencies in the information /documents submitted by the Bidder, affecting the Eligibility Criteria;
- 6.1.2.4 Information not submitted in the formats specified in this DNIE-T;
- 6.1.2.5 Bid being conditional in nature;
- 6.1.2.6 Bid not received by the Bid Deadline;
- 6.1.2.7 Bid having Conflict of Interest;
- 6.1.2.8 Bidder delaying in submission of additional information or clarifications sought by TREDA as applicable;
- 6.1.2.9 Bidder makes any misrepresentation.

6.1.3 SECOND PART (PART-B) containing Price Bid shall be opened (after obtaining clarifications and establishing technical suitability of the offer) as per schedule. Second part (PART B) of only those Bidders shall be opened whose first part (PART-A, technical bid) is found substantially responsive as per terms & conditions of DNIE-T and evaluated as technically qualified.

6.1.4 The date of opening of Price bid of technically qualified bidders would be intimated through e-procurement portal **<https://tripuratenders.gov.in>**.

6.2 CLARIFICATION ON SUBMITTED e-TENDER DOCUMENT:

6.2.1 During the process of evaluation of the tender (Technical & Price bid), TREDA at its discretion may ask the bidder for a clarification of his tender either in written or fixing meeting inviting bidders at TREDA Office, Agartala. The request for clarification and the response shall be in writing & in English only.

6.2.2 Any query regarding any clarification required by TREDA on the information submitted by the bidder, must be replied by the bidder within the allowed time schedule.

6.3 If document(s) / certificate(s) submitted by the bidder(s) is/are found fake/false/fabricated, Bid Security / EMD of the bidder will be forfeited and the bidder will be suspended / debarred / blacklisted for 5(five) years from being eligible to submit Bids / proposals for contracts with TREDA along with other punitive actions deemed found suitable by TREDA will also be initiated.

6.4 TREDA reserves the right to interpret the Bid submitted by the Bidder in accordance with the provisions of this document and make its own judgment regarding the interpretation of the same. In this regard TREDA shall have no liability towards any Bidder and no Bidder shall have any recourse to TREDA with respect to the selection process. TREDA shall evaluate the Bids using the evaluation process specified in this document or as amended, at its sole discretion. TREDA's decision in this regard shall be final and binding on the Bidders.

SECTION-7: PROCEDURE FOR FINALISATION OF BID

7.1 THE PROCEDURE FOR FINALIZATION OF BID WOULD BE AS FOLLOWS:

- 7.1.1 First the Technical bids shall be opened and evaluated by Tender Evaluation Committee of TREDA.
- 7.1.2 Then the price bid of technically qualified bidders shall be opened.
- 7.1.3 The bidder quoting the Overall Lowest Price will be declared as successful bidder (L1) and the Price quoted by successful bidder (L1) shall be considered as "APPROVED LOWEST PRICE" for awarding the contract.
- 7.1.4 If it is found that the bidders have quoted various GST rates, TREDA will consider uniform GST as per GST Rules as applicable on basic rates offered by the bidders and "APPROVED LOWEST PRICE" will be evaluated.

7.2 FINALIZATION OF TENDER:

- 7.2.1 The bidder quoting the Overall Lowest Price (i.e. "APPROVED LOWEST PRICE") will be declared as successful bidder (L1).
- 7.2.2 For allocation of tendered quantity, TREDA shall arrange the bids in ascending order i.e. L2, L3, L4 ---- & so on (L2 bidder being the second lowest quoted) and "Rates of Approved Lowest Price" would be offered to those respective lowest bidders, having quoted price not more than 25% of "APPROVED LOWEST PRICE", to match with "Rates of Approved Lowest Price".
- 7.2.3 In case of L2 bidder does not agree to match the Approved Lowest Price, TREDA will offer the same to L3, L4, L5 ---- & so on for allocating capacity/quantity at the Approved Lowest Price.
- 7.2.4 If two or more bidders quote the same price, the selection of bidder will be in the following order:
 - 7.2.4.1 The bidder whose BID CAPACITY is higher will be selected.
 - 7.2.4.2 In case, the BID CAPACITY is also same, the order of section of bidder would be determined based on the highest Annual Turnover.
 - 7.2.4.3 Even if, the criteria incidentally become the same, highest Net Worth, failing which Higher Profitability will determine the order of selection.
 - 7.2.4.4 Information furnished at FORMAT 3 of the bidder would be considered to determine the selection of order specified under Sub-clause No. 7.2.4.2 and Sub-clause No. 7.2.4.3.
- 7.2.5 TWO (2) bidders in APPROVED LOWEST PRICE may be empanelled against this tender.
- 7.2.6 Allocation of tendered capacity shall be on discretion of TREDA. Decision of TREDA in this regard should be final & binding on successful bidder.
- 7.2.7 In case, the numbers of qualified bidders are less than the required quantity, L1 may get full allocation or as decided by TREDA. However, decision of TREDA in this regard should be final & binding on successful bidder.
- 7.2.8 TREDA reserves the right to increase/decrease the bidder(s) allocated quantity at the sole discretion of TREDA.
- 7.2.9 The completion of above mentioned work is time bound and TREDA will monitor the progress of work by the successful bidder(s) to whom the allocation is made. In case, the progress of work is not at per agreed PERT Network / LoA schedule, TREDA will review the allocated quantity and based on the performance of the bidders, TREDA reserves the right to reallocate the LoA quantity among the successful bidder(s). In such cases, decision of TREDA for revised allocation will be final & binding on the successful bidders.

7.2.10 The selection process shall stand completed once the Tendered Capacity has been achieved through the allocation of the capacity offered to the Successful Bidder(s).

7.3 **ABNORMALLY LOW BIDS:** TREDA may seek written clarification from the bidder, including detailed price analysis of its rates in relation to scope, schedule, resource mobilization, allocation of risks and responsibilities, any other requirements of the bid document. If, after evaluating the price analysis, TREDA determines that the bidder has substantially failed to demonstrate its capability to deliver the contract at the offered price, TREDA may reject the bid / proposal.

7.4 **CONFIDENTIALITY:**

7.4.1 Information relating to the examination, evaluation, comparison and post qualification of the bids and recommendation for issue of Letter of Award, shall not be disclosed to Bidders or any other person not officially concerned with such process until information on the issuance of Letter of Award (LoA) is communicated to all bidders through e-procurement portal, unless it is specifically required by TREDA to do such disclosure as per the specific requirements.

7.4.2 Any attempt by the Bidders to influence TREDA's decision in the evaluation of the bids or issuance of Letter of Award may result in the rejection of its bid.

7.4.3 Notwithstanding Sub-clause No. 7.4.2 above, from the time of bid opening to the issuance of Letter of Award, if any bidder wishes to contact TREDA on any matter related to the bidding process, it may do so in writing only.

7.5 **VALIDITY OF TENDER & FINALIZED RATE:** The APPROVED LOWEST PRICE (excluding GST) will be valid for 18 (eighteen) months from the date of issue of Letter of Award (LoA). However, the validity of L1 rates may be extended on mutual agreement between TREDA & the successful bidder for another period of 6(six) months.

7.6 **LETTER OF INTENT (LoI):**

7.6.1 Prior to the expiration of the period of Bid validity, TREDA will notify successful bidder in writing by issuing Letter of Intent (LoI) as per **FORMAT 11** either through scanned e-mail or through registered / speed post / courier that his bid has been accepted and will request the bidder to submit "Performance Bank Guarantee/ Additional Performance Bank Guarantee (If required)" as Performance Security as per Clause No. 2.9 of SECTION 2 of the DNle-T.

7.6.2 Failure of the successful bidder to complete all the formalities mentioned in the Letter of Intent (LoI) shall constitute sufficient grounds for the annulment of the Letter of Intent (LoI) and forfeiture of the Bid Security / EMD. In this event, TREDA may issue the Letter of Intent to next lowest evaluated (L2, L3 and so on in sequence bidders quoted price not more than 25%) at the Approved Lowest Price. In case any of the bidder from L2, L3 and so on in sequence bidders (quoted price not more than 25%) does not accept the Letter of Intent, then TREDA shall have the right to select the successful bidder or annul the tender process at its own discretion.

7.7 LETTER OF AWARD (LoA):

- 7.7.1 After receipt of Performance Security [Performance Bank Guarantee /Additional Performance Bank Guarantee (if required)] from successful bidder, TREDA will notify the successful bidder(s) in writing through issuing Letter of Award (LoA).
- 7.7.2 **LoA will be uploaded in e-procurement portal <https://tripuratenders.gov.in>.**
- 7.7.3 The LoA will constitute the formation of the contract and the successful bidder has to return the duplicate copy of the LoA with duly signed as acceptance of LoA within 10 (ten) days, otherwise LoA will be deemed to be accepted by the successful bidder on whom the award is given.
- 7.7.4 The Successful Bidder shall not assign or make over the work, the benefit or burden thereof to any other person or persons or body corporate for the execution of the contract or any part thereof without the prior written consent of TREDA. TREDA reserves its right to cancel the Letter of Award (LoA) either in part or full, if this condition is violated.

7.8 CONTRACT AGREEMENT:

- 7.8.1 A contract agreement for execution of the work shall be signed by the successful bidder with TREDA within 20 (twenty) days of issuance of Letter of Award (LoA) by TREDA.
- 7.8.2 In case agreement is not executed within the stipulated time, the LoA will be cancelled and necessary actions as per DNIE-T will be initiated.
- 7.8.3 Successful bidder is to make in his own cost 2 (two) copies of Contract Agreement with TREDA in non-judicial Stamp paper of Rs. 200/- (Rupees two hundred) each and make 2 (two) Original copies of Contract document in bind volume containing Contract Agreement at the top, Letter of Intent (LoI), Letter of Award (LoA), Techno-commercial Offer of the bidder, Copy of Price Bid and copies of all tender documents in support of your proposal submitted against the DNIE-T which are to be jointly signed by TREDA & the successful bidder.
- 7.8.4 7 (seven) more copies of Contract Agreement in bind volume are to be submitted by the successful bidder to TREDA at the cost of the successful bidder.
- 7.8.5 The denial of the successful bidder to undertake the work after issue of Letter of Award (LoA) / execution of Contract Agreement shall be treated as breach of contract and hundred percent (100%) Performance Bank Guarantee (PBG) amount submitted by successful bidder would be forfeited by TREDA.

7.9 INCREASE/DECREASE OF ALLOCATED QUANTITY:

- 7.9.1 TREDA reserves the right to increase or decrease @25% of the total tendered quantity at the time of issuance of Letter of Award.
- 7.9.2 TREDA may place Letter of Award for additional quantity up to 25% of the total tendered quantity within the validity of tender as per approved rates, technical specifications, terms & conditions etc.
- 7.9.3 In case capacity is enhanced by TREDA, successful bidder shall submit the PERFORMANCE SECURITY as per CLAUSE NO. 2.9 of SECTION 2 of the DNIE-T on issuance of such Letter of Intent (LoI).

PART – 3: GENERAL CONDITION OF THE CONTRACT

1 DEFINITIONS:

- 1.1 **“Tendering Authority”** shall mean the undersigned i.e. the Joint Director, Tripura Renewable Energy Development Agency, A constituent organization of Department of Power, Govt. of Tripura with its Head office at Vigyan Bhawan, 2nd Floor, Pandit Nehru Complex, Gorkhabasti, Agartala, West Tripura District, Pin: 799006.
- 1.2 **“TREDA”** shall mean The Director General or his representative of “Tripura Renewable Energy Development Agency, A constituent organization of Department of Power, Govt. of Tripura” with its Head Office at Vigyan Bhawan, 2nd Floor, Pandit Nehru Complex, Gorkhabasti, Agartala, West Tripura District and shall also include its successors in interest and assignees.
- 1.3 The **“Contractor” Or “Successful Bidder”** shall mean shall mean the Firm/ Person (whose tender has been accepted by TREDA) and shall include his legal representatives, successor in interest and assignees.
- 1.4 The **“Contract”** shall mean “Design, survey, manufacture, supply, erection, testing, commissioning of 10 kWp & 5 kWp Solar Photovoltaic Power Plant (Hybrid) at Block Offices including Warranty /Guarantee, Annual Maintenance Contract and insurance coverage for 5 (five) years on turn-key basis throughout the state of Tripura.” from the date of commissioning and shall be valid till the completion of all related works.
- 1.5 **“Annual Maintenance Contract (AMC)”** means routine and on-call maintenance of the systems including supply of all spare parts as required for reliable operation of the systems for 5 (five) years Warranty / Guarantee period.
- 1.6 **“BIS”** shall mean specifications of Bureau of Indian Standards (BIS);
- 1.7 **“IEC”** shall mean specifications of International Electro-technical Commission;
- 1.8 **“Electricity Act 2003”** shall mean the Electricity Act, 2003 and any rules, amendments, regulation, notifications, guidelines or policies issued there under from time to time,
- 1.9 **“CEA”** shall mean Central Electricity Authority.
- 1.10 **“DNIE-T”** shall mean Detailed Notice Inviting e-Tender.
- 1.11 **“Bids”** shall mean the Technical Bid and the Price Bid submitted by the Bidder electronically at the prescribed web portal, in response to the e-bid document, in accordance with the terms and conditions hereof;
- 1.12 **“Bid Deadline”** shall mean the last date and time for submission of Bid in response to the DNIE-T and as may have been extended in accordance with the e-bid document;
- 1.13 **“Bidder”** shall mean a Bidding Company complying with the provisions of **SECTION 3**;
- 1.14 **“Completion of work”** means the Project / Works has been completed operationally, structurally & Commissioning has been attained as per Technical Specifications as per DNIE-T.
- 1.15 **“Electricity”** means the electrical energy in kilowatt hours;
- 1.16 **“Price Bid”** shall mean the e-Bid, containing the Bidder’s quoted cost for different category of power plants in the **PART -B** of the e-bid document;
- 1.17 **“EPC”** shall mean engineering, procurement and construction of a plant or

facility with obligation to meet minimum performance standards along with requisite warranties for the plant or facility.

- 1.18 **“Plant”** shall mean solar photovoltaic power generation plants implemented on the individual site;
- 1.19 **“PV”** shall mean photovoltaic;
- 1.20 **“Solar Company”** shall mean the Selected Bidder who submits the Security deposit /Performance Guarantee and implements the solar photovoltaic power generation plant in accordance with the scope of work as elaborated in this tender.
- 1.21 **“Statutory Auditor”** shall mean the auditor of a company appointed under the provisions of the Companies Act, 1956 or under the provisions of any other applicable governing law;
- 1.22 **“Security Deposit”** shall mean the bank guarantee to be provided from a Selected Bidder to the TREDA in accordance with the prescribed Format;
- 1.23 **“Wp”** shall mean Watt Peak.
- 2** The agreement shall be rate contract basis valid for 18 (eighteen) months. All the allocated work shall be completed on turnkey basis within 12 (twelve) months from the date of issue of Letter of Award (LoA) on turn-key basis. However, “TREDA” may in case of urgency ask the bidder to complete the work earlier, with the mutual consent of the contractor/ bidder.
- 3** In case the contractor/ bidder fails to execute the said work within stipulated time, “TREDA” will be at liberty to get the work executed from the open market without calling any tender/e-tender and without any notice to the contractor/ bidder, at the risk and cost of the contractor/ bidder. Any additional cost incurred by “TREDA” shall be recovered from the contractor/ bidder. If the cost of executing the work as aforesaid shall exceed the balance due to the contractor/ bidder, and if the contractor/ bidder fails to make good the additional cost, “TREDA” may realize it from the contractor/ bidders’ pending claims (PBG), Additional Performance Security or in any lawful manner.
- 4** That on the request of the contractor/ bidder and also in the interest of the organization, “TREDA” is authorized to extend the validity of the agreement, subject to that the request of the contractor/ bidder (with proof of evidence of such delay) is received before the expiry of the agreement period, or any extended period granted to the contractor/ bidder. Maximum period of extension shall be 60 (sixty) days on the same terms and conditions as contained in this agreement.
- 5** The relevant clauses, terms & conditions of Agreement shall remain valid up to the date of completion as per Letter of Award (LoA) or the extended period granted by TREDA.
- 6** It will be the sole responsibility of the contractor/ bidder, to execute orders placed as per time schedule, and to ensure quality parameters, specifications and other requirements provided in the e-tender document and as per agreement.
- 7** In the interest of the work and the programme, agreement executed between the contractor/bidder and the “TREDA” may be extended to a mutually agreed period, if the need so arises. It shall be sole responsibility of the contractor/ bidder to get verified the quality & quantity of the supplied material at the site of delivery.

8 TEST CERTIFICATES / BIS CERTIFICATE:

- 8.1 The successful bidder is required to submit BIS Certificate and Test Report(s) for the following components in compliance with relevant Standards of MNRE,

BIS and as mentioned at PART – 4 of this DNle-T and tested components must be used for entire project without any change:

- 8.1.1 SPV Module: BIS Certificate and Test Reports along with enlistment under “Approved Models and Manufacturers of Solar Photovoltaic Modules (Requirements for Compulsory Registration) Order, 2019 issued by Ministry of New & Renewable Energy, Govt. of India vide Office Memorandum No. 283/54/2018-GRID SOLAR, dated 2nd January, 2019 and further Amendment(s) issued by MNRE.
- 8.1.2 Power Conditioning Unit: BIS Certificate and / or Test Report
- 8.1.3 Tubular GEL Battery: BIS Certificate and / or Test Report
- 8.1.4 Galvanization Report of Module Mounting Structure.
- 8.1.5 Structural Drawing of Module Mounting Structure.
- 8.2 The components and should be tested from SECI / MNRE approved / NABL / IEC Accredited Test Laboratories in compliance with BIS / MNRE specifications and tested components must be used for entire project without any change.
- 8.3 Type Test Reports should not be more than 5 (five) years old as on the date of submission to TREDA and BIS Certificates should be valid as on the date of submission to TREDA.
- 8.4 In case of change of make of components, BIS Certificate and Test Report of individual components have to be submitted for approval of TREDA.
- 8.5 Within **30 (thirty) days** from the issue of the Letter of Award (LoA), the successful bidder has to submit the following documents:
 - 8.5.1 PART 6: TECHNICAL INFORMATION along with required documents specified above.
 - 8.5.2 Declaration in the letter head of concerned component manufacturers for post-installation maintenance and performance warrantee/guarantee of the components as per stipulations of DNle-T.
- 8.6 **DECLARATION** as per **FORMAT 12** in the letter head of the bidder to submit “PART 6: TECHNICAL INFORMATION along with other required documents” within **30 (thirty) days** from the issue of the Letter of Award (LoA) should be uploaded along with the technical bid.
- 8.7 Any bid received without Declaration as mentioned above will be summarily rejected for non-responsive to the tender condition.

9 TYPE AND QUALITY OF MATERIALS AND WORKMANSHIP:

- 9.1 The design, engineering, manufacture, supply, installation, testing and performance of the equipment shall be in accordance with latest appropriate IEC/ Indian Standards (BIS) as detailed in the PART 4 of the bid document. Where appropriate Indian Standards and Codes are not available, other suitable standards and codes as approved by the MNRE shall be used.
- 9.2 The specifications of the components should meet the technical specifications mentioned in PART 4 of the bid document.
- 9.3 Any supplies which have not been specifically mentioned in this Contract but which are necessary for the design, engineering, manufacture, supply & performance or completeness of the project shall be provided by the successful Bidder without any extra cost and within the time schedule for efficient and smooth operation and maintenance of the Power Plants & IE works.

10 SUPPLY OF MATERIALS:

- 10.1 It is a turn-key job, as such no extra payments against any unforeseen items / works required to complete the job will be paid/allowed. The successful bidder should be responsible for packing, forwarding and dispatching, insurance and safe delivery of materials and installation & commissioning including proper civil works, storage & handling as required at specific sites at their quoted price. Temporary storage of materials during transit or at site shall be the responsibility of the successful bidder at his risk and cost.
- 10.2 Delivery of the complete system at site shall commence from 90th day and the work should be completed within the timeline stipulated in the LoA. The date of issue of Letter of Award (LoA) would be considered as zero date.
- 10.3 Maker of the components such as SPV Module, PCU, Battery etc. of Solar Power Plant shall be supplied & installed as per submitted valid test report / BIS & approved by TREDA.
- 10.4 Any delay in delivery except forced majeure shall be viewed seriously and may be linked with LD as found deem fit by TREDA.
- 10.5 The Solar Power Plants are to be supplied, installed & commissioned at specified location at the cost & risk of the successful bidder. TREDA/Users will not provide any transit store during supply / execution of work.
- 10.6 The maintenance of the system may also be taken up by the contractor/bidder after expiry of 5 (five) years Warranty/Guarantee on mutually agreed terms & conditions.

11 WORK EXECUTION:

- 11.1 OFFICER – IN - CHARGE: The TREDA shall appoint & Notify the successful bidder to whom LoA is issued in writing of the name of the Officer-in-charge. The Officer-in-charge shall represent & act for the TREDA at all times during the execution of the contract.
- 11.2 CONTRACTOR'S REPRESENTATIVE / CONSTRUCTION MANAGER: Within 10 (fifteen) days of the issue of LoA, the successful bidder shall appoint the bidder's representative and shall request the TREDA in writing to approve the person so appointed. The bidder's representative shall represent and act for the bidder at all time during the contract period. All Notices, instructions & all other communication made by TREDA or the Officer-in-charge shall be given to the bidder's representative. The bidder shall not revoke the appointment of bidder's representative without the prior written consent of TREDA.
- 11.3 From the commencement of installation of the Solar Plant & IE works at site until operational acceptance, the bidder's representative shall supervise all work done at site and shall be present at site throughout normal working hours except when on leave, sick or absent for reasons connected with the proper performance of the contractor.
- 11.4 In absence of Contractor's representative, as mentioned above, a suitable person shall be appointed to act as his/her deputy.
- 11.5 TREDA may by Notice to the bidder object to any representative / person employed by the bidder in execution of the contract, who in the reasonable opinion of TREDA may behave inappropriately, may be incompetent or negligent or may commit serious breach of site regulations provided under the Contract. The TREDA shall provide evidence of the same whereupon the bidder shall remove such person from the site and promptly appoint a replacement.

12 WORK PROGRAMME:

- 12.1 CONTRACTOR'S ORGANIZATION: The successful bidder shall provide TREDA a chart showing the proposed organization to be established for carrying out the work within 21 (twenty one) days from the date of issue of LoA. The chart shall include the identities of the key personnel to be employed mentioning their roles & responsibilities.
- 12.2 PROGRAMME OF PERFORMANCE: Within 25 (twenty five) days after the date of issue of LoA, the bidder shall prepare & submit to TREDA for approval, a detailed programme of performance of the contract, made in form of **PERT Network** and showing the sequence in which it proposes to design, manufacture, transport, assemble, install and commission the Solar Power Plants to enable him to execute the contract in accordance with the programme and to achieve completion & acceptance of the Power Plant by TREDA in accordance with the contract. The PERT Network submitted showing the time schedule in accordance with the period specified in the contract shall be approved by TREDA.
- 12.3 PROGRESS REPORT: The TREDA will monitor progress of all activities specified in the PERT Network and the bidder shall supply a progress report to TREDA in every week or as decided by Officer-in-charge. The progress report shall be in a form acceptable to TREDA and shall also indicate completion achieved compared with planned completion for each activity and where any activity is behind the programme, giving comments & likely consequence and stating corrective actions being taken.
- 12.4 WORK PROCEDURE: The contract shall be executed in accordance with the contract documents. The bidder shall be required to attend site progress review meeting organised by TREDA or authorised representative. The deliberations in the meetings shall inter alia include progress of work, daily / weekly programme (Including details of manpower, tools & plants deployed by the bidder vis-a-vis agreed schedule, inputs to be provided by TREDA, delays, if any and recovery programme, specific hindrances to work and instruction by TREDA). The Minutes of meeting shall be recorded for reference.

13 PROJECT INSPECTION:

- 13.1 Project(s) shall be got inspected by the Officer – In - Charge, TREDA as per the checklist requirement for release of payment.
- 13.2 TREDA reserves the right to do sample inspection checks of the Power Plants & IE Works commissioned by the Bidder.
- 13.3 TREDA may also depute a technical person(s) from its list of empaneled experts for inspection, Third party verification, monitoring of system installed to oversee, the implementation as per required standards and also to visit the manufactures facilities to check the quality of products as well as to visit the system integrators to assess their technical capabilities as and when required.

14 COMPLETION TIME:

- 14.1 Time of completion for design, survey, manufacture, supply, erection, testing, commissioning of Solar Photovoltaic Power Plants and Internal Electrification Works etc. at Block on turn-key basis shall be maximum 12 (twelve) months from the date of issue of LoA.

- 14.2 In case of splitting of quantity among the bidders, the Completion Time would be decided by TREDA based on the allocated quantity / capacity. Decision of TREDA in this regard would be final & binding on the bidders.

15 COMPLETION TIME GUARANTEE: If the bidder fails to attain completion of work or any part thereof within the time for completion or any extension thereof, the successful bidder on whom LoA is made, the bidder shall pay to TREDA Liquidated Damages (LD) in the amount computed at the rates specified below. However, payment of LD, shall not in any way relieve the bidder on whom award is made from any of its obligations to complete the work or from any other obligations and liabilities under the contract.

16 LIQUIDATED DAMAGES:

- 16.1 If the bidder fails to commission the sanctioned project within specified time, LD on per day basis calculated for the Performance Security or Performance Bank Guarantee (PBG) would be levied. TREDA may increase the project duration at its own discretion as per relevant clauses of DNIE-T. After completion time, the project will get cancelled and total PBG amount shall be forfeited by TREDA.
- 16.2 For example, If the value of Performance Security or Performance Bank Guarantee is Rs. 3,65,000/- for validity period of 365 days, a capacity of 0.5 MW is delayed out of the total capacity to be commissioned by 30 days then the Liquidated Damages will be levied as given below:

$$\text{Liquidated Damages} = [\text{PBG Amount} / \text{PBG validity period (in days)}] * \text{Delayed Days} * \text{Capacity non-Commissioned (MW)} = (\text{Rs. } 3,65,000 / 365) * 30 * 0.5 = \text{INR } 10,000/-.$$

- 16.3 Alternatively, TREDA reserves the right to purchase of the material and completion of the works including maintenance from elsewhere at the sole risk and cost of the successful bidder/ contractor and recover all such extra cost incurred by TREDA in procuring the material from resources available including EMD/Bid security/ encashment of the Performance Bank Guarantee / Additional Performance Security or any other sources etc. Further, if any extra cost is incurred by TREDA due to delay in work completion by the party beyond the completion time as per LoA, the same shall be recovered from the successful bidder's Invoice/EMD/PBG/ Additional Performance Security etc.
- 16.4 TREDA may cancel the order completely or partly without prejudice to his right under the alternatives mentioned above.

17 The contractor/ bidder shall have to comply with all the rules, regulations, laws and by-laws for the time being in force and the instructions if any, of the organization, in whose premises the work has to be done. "TREDA" shall have no liability in this regard.

18 FORCE MAJEURE CONDITIONS:

- 18.1 Notwithstanding the provisions of clauses contained in this DNIE-T; TREDA shall not be liable to forfeit (a) PBG & Additional PBG (If required) for delay and (b) termination of contract; if the contractors unable to fulfill his obligation under this contract due to force majeure conditions.

- 18.2 For purpose of this clause, the term force majeure shall herein mean riots (other than among the contractor's employee), civil commotion, war or revolutions (whether declared or not), invasion, act of foreign enemies hostilities, rebellion, insurrection, military coup to usurp power, Act of God such as earthquake, lightening, floods, fires, epidemics, quarantine restrictions, Grid Problems/ shut downs and fright embargoes etc. not caused by contractor's negligence and other cause which the contractor has no control. Whether a "Force majeure" situation exists or not, shall be decided by TREDA and its decision shall be final and binding on the contractor and all other concerned.
- 18.3 In the event that the contractor is not able to perform his obligations under this contract on account of force majeure, he will be relieved of his obligations during the force majeure period. In the event that such force majeure extends beyond one (1) month, TREDA has the right to terminate the contract in which case, the PBG shall be refunded to him.
- 18.4 If a force majeure situation arises, the contractor shall notify TREDA in writing promptly, not later than 14 days from the date such situation arises. The contractor shall notify TREDA not later than 3 days of cessation of force majeure conditions. After examining the cases, TREDA shall decide and grant suitable additional time for the completion of the work, if required.

19 SUPERVISION / EXECUTION OF WORKS: The works should be done as per direction of Officer-in-charge, Tripura Renewable Energy Development Agency. TREDA shall have at all reasonable time access to the works being carried out by the contractor/ bidder under the contract. All the work shall be carried out by the contractor/bidder to the satisfaction of TREDA.

20 The contractor/ bidder shall not, without the consent in writing of TREDA, transfer, assign or sublet the work under the contract or any substantial part thereof to any other party.

21 If any question, dispute or difference what so ever shall arises between TREDA and the contractor/ bidder, in the connection with the agreement except as to matters, the decisions for which have been specifically provided, either party may forthwith give to the other notice in writing of existence of such question, dispute or difference and the same shall be referred to the sole arbitration of the Principal Secretary/Secretary, Power Department, Govt of Tripura or a person nominated by him not below the rank of Secretary. This reference shall be governed by the Indian Arbitration and Conciliation Act 1996, and the rules made there under. The award in such arbitration shall be final and binding on both proceedings unless the TREDA or the arbitrator directs otherwise.

22 TREDA may at any time by notice in writing to the contractor/ bidder either stops the work all together or reduces or cut it down. If the work is stopped all together, the contractor/bidder will only be paid for work done and expenses distinctly incurred by him as on preparation or the execution of the work up to the date on which such notice is received by him. Such expenses shall be assessed by TREDA, whose decision shall be final and binding on the contractor/ bidder. If the work is cut down the contractor/ bidder will not be paid any compensation what so ever for the loss or profit which he might have made if he had been allowed to complete all the work included in the contract.

23 INSPECTION AND TESTING OF MATERIALS:

- 23.1 All materials / equipments manufactured by the bidder against LoA shall be subject to inspection, check and / or test by TREDA at all stages and place, before, during and after the manufacture. All these tests shall be carried out as per technical specification and bidder shall submit relevant test reports.
- 23.2 The materials will be despatched / installed by the successful bidder after pre-despatch inspection by the Director General, TREDA or his representatives at the factory / works of the successful bidder as per DNIE-T specifications and acceptance of the same.
- 23.3 The successful bidder shall provide without any extra charge, all materials, tools, testing equipments, labour and assistance of every kind which the inspecting officer may consider necessary for any test or examination which he may require to be made at manufacturing unit and State / District Level Godown.
- 23.4 TREDA can also get the components of the systems tested from any SECI / MNRE approved / NABL / IEC Accredited Test Laboratories and the expenses shall be borne by the successful bidder.
- 23.5 All the expenses for the inspection of gadgets (At the bidder's premises / manufacturing unit / site) by the TREDA's representatives like to & fro air fare, boarding and lodging charges etc. will have to be borne by the successful bidder.
- 23.6 The successful bidder, within 10 (ten) days from the date of component approval from TREDA, shall issue request letter for inspection (PDI) to TREDA for testing of any component of the plant which is ready for testing, the inspection call should be issued at least 7 (seven) days in advance from the actual date of testing at the bidder's premises / manufacturing unit.
- 23.7 Director General, TREDA reserves the right to waive out inspection of gadgets at the manufacturing unit due to unavoidable circumstances, in such cases, the materials will be inspected at State / District Level Godown of successful bidder, Lot wise Inspection Charge will be deducted from the bills of successful bidder, the amount of Inspection Charges will be decided by TREDA.
- 23.8 The successful bidder should not despatch any component of the power plant to site without getting despatch clearance of the same from TREDA.
- 23.9 If upon deliver whether inspected and approved earlier or otherwise, the materials / equipment is not in conformity with the specification, the same shall be rejected by TREDA and notification to this effect will be issued to the bidder normally within 7 (seven) days from the date of delivery of materials at site / work.
- 23.10 The bidder shall arrange removal of the rejected items within 10 (fifteen) days from the date of notification and take necessary action for repairs / modification etc. or for replacement at the cost & risk of the bidder.

24 PACKING FORWARDING:

- 24.1 Contractor/ bidders, wherever applicable, shall after proper painting, pack and crate all the equipments in such a manner as to protect them from deterioration and damage during rail and road transportation to the site and storage at the site till time of installation. Contractor/bidder shall be sole responsible for all damage due to improper packing.

24.2 The contractor/ bidder shall inform the TREDA of the date of each shipment from his works, and the expected date of arrival at the site for the information of the TREDA Officer-in-charge at least 7 days in advance.

25 RESPONSIBILITY OF TRANSPORTATION AND STORES: Successful bidder should be solely responsible for stores in transit & installation. Any legal interference of Police / Sales tax / Income tax / Transport / Any other Govt. Agencies will be faced by successful bidder. The transportation delay / non availability of train / truck etc. will never be considered by the tendering authority as reason of delay to supply and naturally no extension of delivery period will be granted on this account.

26 INSURANCE:

- 26.1 The Bidder shall be responsible and take an Insurance Policy for transit-cum-storage-cum-erection for all the materials to cover all risks and liabilities for supply of materials on site basis, storage of materials at site, erection, testing and commissioning. The bidder shall also take appropriate insurance during AMC period.
- 26.2 The Bidder shall also take insurance for Third Party Liability covering loss of human life, engineers and workmen and also covering the risks of damage to the third party/material/equipment/properties during execution of the Contract. Before commencement of the work, the Bidder will ensure that all its employees and representatives are covered by suitable insurance against any damage, loss, injury or death arising out of the execution of the work or in carrying out the Contract. Liquidation, Death, Bankruptcy etc., shall be the responsibility of bidder.
- 26.3 In case of any loss or damage or pilferage or theft or fire accident or combination of the said incidents etc. under the coverage of insurance, the Contractor shall lodge the claim as per rules of insurance. Any FIR required to be lodged to local Police Station shall be the responsibility of the successful bidder.
- 26.4 The bidder shall arrange to supply/rectify/recover the materials even if the claim is unsettled for timely completion of the Project. The final financial settlement with the insurance company shall be rested upon the Contractor.
- 26.5 Insurance as applicable during operation of Project for covering all risks against any mishap to its workmen. The TREDA/User will not be responsible for any such loss or mishap. All other insurances like, insurance against theft and acts of GOD, as required for AMC of the plant and to indemnify the TREDA/User equipment/ material and resources shall be borne by the contractor. Fire insurance is to be arranged by the Contractor up to the five years of AMC of the Contract.

27 TERMINATION OF THE LETTER OF AWARD / CONTRACT: TREDA may at any time terminate the Letter of Award / Contract by giving written notice to the Contractor/successful bidder without compensation to the Contractor/successful bidder. No consequential damages shall be payable to the Contractor in the event of such termination. The reasons for such termination are furnished below:

- 27.1 Non execution of Contract Agreement within specified time period.
- 27.2 Non-fulfilment of Contractual obligations as per terms & conditions of DNIE-T

27.3 **TERMINATION FOR INSOLVENCY:** TREDA may at any time terminate the contract by giving written notice to the contractor/bidder without compensation to the contractor/ bidder, if it becomes bankrupt or otherwise insolvent, provided that such termination will not prejudice or affect any right of action or remedy, which has accrued or will accrue thereafter to the TREDA.

27.4 **TERMINATION FOR CONVENIENCE:** The TREDA, may by written notice sent to the contractor/ bidder, terminate the contract, in whole or in part at any time for its convenience. The notice of termination shall specify that termination is for the purchaser's convenience in the interest of TREDA.

28 INDEMNITY: The successful bidder agrees that it shall indemnify and hold harmless TREDA and its members, officers, employees, casual labours, persons permitted to run any business or service such as canteens, stores, photocopy units, banks, post office, courier service, hospital and any lawful visitors (Collectively, the "TREDA Indemnified Parties") from and against any and all losses incurred by the "TREDA Indemnified Parties" to the extent arising from or out of the following any claim for or arising out of any injury to or death of person or loss or damage to property of any person to the extent arising out of the successful bidder's negligence or willful misconduct of any TREDA Indemnified Party.

29 CONFLICT OF INTEREST:

29.1 TREDA considers a conflict of interest to be a situation in which a party has interests that could improperly influence that party's performance of official duties or responsibilities, contractual obligations, or compliance with applicable laws and regulations. In pursuance of TREDA's procurement ethics, TREDA will take appropriate actions against the bidder, if it determines that a conflict of interest has flawed the integrity of any procurement process. Consequently, all bidders found to have conflict of interest shall be disqualified.

29.2 Conflict of interest will be determined in accordance with the relevant clauses of this bid document, which read as under:

29.2.1 Conflict of Interest among Bidders:

- 29.2.1.1 they have proprietor / partner(s)/Director(s) in common; or
- 29.2.1.2 they receive or have received any direct or indirect subsidy / financial stake from any of them; or
- 29.2.1.3 they have same legal representative / agency for purposes of this bid; or
- 29.2.1.4 they have relationship with each other, directly or through common third parties, that puts them in a position to have access to information about or influence on the bid of another bidder or influence the decisions of TREDA regarding this bidding process; or
- 29.2.1.5 A bidder or any of its affiliates participated as a consultant in the preparation of the design or technical specifications of the Contract that is the subject of the Bid;
- 29.2.1.6 In case of a holding company having more than one independent Consultancy Firm / Agency or more than one unit having common business ownership / management, only one unit should be quoted. Similar restrictions would

apply to closely related sister companies. Bidders must proactively declare such sister / common business/management units in the same / similar line of business.

- 29.2.1.7 Bidder shall not act in contravention / violation to the provisions of Competition Act, as amended from time to time.
- 29.2.1.8 Every bidder should, at the time of submission of bid, give declaration, that bidder shall not have conflict of interest with other bidders for submitted bid, as above.
- 29.2.1.9 Further, it may be considered to be in conflict of interest with one or more parties in the bidding process if:
 - 29.2.1.9.1 they have controlling shareholders in common; or
 - 29.2.1.9.2 it receives or have received any direct or indirect subsidy from any of them; or
 - 29.2.1.9.3 They have the same legal representative for purposes of the bid.
- 29.2.1.10 As may be determined by TREDA on the basis of information received, which decision shall be final.

30 APPLICABLE LAW: The contractor/ bidder shall be interpreted in accordance with the laws of the purchaser's country i.e. India. The station of TREDA Headquarter at Agartala shall have exclusive jurisdiction in all matters arising under this contract.

31 LEGAL CASE: All disputes are to be settled within the jurisdiction of High Court of Tripura, Agartala.

32 SETTLEMENT OF DISPUTE:

- 32.1 To the best of their ability, the parties hereto shall endeavor to resolve amicably between themselves all disputes arising in connection with this work order. If the same remain unresolved within thirty (30) days of the matter being raised by either party, either party may refer the dispute for adjudication by arbitration. The arbitration shall be undertaken by the sole arbitrator jointly appointed by the parties. In case the parties fail to arrive at consensus to appoint the sole arbitrator, either party may approach the Court for appointing an arbitrator under Section 11 of the Arbitration and Conciliation Act, 1996 and the award of the said sole arbitrator, shall be final and binding upon the parties. The arbitration proceeding shall be conducted in accordance with this provisions of the Indian Arbitration & Conciliation Act, 1996 (as amended up to date) and the venue of such arbitration shall be the city of New Delhi only. The Arbitration shall be conducted in English language only. The Tripura High Court, Agartala shall have the exclusive jurisdiction over the subject matter of Arbitration/dispute. The cost of the Arbitration shall be equally shared by the parties as per directions of the Sole Arbitrator.

33 PRE-BID MEETING:

- 33.1 The prospective bidders may attend Pre-bid meeting at the Office of the Director General, Tripura Renewable Energy Development Agency, Vigyan Bhawan (2nd floor), Gorkhabasti, Agartala, West Tripura District, Pin: 799006 on the date as specified in the BID INFORMATION SHEET.

- 33.2 The purpose of the Pre-bid Meeting will be to clarify any issue regarding the tender including the particular issues raised in writing as per **FORMAT 10** and submitted by the interested bidders **only through e-procurement portal <https://tripuratenders.gov.in>** on or before the date & time stipulated in **BID INFORMATION SHEET**.
- 33.3 Queries / Clarifications sought through e-mail of TREDA OR in any other form OR after the deadline for submission through e-procurement portal <https://tripuratenders.gov.in> will not be accepted.
- 33.4 TREDA will not sent the reply of the queries to individual bidders. However, any Amendment / Corrigendum of the tender made in response to queries will be up-loaded in e-procurement portal after pre-bid meeting.
- 33.5 In case, Pre-bid meeting is suspended due to unavoidable circumstances, TREDA will respond to the queries / clarifications of prospective bidder through e-mail or through e-procurement portal.
- 33.6 TREDA is not under any obligation to entertain/respond to suggestions made or to incorporate modifications sought for.

34 NOTICE:

- 34.1 Any notice given by one party to the other pursuant to the contract shall be sent in writing, soft copy should be forwarded through e-mail & hard copy through Speed Post / Courier to the address specified for that purpose.
- 34.2 A notice shall be effective when delivered or on the notice's effective date, whichever is later.

35 OTHERS:

- 35.1 The Contractor/ bidder in consultation with concerned OFFICER – IN - CHARGE of TREDA will conduct training programme for users, focusing on main features, operation and maintenance of the systems.
- 35.2 It shall be the sole responsibility of the contractor/ bidder to get verified the quality & quantity of the supplied material at the site of delivery.

36 PAYMENT & OTHER FINANCIAL TERMS:

- 36.1 No advance payment will be made.
- 36.2 Payment will be made on completion of on successful & satisfactory installation and commissioning of the Power Plant including Internal Electrification works at site.
- 36.3 **SPV POWER PLANT & INTERNAL ELECTRIFICATION (IE) WORKS:**
- 36.3.1 Eighty five percent (85%) payment after statutory deductions will be released to successful bidder(s) on supply, installation & commissioning of the SPV Power Plant including IE Works in all respect duly verified by OFFICER-IN-CHARGE, TREDA and on submission of following documents:
- 36.3.1.1 Three (3) original copies of Bills for 85% payment duly verified/certified by respective District OFFICER-IN-CHARGE, TREDA,
- 36.3.1.2 Documents specified at CLAUSE NO. 4 under PART 4 of the DNIE-T (i.e, Installation & Commissioning, Handing over Certificate, Site test reports and operational parameters of the Plants, Geotagged photographs and an undertaking by successful bidder certifying

that the Module Mounting Structure will withstand the wind speed of 200 Km/hr. in all weather conditions etc.).

- 36.3.2 Balance fifteen percent (15%) amount, after statutory deductions, will be retained as ADDITIONAL PERFORMANCE SECURITY which will be released in phased manner @ three percent (3%) per year on pro-rata basis at the end of each year of the Guarantee / warranty and AMC period (from the date of commissioning) on satisfactory performance of SPV Power Plants including Internal Electrification Works etc. in compliance with PART 5 of the DNIE-T and on submission of following documents:
- 36.3.2.1 Plant wise Three (3) original copies of Bills for 3% payment duly verified/certified by OFFICER-IN-CHARGE, TREDA.
 - 36.3.2.2 Quarterly & Yearly Maintenance Report in compliance with PART 5 of the DNIE-T
 - 36.3.2.3 Insurance copy.
 - 36.3.2.4 Copy of Installation & Commissioning Report / Certificate.
 - 36.3.2.5 Copy of Valid PERFORMANCE SECURITY [Performance Bank Guarantee / Additional Performance Bank Guarantee (If required)].
- 36.4 Running bills may be considered for payment on successful commissioning of at least 5% of the LoA quantity.
- 36.5 All payments shall be made in Indian Rupees through NEFT/RTGS only to the Bank Account linked with GSTIN of successful bidder.

37 INSPECTION AND REPORTING:

- 37.1 The TREDA or any other agency designated by TREDA may inspect the ongoing installations or installed systems of Power Plants & IE Works. In case the installed systems are not as per standards, non-functional on account of poor quality of installation or non-compliance of Warranty/guarantee and AMC obligations, the TREDA reserves the right to blacklist / debar the successful bidder to a period of five (5) years from the date of such decision of debarment. Blacklisting / debarment inter-alia includes the followings:
- 37.1.1 The successful bidder will not be eligible to participate in tenders for Government supported projects.
 - 37.1.2 In case, the concerned Director(s) of the successful bidder joins another existing or start/joins a new firm/company, the company will automatically be blacklisted.

38 LANGUAGE

- 38.1 All documents, drawings, instructions, design data, calculations, operation, maintenance and safety manuals, reports, labels and any other data shall be in English Language. The contract agreement and all correspondence between the TREDA and the bidder shall be in English language. O&M manual and warranty card should be in English & Hindi languages.

39 FAMILIARIZATION OF SITES OF INSTALLATION BY THE BIDDER:

- 39.1 The intending Bidder shall be deemed to have familiarized with site conditions in Tripura while submitting the Bid. Non-familiarity with the site conditions will not be considered a reason either for extra claims or for not carrying out the works in strict conformity with the technical specifications or for any delay in performance.

- 39.2 The bidder, at the bidder's own responsibility and risk is advised to visit site for work (Block and Panchayet Samity / Block Advisory Committee Offices) and its surroundings and obtain all information that may be necessary for preparing the bid for entering into a contract for installation & commissioning of SPV Power Plants & IE Works.
- 39.3 The costs of visiting the site shall be at the Bidder's own expense and the bidder shall submit the Bid on being satisfied by his inspection of the existing site condition including **facilities / hurdles**.
- 39.4 In order to ensure the above, the bidder shall have to submit a declaration in compliance to the site visit in the Site Visit Declaration Form **(FORMAT13)**. This Site Visit Declaration shall be uploaded along with the tender document mandatorily, for determination and making him technically qualified if eligible otherwise.
- 39.5 Site locations, if required, may be collected from the Office of the Director General, TREDA, Vigyan Bhawan, Gorkhabasti, Agartala, West Tripura District.
- 40** Any other conditions which are not mentioned in this document but if needed during execution of work shall be as per latest MNRE guidelines.

PART - 4: SCOPE OF WORK & TECHNICAL SPECIFICATIONS

1. OVERALL SCOPE:

- 1.1. **ITEM I:** Design, survey, manufacture, supply, erection, testing, commissioning of 10 kWp Solar Photovoltaic Power Plant (Hybrid) & Internal Electrification Works at Block Offices etc. including 5 (five) years Warranty /Guarantee, Annual Maintenance Contract and insurance coverage on turn-key basis throughout the state of Tripura.
- 1.2. **ITEM II:** Design, survey, manufacture, supply, erection, testing, commissioning of 5 kWp Solar Photovoltaic Power Plant (Hybrid) & Internal Electrification Works at Block Offices etc. including 5 (five) years Warranty /Guarantee, Annual Maintenance Contract and insurance coverage on turn-key basis throughout the state of Tripura.
- 1.3. The successful bidder shall be responsible for successful data transmission of REMOTE MONITORING UNITS of the SPV Power Plants for monitoring of real time data. Indicative responsibilities are as follows:
 - 1.3.1. SIM Cards (BSNL/other service providers) to be used for data transmission from each site through Remote Monitoring Unit will be under the scope of successful bidder at his own expense, all run time expenditures, such as procurement of SIM Cards, activation / top up charges, charges for suitable SMS packages etc. for 5 (five) years AMC period will also be under the scope of successful bidder.
 - 1.3.2. Automatic hosting of real time data of the SPV Power Plant with nominal 4 to 6 hours interval.
 - 1.3.3. The successful bidder has to conduct the survey of sites for identification of suitable Service Provider for SIM Cards to be used in the RMU
 - 1.3.4. for un-interrupted data transmission. However, decision of TREDA regarding selection of service provider will be final and binding on the successful bidder.
 - 1.3.5. The successful bidder may choose appropriate plan of service provider for procurement of SIM Cards, activation / top up charges of SIM Cards and charges for suitable SMS packages etc.
 - 1.3.6. The successful bidder has to return back the SIM Cards to TREDA after completion of AMC period.
 - 1.3.7. The successful bidder has to maintain HTTP protocol for storage of all RMU data of SPV Power Plants and should give access to TREDA for monitoring data of SPV Power Plants Day to day basis.
- 1.4. The successful bidder shall have to made site survey in close coordination with Officer-in-charge of TREDA to identify a suitable location within the Block/Panchayat Samiti Offices/Block Advisory Committee premises for the installation of the components of the SPV Power Plant i.e. Module Mounting Structure, Battery, Power Conditioning Unit, ACDB, DCDB, LA, earthing etc. & components required for internal electrification works including cable routing of PV Systems, cable routing for grid connectivity of PCU, Solar Energy meter & Net-meter etc. The Scope of Work includes AC cabling for a distance between the PCU and grid interconnection point.
- 1.5. The successful bidder shall submit SITE SURVEY REPORT comprising of Single Line Diagram showing the position of components SPV Power Plant (PV Module, Battery, PCU, DCDB, ACDB, AJB, Earthing, LA etc.)

- position of components of internal electrification works, Solar Energy meter & Net-meter of each site along with BOM.
- 1.6. The SITE SURVEY REPORT should be duly signed by the representative of successful bidder, User and representative of TREDA and should be get approved by TREDA / Officer-in-charge of TREDA before execution of work.
 - 1.7. Simultaneously with the installation of Power Plants, the successful bidder shall execute the Internal Electrification works at Block/Panchayat Samiti Offices/Block Advisory Committee Offices as per site requirement and separate LoA / Work Order on actual measurement basis of IE Items will be issued.
 - 1.8. The scope of work includes Warranty / Guarantee & Annual Maintenance Contract of the Solar Power Plant & components of IE works etc. and Insurance Coverage of all the components for 5 (five) years as detailed at PART 5 of the DNIE-T.
 - 1.9. All components and works shall adhere to relevant Indian Standards, State and Central policies and regulations and all statutory provisions under the Indian Law.
 - 1.10. The successful bidder shall be entirely responsible for the execution of scope of work in accordance to this tender document including but not limited to its specifications, schedule and annexure. The successful bidder shall further provide guarantee and be responsible for the quality and workmanship of all materials and completed works, survey, correct designs and drawings, correct delivery of materials, erection, testing, commissioning and mandatory maintenance.
 - 1.11. Any other works which is not mentioned but necessary for successful commissioning, Guarantee / Warranty and Annual Maintenance Contract for 5 (five) years would be in the scope of bidder. All approvals required for successful commissioning of the systems shall be in the scope of successful bidder.
 - 1.12. The successful bidder shall repair at its own cost any damages caused to premises of installation during implementation of the work.
 - 1.13. The successful bidder shall maintain an ACTIVE LOCAL OFFICE CUM SERVICE CENTRE in each district where work is carried out under this tender to rectify faults within the stipulated timeframe described in the tender document.
 - 1.14. Systems installed under this scheme shall meet technical specifications and construction standards as specified by BIS and MNRE from time to time / Standards mentioned at PART 7 of the DNIE-T. Non-compliance will be taken seriously to the extent of blacklisting of successful bidder apart from taking action under any other law in force.
 - 1.15. In line with maintenance and servicing report requirement, quarterly cleaning of dust from SPV panel shall be in the scope of successful bidder.
 - 1.16. All final specifications (In line with MNRE specifications), bill of materials, quality assurance plan etc. shall be inspected, vetted and approved by the TREDA.
 - 1.17. The successful bidder shall submit technical details of module, PCU etc. and its test report, testing and commissioning report & handing over taking over certificate of Plant, internal electrification works, installed system photographs, and bill of material to TREDA for future reference.

- 1.18. Leaflet containing the details of operation & maintenance practices, precautions etc. and the details of ACTIVE LOCAL OFFICE CUM SERVICE CENTRE, particulars of service personals shall be provided to each user / beneficiary and TREDA.
- 2. INSTALLATION, COMMISSIONING OF THE PLANT:**
 - 2.1. The installation & commissioning of the PLANT shall include successful installation & commissioning of SPV Power Plant including Internal Electrification works etc. at Block / Panchayat Samiti Offices/Block Advisory Committee / Other Govt. Offices specified by TREDA.
 - 2.2. The PLANT shall be deemed to be successfully installed & commissioned after 72 (seventy two) hours of trouble free operation of the Power Plant including internal electrification works in compliance with IEC/BIS/MNRE standards.
- 3. HANDING OVER ASSETS:**
 - 3.1. After successful installation, commissioning, testing with User Acceptance Test (UAT) of complete SPV Power Plant & IE works, the asset is to be jointly handed over to the USER for running of the plant. The Commissioning Certificate and Handing over Certificate covering the details of all the materials used and total work executed must be signed jointly by the authorized representative of USER, representative of successful bidder and TREDA. The format for commissioning certificate & handing over certificate will be provided by TREDA.
 - 3.2. User Acceptance Test (UAT) shall comprise of measurement and reporting of all technical parameters of the system with jointly signed by successful bidder, representative of TREDA and authorized representative of USER. All BIS/MNRE technical standards shall be applicable for such UAT including inspection of support structures, modules, mounting, cabling, equipment levels, markings, placards, manufacturer identification, specifications and ratings. Relevant IEC/BIS/MNRE standards shall be guiding standards for such tests or inspections.
 - 3.3. Taking over certificate & Operational Acceptance Certificate as well as completion certificates (As per approved formats of TREDA) shall be issued accordingly on completion of requisite works and completion of Contract Period, if the complete scope of work is completed.
- 4. POST COMMISSIONING ACTIVITIES:** On completion of work, the successful bidder shall submit commissioning report (Minimum 2 copies), all the documents, test reports etc. related to the execution of contract, the commissioning report shall comprise of:
 - 4.1. Commissioning and Handing over Certificate duly signed by user, Officer-in-charge of TREDA & the representative of successful bidder.
 - 4.2. Layout and as built drawings of the plant including IE Works.
 - 4.3. Site test reports and operational parameters of the Plants.
 - 4.4. Minimum 20 (twenty) Laval Geotagged photographs of SPV Power Plant including Internal Electrification Works etc. along with 3 (three) Laval geotagged photographs of sites (i) before installation, (ii) during installation and (iii) during imparting training to the user of every Block / Panchayat Samiti Offices/Block Advisory Committee / Other Govt. Offices. The successful bidder shall not display the photograph of the

work and shall not take advantage through publicity of the works without written permission of TREDA.

- 4.5. Performance report for 1 (one) day after commissioning based on data received from Remote Monitoring system or data logger in cases where internet service is not available.
- 4.6. An undertaking by successful bidder certifying that the civil work will withstand the **wind speed of 200 Km/hr. in all weather conditions.**
- 4.7. Any other information as per direction of Officer-In-Charge, TREDA

5. **DRAWINGS AND MANUALS:**

- 5.1. Two sets of engineering, electrical drawings, Installation & O&M manuals are to be submitted by the successful bidder to TREDA.
- 5.2. Successful bidder shall provide complete technical data sheets for each equipment giving details of technical specifications along with make / makes basis design of the PV Installation & IE works etc. setup along with protection equipment.
- 5.3. **The successful bidder shall furnish site wise following drawings after issuance of LoA and should obtain approval of TREDA before PDI:**
 - 5.3.1. General arrangement and dimensioned layout of PV Power Plant & internal electrification works.
 - 5.3.2. Schematic drawings showing the PV panels, design of PCU, Junction Boxes, AC and DC Distribution Boards, Lightning Arrester, Earthing, Meters, components of internal electrification works, Solar Energy Meter, Net-meter etc.
 - 5.3.3. Routing diagram of cables and wires.
 - 5.3.4. Data sheets and user manuals of the Solar PV Panels and Power Conditioning Unit.
 - 5.3.5. Structural drawings along with foundation details of MMS
 - 5.3.6. Itemized bill of materials for complete SPV Plant, internal electrification works etc. covering all components and associated accessories.
 - 5.3.7. Layout of Solar PV Array
 - 5.3.8. Maintenance Register
 - 5.3.9. Approved ISI and reputed makers for equipment shall be used.

6. **PLANNING & DESIGN:**

- 6.1. TREDA reserves the right to modify layout and specification of sub-systems and components at any stage as per local site conditions / requirements.
- 6.2. The successful bidder shall submit preliminary drawing for approval and based on any modification or recommendation, if any, the successful bidder shall submit 3 (three) sets and soft copies of final drawing for formal approval to proceed with work.

7. **INSURANCE DURING INSTALLATION:**

- 7.1. During installation period i.e. before commissioning of the PV Plant & IE Works, all insurance – related expenses shall be borne by the successful bidder. The goods supplied by the successful bidder shall be fully insured against the loss or damage incidental to manufacture or acquisition, transportation, storage, deliver, theft, natural or other disaster / calamity etc.



- 7.2. In case of any loss or damage or pilferage or theft or fire accident or natural calamity or combination of the said incidents under the coverage of insurance, the successful bidder shall lodge the claim as per rules of insurance. Any FIR required to be lodged at the local Police Station shall be responsibility of the successful bidder.
- 7.3. The successful bidder shall arrange to supply / rectify / recover the materials even if the claim is unsettled for timely completion of the project. The final financial settlement with the insurance company shall be rested upon the successful bidder.
- 8. INSURANCE DURING OPERATION:**
- 8.1. Upon commissioning of the PV Plant & internal electrification works the successful bidder shall undertake insurance on the installed systems against theft and vandalism and natural calamities for a minimum period of 5 (five) years and include the same within the quoted cost.
- 8.2. At least 2 (two) months before the end of the term of Insurance undertaken by the successful bidder, the successful bidder shall ensure to provide all the necessary documents and guidance to the satisfaction of the Panchayat/PRI Body in order to enable the Panchayat/PRI Body, in case he wants to renew or to take up the insurance of the PV Systems & internal electrification works at the end of the term of 5 (five) years insurance of successful bidder.
- 9. SAFETY MEASURES:** The successful bidder shall take entire responsibility for electrical safety of the installation (s) including connectivity with the grid and follow all the safety rules and regulations applicable as per Electricity Act, 2003 and CEA Guidelines, as well as applicable rules and regulations of Tripura State etc.
- 10. COMPLIANCE WITH LABOUR REGULATIONS:** During execution of the Contract, the successful bidder shall abide at all times by all existing labour enactments and rules made there under, regulations, notifications and bye laws of the State Government or Central Government or local authority and any other labour law (including rules), regulations, byelaws that may be passed or notification that may be issued under any labour law in future either by the State Government or the Central Government or the local authority.
- 11. STATUTORY RESPONSIBILITY:**
- 11.1. The entire responsibility and risk relating towards the workforce working at the site, and compliance of different statutory regulations like Workman Compensation Act, 1923, Employment of Children Act XXVI of 1938 and any enactment or modification of the same Act, Payment of Gratuity Act, 1972, Employees P.F. and Miscellaneous Provision Act, 1952, Maternity Benefit Act, 1948, Contract Labour (Regulation & Abolition) Act, 1970, Minimum Wages Act, 1948, Payment of Wages Act, 1936, Equal Remuneration Act, 1979, Payment of Bonus Act, 1965, Industries Dispute Act, 1947, Industries Employment (Standing Orders) Act, 1946, Trade Union Act, 1926, Child Labour (Prohibition & Regulation) Act, 1986, Inter-state Migrant Workmen's (Regulation of Employment & Conditions of Service) Act, 1979, The Building and other Construction Workers (Regulation of

Employment and Conditions of Service) Act, 1996 and the Cess Act, 1996, Employees State Insurance Corporation (ESIC), Factory Act 1948, Shop and Establishment Act 1948, and other Statutory regulatory bodies shall solely lie with the Contractor/ Bidder(s).

- 11.2. The Contractor/ Bidder(s) shall also be solely responsible for payment of wages, provident fund, bonus, retrenchment compensation leave, etc. applicable as per various statutory regulations to their entire workforce.
- 11.3. All equipment, accessories, materials, civil construction & erection works should comply with statutory requirements, IS and IEC standards.
- 11.4. Statutory requirements for working at the site like labour registration, workman compensation policy, ESIC etc. to be complied with by the vendor before deployment of resources at the Site.
- 11.5. The Contractor /Bidder shall obtain required license/registrations, statutory compliances from the State Government or Central Government or local authority.

12. TECHNICAL SPECIFICATIONS:

- 12.1. The proposed projects shall be commissioned as per the technical specifications given below. Any shortcomings will lead to cancelation of LoA as decided by TREDA and the agreement will be terminated. Competent Authority's decision will be final and binding on the bidder.
- 12.2. SPV POWER PLANT: SPV Power Plant consists of SPV array, Module Mounting Structure, Power Conditioning Unit (PCU) consisting of Maximum Power Point Tracker (MPPT), Battery and Controls & Protections, interconnecting cables, Junction boxes, Distribution boxes and switches. PV Array will be mounted on a suitable structure. Components and parts used in the SPV power plants including the PV modules, metallic structures, cables, junction box, switches, PCUs, Batteries etc., should conform to the BIS or IEC or international specifications as mentioned at PART 7 of the DNIE-T / latest MNRE Standards, wherever such specifications are available and applicable. Solar PV system shall consist of following major equipment/components.
 - i. Solar PV modules consisting of required number of Crystalline PV cells.
 - ii. Power Conditioning Unit Plant (Hybrid) having Remote Monitoring Unit
 - iii. Tubular GEL Battery
 - iv. Module Mounting structures
 - v. Junction Boxes.
 - vi. Earthlings and lightening protections.
 - vii. IR/UV protected PVC Cables, pipes and accessories
- 12.3. COMPONENTS OF INTERNAL ELECTRIFICATION WORKS: Cables, junction box, switches, etc., should conform to the latest BIS or IEC or international specifications / standards as mentioned at PART 7 of the DNIE-T, wherever such specifications are available and applicable. ISI Marked components should be used.
 - i. Wiring for light / fan / exhaust fan / call bell point with 2 x1.5 sq. mm PVC insulated Copper conductor cable (FR) in PVC casing and capping complete with 6/10 amps modular type

- switch, cover plate, surface modular switch board including connection etc as required:-
- a) SHORT POINT (Upto 3 meters)
 - b) MEDIUM POINT (Above 3 meters to 6 meters)
 - c) LONG POINT (Above 6 meters to 10 meters)
- ii. Providing and fixing of 3 pin 6/10 amps plug point with modular type 6/10 amps switch and socket including connection with 2x1.5 sq. mm PVC insulated Copper conductor cable (FR) on surface modular switch board, earthing the 3rd pin etc. as required:-
 - iii. Wiring for Circuit/Sub-main with following sizes PVC insulated Copper conductor cable (FR) in PVC casing and capping etc. as required:-
 - a) 2 X 2.5 sq. mm
 - b) 2 X 4 sq. mm
 - iv. Supplying and drawing 2.5 sq. mm Copper conductor wire along with other wire for loop earthing including connection etc. as required.
 - v. Supplying and fixing of following surface/recess mounted LED luminaries in powder coated die cast housing complete with high efficiency polycarbonate diffuser and necessary connection etc. as required (with 50000 burning hours and minimum lumen output of 100 lm/watt).
 - a) Surface mounted integrated 10-watt LED batten with polycarbonate diffuser:-
 - b) Surface mounted integrated 20-watt LED batten with polycarbonate diffuser.
 - vi. Supplying and installation of Bus bar chamber of the following ratings and sizes made of steel sheet enclosure with all accessories including necessary connection etc. as required.
 - a) 32 amps 415 volts four strip Copper busbar chamber 4 ways of size (300mmx240mmx80mm):-
 - vii. Supplying and installation of Switch disconnecter fuse unit (main switch) of the following ratings with all accessories including necessary connection etc. as required:-
 - a) 32 amps 240 volts DP main switch (rewireable):-
 - b) 32 amps TPN 415 volts front handle cubical main switch (HRC):-
 - viii. Supplying & installation of 10 KA breaking capacity, B/C curve MCB of following ratings in the existing MCB DB including connection as required:-
 - a) 6 to 32 amps DP MCB:-
 - b) 6 to 32 amps TP MCB:-
 - ix. Supplying and installation of powder coated steel sheet enclosure (MCB DB) suitable for 240/415 volts MCB/Isolator/RCCB (SP/SPN/DP/TP/TPN/FP) complete with necessary bus bar, neutral link, din rail, earth bar etc i/c connection as required.
 - a) 2-way SPN MCB DB (single door):-
 - b) 4-way SPN MCB DB (single door):-
 - x. Supplying and installation of Automatic Transfer Switch (ATS) of following the ratings with enclosure along with UV/OV protection controller including necessary connections etc. as required:-

- a) 32 amps, 240volt Two Pole ATS : -
- b) 32 amps, 440volt Four Pole ATS : -
- xi. Supplying and fixing of following outdoor LED luminaries in powder coated die cast housing complete with heat resistant tempered glass/diffuser including necessary connection etc. as required (50000 burning hours and minimum lumen output of 100 lm/watt):-
 - a) 25-watt LED street light fixture with integral driver and secondary lens (IP 66):-
- xii. Drawing of armoured/un-armoured cable with proper saddling etc as required:-
- xiii. Supplying and laying of the following sizes XLPE insulated PVC sheathed aluminium conductor armoured cable with necessary connection etc as required:-
 - a) 2 core 10 Sq mm
 - b) 4 core 16 Sq mm
- xiv. Excavation of earth for cable trench of 800mm depth & 300mm width in all kinds of soil including back filling of cable trench with the excavated soil with 20cm layer of sand, 3.5 nos. bricks per metre for protection including ramming, consolidating to the original shape as per direction of Site Engineer :-
- xv. Supplying and fixing of 1 phase Net Meter: 2 Wire, 5-30 Amp, 240 V with meter box for net metering (Solar Application)
- xvi. Supplying and fixing of Net Meter: 3 Phase, 4 wire, 3 x 10-60 Amp, 3 x 240 V Meter with meter box for net metering (Solar Applicable)

12.4. **TECHNICAL SPECIFICATIONS OF SPV POWER PLANTS**

Solar Photovoltaic Power Plant consists of SPV Array, Module Mounting Structure, Power Conditioning Unit consisting of Maximum Power Point Tracker (MPPT), Control & Protections, interconnection cables, Junction Boxes, switches etc. The SPV power Plants are to supplement grid power, but the PCU should have the facility for grid connection to cater the load during non-sunny days / rainy seasons and having Remote Monitoring Unit. All the components of the Power Plant should confirm to latest BIS or IEC or international specifications wherever such specifications are available and applicable.

12.4.1. **SOLAR PHOTOVOLTAIC MODULE:**

Sl. No.	Item	CAPACITY OF SOLAR POWER PLANT AND TECHNICAL STANDARDS	
		10 kWp	5 kWp
a)	Type of material	Mono/ Multicrystalline	
	Manufacturer and Model of Module	Enlisted under LIST I (Manufacturer of Models & Solar PV Module) of Approved Models and Manufacturers of Solar Photovoltaic Modules (Requirements for Compulsory Registration) Order, 2019 issued by Ministry of New & Renewable Energy, Govt. of India vide Office Memorandum No. 283/54/2018-GRID SOLAR, dated 2 nd January, 2019 including further Amendment(s) issued by MNRE.	

b)	Module efficiency	$\geq 19.50\%$	
c)	Rating of individual module	≥ 500 Wp under STC	
d)	Minimum PV Array Capacity	10000 Wp	5000 Wp
e)	Solar Module frame material	Corrosion resistant materials, preferably anodized Aluminium.	
f)	Protection devices	a) Surge protection b) Low voltage drop by-pass diode	
g)	Junction Box	The module should be provided with flame proof Junction Box with either provision of external screw terminal connection or sealed type and with arrangement for provision of by-pass diode. The box shall have hinged, weather proof lid with captive screws and cable gland entry points or may be sealed type and should confirm IP 65 enclosure.	
h)	IV Curve under STC	Should be laminated at the back side of the module.	
i)	Identification & traceability:	A strip containing the following details should be laminated inside the module so as to be clearly visible from the front side: a) Name of the Manufacturer or distinctive Logo b) Model or Type No. c) Serial No. d) Year of make: e) TREDA	
j)	RFID	In addition, each SPV module should have a RF Identification (RFID) tag containing following information inside the module laminate: a) Name of the manufacturer of PV Module b) Name of the Manufacturer of Solar cells c) Month and year of the manufacture (separately for solar cells and module) d) Country of origin (separately for solar cells and module) e) I-V curve for the module f) Peak Wattage, I_m , V_m and FF for the module g) Unique Serial No and Model No of the module h) Date and year of obtaining IEC PV module qualification certificate i) Name of the test lab issuing IEC certificate j) Other relevant information on traceability of solar cells and module as per ISO 9000 series.	
k)	Warranty / Guarantee:	A. <u>Material Warranty/Guarantee:</u> Material Warranty/Guarantee is defined as: The manufacturer should warrant the Solar Module (s) to be free from the following defects and / or failures for a period not less than twenty five (25) years from the date of commissioning of the Plant. i. Defects and / or failures due to manufacturing	

		ii. Defects and / or failures due to quality of materials. iii. Non conformity to specifications due to faulty manufacturing and / or inspection processes. If the solar Module(s) fails to conform to this warranty, the manufacturer will repair or replace the solar module(s), at the owner's sole option. B. <u>Performance Warranty: The predicted electrical degradation of power generated not exceeding 20% of the minimum rated power over the 25 (twenty five) year period and not more than 10% after 10 (ten) years period of the full rated original output.</u>
--	--	--

12.4.2. **ARRAY JUNCTION BOXES (AJBs):**

12.4.2.1. The array junction boxes are to be provided in the PV array for termination of connecting cables. The AJBs shall be made of GRP/FRP/ Powder Coated Aluminium / cast aluminium alloy with full dust, water, vermin proof & flame proof arrangement. All wires/ cables must be terminated through cable lugs. The AJBs shall be such that input & output termination can be made through suitable cable glands. Single / double compression cable glands should be used.

12.4.2.2. Copper bus bars / terminal blocks housed in the AJB with suitable termination threads conforming to IP65 standard and IEC 62208 Hinged door with EPDM rubber gasket to prevent water entry. Provision of earthings should be provided. It should be placed at 5 feet height or above for ease of accessibility.

12.4.2.3. Each Array Junction Box shall have High Quality suitable capacity Metal Oxide Varistors (MOVs)/ SPDs, suitable Reverse Blocking Diodes. The Array Junction Boxes shall have suitable arrangement for monitoring and disconnection for each of the groups.

12.4.2.4. Suitable markings shall be provided on the bus bar for easy identification and the cable ferrules must be fitted at the cable termination points for identification.

12.4.2.5. All fuses shall have DIN rail mountable fuse holders and shall be housed in thermoplastic IP 65 enclosures with transparent covers.

12.4.2.6. "PIA:TREDA" should be embossed/laminated at the front side of AJBs.

12.4.3. **ARRAY / MODULE MOUNTING STRUCTURE:**

a)	Make	Reputed
b)	Country of origin	India
c)	Location	RCC Roof / Ground / GCI Sheet mounting
d)	Materials & Hardware	a) Hot dip galvanized MS mounting structures shall be used for mounting the modules /panels/arrays. Each structure should have angle of inclination as per the site conditions to take maximum solar insolation. However, to accommodate more capacity the angle inclination may be reduced until the plant meets the specified

		performance ratio requirements. Minimum thickness of galvanization should be 85 microns. b) The mounting structure & foundation shall be so designed to withstand the wind speed of maximum 200 Km per hour. It may be ensured that the design has been certified by a recognized lab/Institution in this regard. Suitable fastening arrangement such as grouting and clamping should be provided to secure the installation against the specific wind speed. c) The mounting structure steel should be as per latest IS 2062: 2011 and galvanization of the mounting structure should be in compliance with latest IS 4759. d) Structural material shall be corrosion resistant and electrolytically compatible with the materials used in the module frame, its fasteners, nuts and bolts. Necessary protection towards rusting need to be provided either by coating or anodization. e) The fasteners used should be made up of stainless steel. The structures shall be designed to allow easy replacement of any module. f) Regarding civil foundation, grouting of the structure should be made with 1:2:4 (1 cement: 2 river sand: 4 stone aggregate of 20 mm. nominal size) cement concrete of size not less than 500 mm X 500 mm X 750 mm. g) The minimum clearance of the structure after mounting the Solar modules should be 500 mm from ground. h) The array structure shall be grounded properly using maintenance free earthing kit.
e)	Galvanization Report / Certificate	Should be produced before commencement of supply of materials.

12.4.4. **DC DISTRIBUTION BOARD (DCDB):**

12.4.4.1. DC Distribution panel to receive the DC output from the array field.

12.4.4.2. Enclosure of DC DPBs should be dust free & vermin proof conform to IP65 protection.

12.4.4.3. The bus bars should be made of copper of desired size.

12.4.4.4. Suitable capacity MCBs / MCCBs shall be provided for controlling the DC power output to the PCU along with necessary surge arrestors.

12.4.4.5. Ammeter & Voltmeter should be provided in the DCDB.

12.4.4.6. All components should be of ISI Marked.

12.4.4.7. "PIA: TREDA" should be embossed/laminated at the front side of DCDB.

12.4.5. AC DISTRIBUTION BOARD (ACDB):

- 12.4.5.1. AC Distribution Panel Board (ACDPB) shall control the AC power from PCU/Inverter, and should have necessary surge arrestors. Interconnection from ACDB to mains at LT Bus bar while PCU is connected with the grid.
- 12.4.5.2. Suitable capacity MCBs / MCCBs shall be provided for controlling the AC power output along with necessary surge arrestors.
- 12.4.5.3. Ammeter, Voltmeter should be provided in the ACDB.
- 12.4.5.4. All switches and the circuit breakers, connectors should conform to IEC 60947, part I, II and III/ IS60947 part I, II and III.
- 12.4.5.5. The provision of changeover switches of required capacity & cabling works as required for SPV Plant will be under the scope of bidder.
- 12.4.5.6. All the Panel's shall be metal clad, totally enclosed, rigid, floor / wall mounted, air- insulated, cubical type suitable for operation on three phase, 410 volts / single phase, 230 volts, 50 Hz as required.
- 12.4.5.7. The panels shall be designed for minimum expected ambient temperature of 45 degree Celsius, 80 percent humidity and dusty weather.
- 12.4.5.8. All indoor panels will have protection of IP54 or better. All outdoor panels will have protection of IP65 or better.
- 12.4.5.9. Should conform to Indian Electricity Act and rules (till last amendment).
- 12.4.5.10. All the 410 volt / 240 volt devices/ equipment like bus support insulators, circuit breakers, SPDs, VTs etc., mounted inside the switchgear shall be suitable for continuous operation and satisfactory performance under $\pm 10\%$ variation in supply voltage & ± 3 Hz variation in supply frequency.
- 12.4.5.11. All components should be of ISI Marked.
- 12.4.5.12. "PIA: TREDA" should be embossed/laminated at the front side of ACDB.

12.4.6. POWER CONDITIONING UNIT (INVERTER PLUS MPPT CHARGE CONTROLLER):

SL. NO.	ITEM	TECHNICAL STANDARDS	
A.	Make	Reputed	
B.	Country of origin	India	
C.	Nominal Capacity	CAPACITY	PARAMETERS
		5 kWp	Minimum 5 kW / 6.25 kVA, single phase
		10 kWp	Minimum 10 kW / 12.50 kVA, 3 Phase
D.	Switching element	IGBT/MOSFET	
E.	Control	Microprocessor / DSP	
F.	Input voltage:		
CAPACITY OF POWER PLANT		PARAMETERS	
5 kWp		a) <i>From PV Array:</i> 96 V nominal DC OR DC input required for PCU from Solar PV Array (500 Wp or higher capacity Solar	

	Modules be connected in appropriate series & parallel combinations so that the Array capacity is minimum 5 KWp.) b) <u>From Battery Bank:</u> 96 Volt, 400 Ah T-GEL Battery OR As required for PCU from DC input c) <u>From AC Source:</u> 240 V (± 10 %), 1 phase, 50 Hz (± .5 Hz).								
10 kWp	a) <u>From PV Array:</u> 180 V nominal DC OR DC input required for PCU from Solar PV Array (500 Wp or higher capacity Solar Modules be connected in appropriate series & parallel combinations so that the Array capacity is minimum 10 kWp.) b) <u>From Battery Bank:</u> 180 volt, 400 Ah T-GEL Battery OR As required for PCU from DC input c) <u>From AC Source:</u> 415 V (± 10 %), 3 phase, 50 Hz (± .5 Hz).								
G. Nominal DC output voltage from Charge Controller Unit:									
<table><tr><td>CAPACITY</td><td>PARAMETERS</td></tr><tr><td>5 kWp</td><td>Suitable for charging 96 V, 400 Ah OR minimum 38400 VAh @ C/ 10 hr. Tubular Plate GEL type battery bank from SPV array.</td></tr><tr><td>10 kWp</td><td>Suitable for charging 180 V, 400 Ah OR minimum 72000 VAh @ C/ 10 hr. Tubular Plate GEL type battery bank from SPV array.</td></tr></table>				CAPACITY	PARAMETERS	5 kWp	Suitable for charging 96 V, 400 Ah OR minimum 38400 VAh @ C/ 10 hr. Tubular Plate GEL type battery bank from SPV array.	10 kWp	Suitable for charging 180 V, 400 Ah OR minimum 72000 VAh @ C/ 10 hr. Tubular Plate GEL type battery bank from SPV array.
CAPACITY	PARAMETERS								
5 kWp	Suitable for charging 96 V, 400 Ah OR minimum 38400 VAh @ C/ 10 hr. Tubular Plate GEL type battery bank from SPV array.								
10 kWp	Suitable for charging 180 V, 400 Ah OR minimum 72000 VAh @ C/ 10 hr. Tubular Plate GEL type battery bank from SPV array.								
H.	Nominal AC Output Voltage and frequency from Inverter	<table><tr><td>CAPACITY</td><td>PARAMETERS</td></tr><tr><td>5 kWp</td><td>240 V (± 10 %), 1 phase, 50 Hz (± .5 Hz).</td></tr><tr><td>10 kWp</td><td>415 V (± 10 %), 3 phase, 50 Hz (± .5 Hz).</td></tr></table>		CAPACITY	PARAMETERS	5 kWp	240 V (± 10 %), 1 phase, 50 Hz (± .5 Hz).	10 kWp	415 V (± 10 %), 3 phase, 50 Hz (± .5 Hz).
CAPACITY	PARAMETERS								
5 kWp	240 V (± 10 %), 1 phase, 50 Hz (± .5 Hz).								
10 kWp	415 V (± 10 %), 3 phase, 50 Hz (± .5 Hz).								
I.	Ambient temperature	-20°C to +50°C							
J.	Humidity	0-95 % non-condensing							
K.	Protection of enclosure	a) Indoor: IP 21 as per IEC 529 b) Outdoor: IP 65 as per IEC 529							
L.	Grid frequency Tolerance range	± 3 Hz or more							
M.	Grid voltage tolerance	As per IE Code							
N.	No-load losses	<1% of the rated power							
O.	Inverter efficiency	>95%							
P.	Total Harmonic Distortion (THD)	<3%							
Q.	Power Factor	>0.90							
R.	Anti-islanding (Protection against Islanding of	Shall have anti-islanding protection in conformity to IEEE 1547/UL 1741/ IEC 62116 or equivalent BIS standard.							

	grid)	
S.	Protection	a) Short Circuit b) Deep discharge of batteries c) Over charging (Automatic trickle charge mode on full charge) d) Input surge voltage e) Over current (Load) f) Battery reverse polarity g) Solar array reverse polarity
T.	Indication (LED / LCD indication)	a) String on b) Mains on c) Input on d) Control on e) Charger on f) Battery charge percentage g) Charger overload h) Battery on trickle i) Battery disconnected/ Fault battery j) Low Solar Power k) System fault l) Charger over temperature m) Input over/ under voltage (for AC)
U.	Standard front LED display (minimum but not exhaustive one)	a) Load, voltage, current, PF, frequency of grid and Inverter b) Solar Charging Current c) Battery Voltage, current, temperature d) Instantaneous and cumulative energy output from Inverter e) Instantaneous and cumulative energy drawn from grid f) Different status of Inverter & Grid g) Kind of fault with audio signal
V.	Front panel metering (minimum but not exhaustive one)	a) Solar ampere meter b) Battery amps meters c) Solar volt meter d) Battery volt meter e) Energy meter (Inverter output)
W.	Controls	Auto / Manual mode selection for Inverter / Grid
X.	Overload capacity	150% of the continuous rating for 30 seconds
Y.	Communication interface	RS 485 / RS 232 / RJ45/USB
Z.	Automatic mode of operation	a) Under light load conditions and with the battery in a full state of charge, the available solar power will supply the load via Inverter with excess solar power (if any) to be exported to grid, if required by user. b) Under medium and heavy load conditions, solar power is used to charge the battery as fast as required and the excess solar power feeds the load via Inverter. c) Under low /no solar power, batteries will feed the load. d) In case of no grid / unstable grid, grid will be disconnected and battery will feed the load. e) In case of low battery and no solar power, grid will feed the load as well as charge the battery.

AA.	Other features	a) PCU / Inverter shall be capable of complete automatic operation including wake-up, synchronization and shutdown. b) The output power factor of PCU/Inverter should be suitable for all voltage ranges or sink of reactive power, Inverter should have internal protection arrangement against any sustainable fault in feed line and against the lighting on feeder. c) Built-in meter and data logger to monitor plant performance through external computer shall be provided. Additionally, facilities for remote monitoring and data acquisition through Remote Monitoring System software at the Owner/TREDA location with latest software/hardware configuration and service connectivity for online/ real time data monitoring of the power plant using available standard broadband/GSM-GPRS mode of internet connectivity. d) In PCU/ Inverter, there shall be a direct current isolation provided at the output by means of a suitable isolating transformer. If isolation Transformer is not incorporated with PCU/ Inverter, there shall be a separate Isolation Transformer of suitable rating provided at the output side of PCU. e) The PCU/ Inverter generated harmonics, flicker, DC injection limits, Voltage Range, Frequency Range and Anti-Islanding measures at the point of connection to the utility services should follow the latest CEA (Technical Standards for Connectivity Distribution Generation Resources) Guidelines. f) The junction boxes/ enclosures should be IP 65 (for outdoor application) / IP 54 (for indoor application) and as per IEC 529 specifications.
BB.	Embossing / Lamination	"PIA: TREDA" at the front side.

12.4.7. **REMOTE MONITORING UNIT:** The Power Plant should have suitable in-built instrumentation for remote monitoring of its Real Time Status. Power Plants shall be capable of transmitting its monitoring parameters over GSM/CDMA/GPRS/TCP IP Network and conform to respective standard protocols. The data from the RMU at the inverter will be transmitted to a central server placed at vendor end and the processed data should be accessible to TREDA through password protected internet gateway. The data transmission shall take place at nominal 4 (four) to 6 (six) hours interval. It will be the responsibility of the vendor to automatically host the real time data of SPV Power Plant.

12.4.7.1. Suggestive list of the events/parameters for monitoring:

- a) Site Name
- b) IP Address
- c) Firmware Version
- d) Inverter Capacity & Battery Bank Voltage
- e) Inverter Status
- f) Date & time (Date in 24 hrs. format)
- g) PV Voltage

- h) PV Current
- i) Cumulative Solar Energy,
- j) Input Voltage
- k) Input Frequency
- l) Output Voltage
- m) Output Frequency
- n) Output Current
- o) Output Power factor
- p) Output Power
- q) Load %
- r) Cumulative Output Active kWh,
- s) Cumulative Output apparent kVAh.

12.4.8. DATA ACQUISITION SYSTEM / PLANT MONITORING:

- 12.4.8.1. Data Acquisition System shall be provided for each of the solar PV plant.
- 12.4.8.2. Data Logging Provision for plant control and monitoring, time and date stamped system data logs for analysis with the high quality, suitable PC. Metering and Instrumentation for display of systems parameters and status indication to be provided.
- 12.4.8.3. The system should have feature of open Application Programming Interface for inbuilt or external data loggers is mandatory for all plants above 25 KW.
- 12.4.8.4. Weather Monitoring for Solar Plant capacity 50kW and above:
 - a) Solar Irradiance: An integrating Pyranometer / Solar cell-based irradiation sensor (along with calibration certificate) provided, with the sensor mounted in the plane of the array. Readout integrated with data logging system.
 - b) Temperature: Temperature probes for recording the Solar panel temperature and/or ambient temperature to be provided complete with readouts integrated with the data logging system.
- 12.4.8.5. The following parameters are accessible via the operating interface display in real time separately for solar power plant:
 - i. AC Voltage.
 - ii. AC Output current.
 - iii. Output Power
 - iv. Power factor.
 - v. DC Input Voltage.
 - vi. DC Input Current.
 - vii. Time Active.
 - viii. Time disabled.
 - ix. Time Idle
 - x. Power produced
 - xi. Protective function limits (Viz. -AC Over voltage, AC Under voltage, over frequency, under frequency ground fault, PV starting voltage, PV stopping voltage.
- 12.4.8.6. All major parameters available on the digital bus and logging facility for energy auditing through the internal microprocessor and read on the digital front panel at any time) and logging facility (the current values, previous values for up to a month and the average values) should be made available for energy auditing through the internal microprocessor and should be read on the digital front panel.

- 12.4.8.7. PV array energy production: Digital Energy Meters to log the actual value of AC/ DC voltage, Current & Energy generated by the PV system provided. Energy meter along with CT/PT should be of 0.5 accuracy class.
- 12.4.8.8. Computerized DC String/Array monitoring and AC output monitoring shall be provided as part of the inverter and/or string/array combiner box or separately.
- 12.4.8.9. String and array DC Voltage, Current and Power, Inverter AC output voltage and current (All 3 phases and lines), AC power (Active, Reactive and Apparent), Power Factor and AC energy (All 3 phases and cumulative) and frequency shall be monitored.
- 12.4.8.10. Computerized AC energy monitoring shall be in addition to the digital AC energy meter.
- 12.4.8.11. The data shall be recorded in a common work sheet chronologically date wise. The data file shall be MS Excel compatible. The data shall be represented in both tabular and graphical form.
- 12.4.8.12. All instantaneous data shall be shown on the computer screen.
- 12.4.8.13. Software shall be provided for USB download and analysis of DC and AC parametric data for individual plant.
- 12.4.8.14. Provision for instantaneous Internet monitoring and download of historical data shall be also incorporated.
- 12.4.8.15. Remote Server and Software for centralized Internet monitoring system shall be also provided for download and analysis of cumulative data of all the plants and the data of the solar radiation and temperature monitoring system.
- 12.4.8.16. Ambient / Solar PV module back surface temperature shall be also monitored on continuous basis.
- 12.4.8.17. Simultaneous monitoring of DC and AC electrical voltage, current, power, energy and other data of the plant for correlation with solar and environment data shall be provided.
- 12.4.8.18. Remote Monitoring and data acquisition through Remote Monitoring System software at TREDA's location with latest software/hardware configuration and service connectivity for online / real time data monitoring / control complete to be supplied and operation and maintenance / control to be ensured by the bidder.
- 12.4.8.19. The bidders shall be obligated to push real-time plant monitoring data on specified intervals through open protocol at receiver location (cloud server) in XML/JSON format, preferably. Suitable provision in this regard will be intimated to the bidders.
- 12.4.8.20. The bidder shall be obligated to measure the harmonic current injection, Direct Current injection and flicker with calibrated meters before the commissioning of the project once in a year in the presence of the TREDA & DISCOM official at a mutually convenient time.

12.4.9. **BATTERY BANK:**

SL. NO.	ITEM	TECHNICAL STANDARDS
a)	Type	Tubular GEL Type.
b)	Country of origin	India
c)	Standards	As per relevant IS Code

d)	Nominal Voltage	2.0 V/cell						
e)	End cell voltage	1.85 V/ cell (27 Deg C)						
f)	Individual Battery / Cell capacity:							
<table><tr><td>CAPACITY</td><td>PARAMETERS</td></tr><tr><td>10 kWp</td><td>400 Ah (at c/10 discharge rate) OR as required to achieve minimum 72000 VAH capacity.</td></tr><tr><td>5 kWp</td><td>400 Ah (at c/10 discharge rate) OR as required to achieve minimum 38400 VAh capacity.</td></tr></table>			CAPACITY	PARAMETERS	10 kWp	400 Ah (at c/10 discharge rate) OR as required to achieve minimum 72000 VAH capacity.	5 kWp	400 Ah (at c/10 discharge rate) OR as required to achieve minimum 38400 VAh capacity.
CAPACITY	PARAMETERS							
10 kWp	400 Ah (at c/10 discharge rate) OR as required to achieve minimum 72000 VAH capacity.							
5 kWp	400 Ah (at c/10 discharge rate) OR as required to achieve minimum 38400 VAh capacity.							
g)	No. of total Cells & combination / Battery Bank:							
<table><tr><td>CAPACITY</td><td>PARAMETERS</td></tr><tr><td>10 kWp</td><td>90 nos. OR as required to achieve minimum 72000 VAh capacity</td></tr><tr><td>5 kWp</td><td>48 nos. OR as required to achieve minimum 38400 VAh capacity</td></tr></table>			CAPACITY	PARAMETERS	10 kWp	90 nos. OR as required to achieve minimum 72000 VAh capacity	5 kWp	48 nos. OR as required to achieve minimum 38400 VAh capacity
CAPACITY	PARAMETERS							
10 kWp	90 nos. OR as required to achieve minimum 72000 VAh capacity							
5 kWp	48 nos. OR as required to achieve minimum 38400 VAh capacity							
h)	Self-discharge	4% @ 30 Deg. C per month						
i)	Charging Efficiency	93% @ 27 Deg. DOD (i.e. 80 % SOC)						
j)	Battery Capacities	Measured at 27 Deg. C						
k)	Battery Accessories	As required under relevant IS code						
l)	Marking	+ ve & - ve terminal shall be marked boldly on the terminals along with color indication Black- negative Red – positive						
m)	Terminals	1 No. of positive 1 No. of negative						
n)	Embossing / Lamination / Sticker	“PIA: TREDA” at front side.						

12.4.10. **BATTERY RACK:** Batteries of the battery bank should be housed in racks of suitable dimensions made of Galvanized MS angle or other suitable and durable material. The racks should be strong enough to bear the load of the batteries. The tiers and rows of the racks for placing the batteries should be made as per site conditions and should facilitate easy monitoring & maintenance of the batteries.

12.4.11. **SOLAR METER:** Uni-directional Solar Meters (as per relevant specifications by MNRE/TREDA) to log the actual value of the Energy generated by the PV System shall be supplied & installed by the successful bidder at suitable location near LT Panel of the USER. Detailed indicative specifications shall be shared with the successful bidder.

12.4.12. **NET-METER:**

12.4.12.1. Bi-directional Smart Energy Meter (as per relevant specifications by MNRE/TREDA/DISCOM) shall be supplied &

installed by successful bidder for measurement of import / export of energy. Detailed indicative specifications shall be shared with the successful bidder.

12.4.12.2. The User shall be responsible for undertaking all applications and obtaining all approvals, permissions etc. for grid connectivity of the Plant.

12.4.12.3. The successful bidder shall be responsible for supply & installation of Net Meter (1 phase / 3 phase as per site requirement) and synchronization of the PCU with the local grid as and when asked for.

12.4.12.4. Grid connectivity of the Net meters will be under the scope of the DISCOM.

12.4.12.5. In cases where TREDA's certification is required, the Contractor shall inform the TREDA and provide the required formats for such certification well in advance.

12.4.12.6. Supply & installation of Net-meters including CTs & PTs, wherever applicable, shall be carried out by the successful bidder as per procedures in line with TREDA & DISCOM guidelines and with their permissions.

12.4.13. **PROTECTIONS:** The system should be provided with all necessary protections as follows:

12.4.13.1. **LIGHTNING PROTECTION:**

12.4.13.1.1. The SPV power plants shall be provided with lightning & overvoltage protection.

12.4.13.1.2. The main aim in this protection shall be to reduce the over voltage to a tolerable value before it reaches the PV or other sub system components. The source of over voltage can be lightning, atmosphere disturbances etc. The entire space occupying the SPV array shall be suitably protected against Lightning by deploying required number of Lightning Arrestors. Lightning protection should be provided as per IEC 62305 standards. The protection against induced high-voltages shall be provided by the use of Metal Oxide Varistors (MOVs) and suitable earthing such that induced transients find an alternate route to earth.

12.4.13.2. **SURGE PROTECTION:** Internal surge protection shall consist of three MOV type surge-arrestors connected from +ve and -ve terminals to earth.

12.4.13.3. **EARTHING PROTECTION:**

12.3.12.3.1 Earthing system shall be in accordance with IS:3043:2018 and IEEE 80-2000, Indian Electricity Rules, Codes of practice and regulations existing in the location where the system is being installed.

12.3.12.3.2 LT power systems, switchyard, PCU and Junction Boxes etc. should be grounded/ earthed properly and earth resistance shall not be more than 5 ohms. It shall be ensured that all the earthing points are bonded together to make them at the same potential. Minimum 4 (four) earthlings as listed below should be provided:

- i. Array Field & Array Junction Box
 - ii. Lightning Arrester
 - iii. DC Distribution Board and Battery Bank
 - iv. AC Distribution Board & Power Conditioning Unit.
- 12.3.12.3.3 The PV module structure components shall be electrically interconnected and shall be grounded.
- 12.3.12.3.4 Earthing conductors shall have a minimum size of 6.0 mm² copper or 16 sq. mm Aluminum or 70 sq. mm hot dip galvanized steel. Unprotected aluminum or copper-clad aluminum conductors shall not be used for final underground connections to earth electrodes.
- 12.3.12.3.5 Separate dedicated and interconnected earth electrodes (DC/AC) must be used for the earthing of the solar PV system support structure with a total earth resistance not exceeding 5 Ohm.
- 12.3.12.3.6 The earth electrodes shall have a precast concrete enclosure with a removable lid for inspection and maintenance. The entire earthing system shall comprise non-corrosive components.
- 12.3.12.3.7 In case LA earthing conductors made up of dip galvanized steel, the conductor should have GI Strip Insulator.
- 12.3.12.3.8 All metal casing/shields of the system shall be earthed properly and all electrical circuits should be thoroughly grounded in accordance with Indian Electricity Act./ Rules.
- 12.3.12.3.9 The earthing shall be maintenance free Gel with Pipe in pipe / Pipe in strip technology filled with anti-corrosive conductive compound (CPRI Tested) below the ground in 100-200 mm dia.
- 12.3.12.3.9.1 50 mm dia. GI pipe should be class B and strip of 25 mm X 6 mm should be inserted top to bottom with no joints.
 - 12.3.12.3.9.2 GI strip should be top to bottom inserted & welded on bottom part of electrode.
 - 12.3.12.3.9.3 GI strip should be hot dip galvanized as per IS-3043 standard 80-100 micron zinc coated.
 - 12.3.12.3.9.4 CCM (Crystalline Conductive Mixture) should be anti-corrosive & to be filled top to bottom in electrode.
 - 12.3.12.3.9.5 Chemical compound PH value should not be less than 8.
 - 12.3.12.3.9.6 With 3 meters electrode 2 bags (25 kg each)/4 bags chemical compound has to be filled in bore.
 - 12.3.12.3.9.7 For semi rocky & rocky area, 8 bags (25 kg each) chemical compound has to be filled in open bore.
 - 12.3.12.3.9.8 Chemical compound has to be mixed with water to make it in paste form and pour surrounding area of electrode.
 - 12.3.12.3.9.9 For 50 mm dia. electrode, bore size should be 6 inch X 3.5 meter depth.
 - 12.3.12.3.9.10 All body connectors should be earthed with GI earthing.
 - 12.3.12.3.9.11 Supply & erection of maintenance free Gel earthing with strip in pipe technology filled with anti-corrosive conductive compound (CPRI tested) below the ground in

100-200 mm dia. Earth pit & surrounding filled with required Mineral Filling Compound (MFC should have hygroscopic property to retain the moisture for long time to create low resistance zone) and C.C. finished chamber covered with hinged type with locking arrangement C.I. Cover, C.I. Frame of size 300 mm X 300 mm complete testing of earth resistance as required GI pipe (IS:1239 marks) GI pipe 3000 mm long, 50 mm dia., GI Strip. Earthing as per IS:3043-1987 and its latest amendments.

- 12.3.12.3.10 Gel earthing & Chemical compounds should be type tested from Govt. approved / Govt. recognized / NABL Accredited Laboratory / ILAC i.e. International Laboratory Accredited Laboratory (in case of foreign laboratory).

12.4.14. GRID ISLANDING:

12.4.14.1. In the event of a power failure on the electric grid, it is required that any independent power-producing Inverters attached to the grid turn off in a short period of time. This prevents the DC-to-AC Inverters from continuing to feed power into small sections of the grid, known as “islands.” Powered islands present a risk to workers who may expect the area to be unpowered, and they may also damage grid-tied equipment. The PV system shall be equipped with islanding protection. In addition to disconnection from the grid (due to islanding protection) disconnection due to under and over voltage conditions shall also be provided.

12.4.14.2. A manual disconnect 2 – pole / 4-pole isolation switch of appropriate capacity beside automatic disconnection to grid would have to be provided at utility end to isolate the grid connection by the utility personnel to carry out any maintenance. This switch shall be locked by the utility personnel.

12.4.15. **CABLES:** Cables of appropriate size to be used in the system shall have the following characteristics:

12.4.15.1. Standard: 1.1 KV grade confirming IEC 60227/IS694, IS7098 Part 1 and IEC60502/IS1054 (Pt. I & II).

12.4.15.2. Temp. Range: -10°C to +80°C

12.4.15.3. Excellent resistance to heat, cold, water, oil, abrasion, UV radiation

12.4.15.4. Flexible

12.4.15.5. All cables and connectors for use for installation of solar field must be of solar grade which can withstand harsh environment conditions for 25 years and voltage level as per latest IEC standards.

12.4.15.6. The cables of 1000 volts DC outdoor installations shall comply with the draft EN 50618 for service life expectancy of 25 years.

12.4.15.7. Sizes of cables between array interconnections, array to junction boxes, junction boxes to Inverter etc. shall be so selected to keep the voltage drop (power loss) of the entire solar system to the minimum (2%).

12.4.15.8. For the DC cabling, XLPE or XLPO insulated and sheathed, UV-stabilized single core multi stranded flexible copper cables shall be used. Multi-core cables shall not be used.

- 12.4.15.9. For the AC cabling, PVC or XLPE insulated and PVC sheathed single or multi-core multi stranded flexible copper cables shall be used. Outdoor AC cables shall have a UV-stabilized outer sheath.
- 12.4.15.10. The cables (as per IS) should be insulated with a special grade PVC compound formulated for outdoor use. Outer sheath of cables shall be electron beam cross linked XLPO type and black in colour.
- 12.4.15.11. The DC cables from the SPV module array shall run through a UV-stabilized PVC conduit pipe of adequate diameter with a minimum wall thickness of 1.5 mm.
- 12.4.15.12. Cables and wires used for the interconnection of Solar PV modules shall be provided with solar PV connectors (MC4) and couplers.
- 12.4.15.13. All cables and conduit pipes shall be clamped to the rooftop, walls and ceilings with thermo-plastic clamps at intervals not exceeding 50 cm; the minimum DC cables size shall be 4.0 mm² copper, the minimum AC cable size shall be 4.0 mm² copper. In three phase systems, the size of the neutral wire size shall be equal to the size of the phase wires.
- 12.4.15.14. Cable Routing/ Marking: All cable/wires are to be routed in a GI / PVC cable tray and suitably tagged and marked with proper manner by good quality ferule or by other means so that the cable easily identified. In addition, cable drum no./ Batch no. to be embossed / printed at every one meter.
- 12.4.15.15. Cable Jacket should also be electron beam cross – linked XLPO, flame retardant, UV resistant and black in colour.
- 12.4.15.16. All cables and connectors for use for installation of solar field must be of solar grade which can withstand harsh environment conditions including high temperatures, UV radiation, rain, humidity, dirt, salt, burial and attack from moss, microbes for 25 years and voltages as per IEC standards. DC cables used from solar modules to array junction box shall be solar grade copper with XLPO insulation and rated for 1.1KV as per relevant standards only.
- 12.4.15.17. The total voltage drop on the cable segments from the solar PV modules to the solar grid Inverter and to battery bank shall not exceed 2.0%.
- 12.4.15.18. The total voltage drop on the cable segments from the solar Inverter to the distribution board shall not exceed 2.0%.
- 12.4.16. TOOLS & TACKLES AND SPARES:**
- 12.4.16.1. After completion of installation & commissioning of the solar power plant & internal electrification works, necessary tools & tackles are to be provided free of cost by the bidder for maintenance purpose. Digital Multi meter, Cell Tester, set of Hand Tools, Magnetic Compass etc. to be supplied by the bidder shall be approved by TREDA for specifications and make.
- 12.4.16.2. A list of recommended spares in case of PCU/Inverter comprising of a set of control logic cards, IGBT driver cards etc., fuses, MOVs / arrestors etc. shall be indicated and supplied along with the equipment. A minimum set of spares shall be maintained in the plant itself for the entire period of warranty and Annual Maintenance Contract Period which upon its use shall be replenished.

- 12.4.17. **DANGER BOARDS AND SIGNAGES:** Required quantity of Danger boards should be provided as and where necessary as per IE Act. /IE rules as amended up to date. Text of the signage to be finalized in consultation with TREDA.
- 12.4.18. **FIRE EXTINGUISHERS:** The fire fighting system for the proposed power plant for fire protection shall be consisting of:
- 12.4.18.1. Portable fire extinguishers in the control room for fire caused by electrical short circuits
 - 12.4.18.2. Sand buckets in the control room
 - 12.4.18.3. The installation of Fire Extinguishers should confirm to TAC regulations and BIS standards. The fire extinguishers shall be provided in the control room housing PCUs as well as on the Roof or site where the PV arrays have been installed.
- 12.4.19. **SAFETY MEASURES:** The bidder shall take entire responsibility for electrical safety of the installation(s) including connectivity with the grid and follow all the safety rules & regulations applicable as per Electricity Act, 2003 and CEA guidelines etc.
- 12.4.20. **DISPLAY BOARD:** Sign Board on Aluminium sheet in MS Angle frame not less than of size 3 ft. X 2 ft. should be supplied and installed at a suitable location at the site of installation of system as per direction of the Project Manager, TREDA. The sign board should contain the information as follows:
- i. Name of the site:-
 - ii. Capacity of SPV Power Plant:
 - iii. Implemented by: Tripura Renewable Energy Development Agency (TREDA), (Department of Power, Govt. of Tripura).
 - iv. Funded by:
 - v. Date of commissioning: ____/____/20____,
 - vi. Contact Person details:
- 12.4.21. **DISPLAY OF SINGLE LINE DIAGRAM:** SLD of Power Plant of suitable size should be displayed in the Control Room as per direction of the Officer-in-charge, TREDA.

PART 5: DETAILS OF SCOPE OF WARRANTEE/GUARANTEE AND ANNUAL MAINTENANCE CONTRACT (AMC):

1. WARRANTEE/GUARANTEE:

- 1.1 The SPV Power Plants & Internal Electrification Works shall be warranted /guaranteed for 5 (FIVE) years for trouble free operation from the date of commissioning. The mechanical structures, electrical works including power conditioners /inverters/ charge controllers/maximum power point tracker units/distribution boards/digital meters/switchgears, storage batteries and overall workmanship of the SPV power plants/systems and internal electrification works must be Warranted / Guaranteed against any manufacturing/ design/ installation defects for a minimum period of 5 (FIVE) years. Materials found damaged during transit or erection shall be replaced by the supplier immediately without waiting for settlement of their claim with their Insurance Company.
- 1.2 The Warrantee/Guarantee shall be against breakages, malfunctions, non-fulfilment of guaranteed performance and breakdowns due to manufacturing defects or defects that may arise due to improper operation of electrical or electronics components of the system but do not include physical damages by end users.
- 1.3 The Warrantee/Guarantee will cover all the materials and components involved in the installation & commissioning of the SPV Power Plants & internal electrification works by the successful bidder under this contract irrespective of the fact whether these have been manufactured by the successful bidder or not. The decision of TREDA in this regard should be final & binding on successful bidder.
- 1.4 PV modules must be warranted for their output wattage, which should not be less than 90% at the end of 10 years, 80% at the end of 25 years from the date of commissioning of the Plant.
- 1.5 During the warranty / guarantee period, defective/ non-functioning SPV System including Solar Modules, Power Conditioning Unit, storage batteries & internal electrification works etc. and other spare parts of the systems shall be repaired / replaced free of cost by the successful bidder at his own cost for proper, sustained & reliable operation of the systems.
- 1.6 Minimum 2 (two) staffs of Block/Panchayat Samiti Offices/Block Advisory Committee / Other Government Offices should be trained by the successful bidder for running of SPV Power Plant.
- 1.7 During the Warranty / Guarantee period, TREDA / District Project Director Officer / District Project Officers of TREDA, / User reserves the right to cross check the performance of the systems with the minimum performance levels specified at PART 4 of this DNIE-T / MNRE latest Guidelines for off-grid and decentralized solar applications.
- 1.8 Warrantee/guarantee Card is to be supplied with each system. The format of Warranty / Guarantee card should get approved from TREDA.

2 ANNUAL MAINTENANCE:

- 2.1 The successful bidder should provide 5 (five) years maintenance of the SPV Power Plants & internal electrification works from the date of

- commissioning which shall include corrective & preventive maintenance as well as routine service visit to ensure satisfactory performance of the systems.
- 2.2 Periodic cleaning & check of SPV Modules, PCUs, Batteries, BoS & internal electrification works etc. shall be carried out by the successful bidder as a part of the routine prevention and breakdown maintenance. Annual Maintenance Guideline is given at Clause No. 6 below. The Quarterly & Yearly Annual Maintenance Report (As per format of TREDA) should be duly signed by Bidder's Representative / Construction Manager, User & TREDA / respective District Officer-in-charge and are to be submitted to TREDA.
 - 2.3 Proper repainting, recoating of exposed surfaces to prevent rusting & replacement of worn out parts shall be carried out along with the maintenance of the Inverter/PCU and battery bank.
 - 2.4 Internal electrifications should be periodically checked & any defect should be removed to protect from sudden breakdown.
 - 2.5 The successful bidder shall attend to any complain from User and rectify the faults or breakdowns within a time frame of **72 (Seventy two) hours** from such complain. In case of theft, the complaint is to be resolved within 10 (Ten) calendar days provided that the theft is duly certified by Officer-in-charge, TREDA. In case of any part to be imported from outside the state, the complaint is to be resolved within 15 (Fifteen) calendar days. Maintenance Report in prescribed format duly signed by Project Manager of successful bidder, User and District Officer –in-charge of TREDA are to be submitted to TREDA.
 - 2.6 If the down time period for any beneficiary complaint exceeds 72 hours (10 calendar days in case for theft cases) and the successful bidder fails to make the plant operational, a penalty for time period exceeding 72 hours (10 calendar days in case for theft cases), penalty of Rs. 300/- (Rupees three hundred) per day per plant for first 3 (three) days and thereafter Rs. 500/- (Rupees five hundred) per Plant per day shall be imposed till SPV Power Plants including IE works puts back to satisfactory working condition.
 - 2.7 The successful bidder shall be issued a Notice to pay the applicable penalty within 7 (seven) days to TREDA. If the successful bidder fails to pay the penalty within notice period, the TREDA shall recover from running bills or Additional Performance Security or will encash the Performance Bank Guarantee / Additional Performance Security immediately.
 - 2.8 In case the successful bidder fails to repair any fault of the systems **within 10 calendar days**, TREDA may make the system functional and debit the actual expenditure to the party which will be realized from the Performance Bank Guarantee / Additional Performance Security of the successful bidder and the defaulter bidder will be blacklisted. The expenditure towards the repairing of the system(s) determined by TREDA should be final & binding on the successful bidder.
 - 2.9 Under Force Majeure circumstances, penalty can be waived off.
 - 2.10 Quarterly Maintenance Report as per prescribed format duly filled in and signed by Bidder's representative / Construction Manager, User & TREDA / respective District Officer-in-charge of TREDA shall be submitted within **7th day** of the following month. The Yearly Maintenance Report as per prescribed format duly filled in and signed by Bidder's representative / Construction Manager, User & TREDA /

- respective District Officer-in-charge of TREDA along with Quarterly Maintenance Reports shall be submitted along with Invoice of AMC for processing of payment, failing which, the claim for AMC will not be entertained.
- 2.11 The service personal of the successful bidder shall make quarterly routine maintenance visits till the completion period of warranty / guarantee and Annual Maintenance Contract period apart from on-call maintenance for troubleshooting. The successful bidder shall provide site wise Schedule of quarterly visits for routine maintenance. The above schedule shall be submitted at the beginning of each year of AMC for acceptance & approval of TREDA.
 - 2.12 The deputed personal shall be in a position to check and test all the equipments regularly, so preventive action, if any, could be taken well in advance to save any equipment from damage/outage.
 - 2.13 During operation and maintenance period of the SPV Power Plants & internal electrification components, if there is any loss or damage of any component due to miss management or miss handling or due to any other reasons pertaining to the deputed personal, what-so-ever, the successful bidder shall be responsible for immediate replacement or rectification. The damaged component may be repaired or replaced by new component.
 - 2.14 The maintenance shall include replacement of any component irrespective of whether the defect was a manufacturing defect or due to wear and tear.
 - 2.15 Successful bidder is to submit inter changeability certificate for its product supplied for replacement during warranty/guarantee and AMC period and even when it is purchased from open market. In case due to change in technology, the supplied product is not available during warranty / guarantee & AMC period, than the improved version of product can be used in warranty/guarantee and AMC period with same or improved technical parameters or the combination thereof after written communication of TREDA at the same cost & terms and conditions. Successful bidder, on whom LoA has been placed, has also to confirm that the prices of improved version of product are not lesser than the original product or its parts in comparison.
 - 2.16 TREDA reserves the right to ascertain the extent of damages due to outage of SPV Plant & internal electrification components and claim the cost of such damages for non-fulfilment of warranty/guarantee/AMC obligations, apart from forfeiture of 100% of Performance Security with Additional Performance Security deposits in the event of unsatisfactory performance due to lack of maintenance.
 - 2.17 Successful bidder shall carry sufficient inventories / spares (At least 5% of the LoA quantity) to assure ex-stock supply of spares for consumables such as cable plugs, nut & bolts, screws, switches, fuses, indicating lamps & blocking diodes, spare modules, spare Luminaries, spare batteries etc. Other mandatory spare parts & components shall be supplied as promptly as possible.
 - 2.18 5% of the spare parts specified at Sub-clause 2.17 above in good running condition have to be supplied by the each successful bidder within their quoted price & should be kept in the Head Quarter of TREDA/ District Offices of TREDA to be used for immediate replacement of any defective components at the site of installation. The cost of transportation of the spares from TREDA/ District Offices of

TREDA to the respective site is to be borne by the successful bidder. Spares as specified at Sub-clause No. 2.17 and Sub-clause No. 2.18 above have to be supplied by the successful bidders along with the supply of 1st lot of materials.

- 2.19 The defective PCU shall be repaired, tested and re-commissioned at site or otherwise replaced by a new PCU by the successful bidder free of cost within 45 (FORTY FIVE) days from the date of replacement of spare PCU. The spare PCU supplied from TREDA shall be returned back to Head Quarter of TREDA/Office of the District Project Officer/ Project Director, TREDA at the cost & risk of the bidder.
- 2.20 The successful bidder shall educate the beneficiaries on the process for registering the complaints on Call Centre / Toll Free Nos.
- 2.21 At the end of each AMC year, the Additional Performance Security should be claimed within 60 (sixty) days of the last date of 1st / 2nd / 3rd / 4th / 5th year of AMC period, failing which, it will be lapsed and the amount should not be carry forward to the next year.

3 FAILURE TO RECTIFY THE PROBLEM:

- 3.1.1 If the Contractor/ Bidder(s) fails to rectify the plant downtime within **15 (fifteen) calendar days** from the date of identification of such defect, unless the extension of time is mutually discussed and agreed between the bidder and the respective Beneficiary.
- 3.1.2 If the Contractor/Bidder(s) fails to rectify the problem, the respective Beneficiary shall/may rectify the problem at the expense of the Contractor/ Bidder(s), in such case on genuine complaint TREDA will take appropriate action including forfeiture of Additional Performance Security & PBG and the bidder will be suspended/ debarred / blacklisted for 5(five) years from being eligible to submit Bids / proposals for contracts with TREDA along with other punitive actions deemed found suitable by TREDA will also be initiated.

- 4 **FAILURE TO PERFORM AMC OBLIGATIONS:** If the bidder fails to perform AMC obligations for any 2 (two) years within 5 (five) years AMC Period, TREDA may take appropriate action including forfeiture of Additional Performance Security & PBG and the bidder will be suspended/ debarred / blacklisted for 5(five) years from being eligible to submit Bids / proposals for contracts with TREDA along with other punitive actions deemed found suitable by TREDA will also be initiated.

5 OFFICE CUM SERVICE CENTRE FOR INSTALLATION, COMMISSION & AFTER SALES SERVICE DURING WARRANTY / GUARANTEE AND ANNUAL MAINTENANCE CONTRACT (AMC) PERIOD:

- 5.1 To ensure timely maintenance of the systems, the successful bidder shall have an ACTIVE OFFICE at Agartala and at least one ACTIVE OFFICE – CUM - SERVICE CENTRE in each Districts where the Power Plant have been installed and must have helpline number in local language to lodge complain by the USERS.
- 5.2 OFFICE – CUM – SERVICE CENTRE must have skilled technical persons with full support of infrastructure & sufficient spare parts as required for installation, commissioning & for providing after sales services during the warranty/guarantee and Annual Maintenance Contract period.

- 5.3 The successful bidder shall submit to TREDA the details of OFFICE – CUM – SERVICE CENTRE, the name of skilled technical persons with T & P, complete addresses, contact number, list of spares etc. along with the list of technicians with address and telephonic numbers to the office of the undersigned before commencement of supply of materials.

6 OPERATION AND MAINTENANCE GUIDELINES:

For the optimal operation of a PV Plant and internal electrification components, preventive maintenance must be carried out on a regular basis. All the components should be kept clean. It should be ensured that all the components are fastened well at their due place. Maintenance guidelines for various components viz. Solar PV Modules, Power Conditioning Unit, Battery, cabling & Internal electrification works etc. are given below:

A. SOLAR PV MODULES:

- A.1. Although the cleaning frequency for the modules will vary from site to site depending on soiling, it is recommended that -- The SPV modules are cleaned at least once in every month.
- A.2. Any bird droppings or spots should be cleaned immediately.
- A.3. Use water and a soft sponge or cloth for cleaning.
- A.4. Do not use detergent or any abrasive material for module cleaning.
- A.5. Iso-propyl alcohol may be used to remove oil or grease stains.
- A.6. Do not spray water on the module if the module glass is cracked or the back side is perforated.
- A.7. Wipe water from the module as soon as possible.
- A.8. Early morning is particularly good time for module cleaning.
- A.9. Check if there is any shade problem due to vegetation or new building.
- A.10. If there are, arrange for removing vegetation or shift panel to shadow free place.
- A.11. Ensure that the module terminal connections are not exposed while cleaning.

B. CABLES AND CONNECTION BOXES:

- B.1. Check the connections for corrosion and tightness.
- B.2. There should be no vermin inside the box.
- B.3. Check the cable insulating sheath for crack, break or burn. If found damaged, replace the wire.
- B.4. Ensure proper clamping of the wire.

C. POWER CONDITIONING UNIT (CHARGE CONTROLLER AND INVERTER):

- C.1. The PCU / Inverter area should be clean, dry and ventilated.
- C.2. Remove any excess dust in heat sink and ventilation with only dry cloth or brush.
- C.3. Check for vermin infestation and clean if found any.
- C.4. Check functionality e.g., automatic disconnect upon loss of grid power supply, at least once a month, if the PV Plant is in operation with grid.
- C.5. Verify the state of DC/AC surge arrestors, cable connections and circuit breakers.

D. BATTERY

- D.1. Batteries should remain in dry and ventilated area.
- D.2. Check the battery terminals for any loose connection, corrosion or dust.
- D.3. Check for electrolyte leaks and integrity.
- D.4. Check for corrosion of racks /cabinets.
- D.5. Check for ambient temperature.

E. SHUTTING DOWN THE SYSTEM:

SIGNATURE OF BIDDER WITH SEAL & DATE



Page 76 of 105



- E.1. Disconnect system from all power sources in accordance with instructions for all other components used in the system.
- E.2. Completely cover system modules with an opaque material to prevent electricity from being generated while disconnecting conductors.
- E.3. To the extent possible, system shutdown shall not be done during day time or peak generation period.

F. INTERNAL ELECTRIFICATION WORKS: Periodic maintenance shall be carried out for safe & trouble free running of electrical installations including replacement of defective components as & when required.

SIGNATURE OF BIDDER WITH SEAL & DATE

do
15/11/2016

PART 6: TECHNICAL INFORMATION
(TO BE SUBMITTED AFTER ISSUANCE OF LETTER OF AWARD)

Sl. No.	Description	NAME OF MANUFACTURER
		Capacity of Plant: 10/5 kWp
6.1	Solar PV Module	
6.1.1	Type of Module: Crystalline/ multi crystalline	
6.1.2	Make & Model	
6.1.3	Max power at STC P_{max} (W)	
6.1.4	Voltage at Max power V_{mp} (V)	
6.1.5	Current at Max power I_{mp} (A)	
6.1.6	Open circuit voltage V_{oc} (V)	
6.1.7	Short circuit current I_{sc} (A)	
6.1.8	Module efficiency	
6.1.9	Fill factor	
6.2	Solar Power Conditioning Unit	
6.2.1	Make & Model	
6.2.2	Operating voltage (DC)	
6.2.3	Operating voltage AC (pure sine wave)	
6.2.4	Switching devices : IGBT/MOSFET	
6.2.5	Control : Microprocessor /DSP	
6.2.6	Details of Indicators provided	
6.3	Tubular GEL Battery	
6.3.1	Make & Model of the offered cells:	
6.3.2	Capacity of the battery	
6.3.3	The type and material of separators	
6.3.4	Material and thickness of container	
6.3.5	Ah efficiency (Min.)	
6.3.6	Watt hour efficiency (Min.)	
6.3.7	End cell voltage at 10 hr. discharge rate	
6.3.8	Short Circuit current of battery bank in Ampere	
6.3.8	Are the offered batteries eco-friendly: YES/NO	
6.4	Other components as per DNIe-T	
[NOTE: TECHNICAL DETAILS ALONG WITH TEST CERTIFICATES / BIS CERTIFICATES AS PER DNIe-T SHALL BE SUBMITTED]		

(Signature of the authorized signatory)
with seal & date

PART – 7: QUALITY CERTIFICATION, STANDARDS AND TESTING FOR SOLAR PV POWER PLANTS:

7.1 Quality certification and standards for solar PV systems are essential for the successful implementation of this technology. It is also imperative to put in place an efficient and rigorous monitoring mechanism, adherence to these standards. Hence, all components of solar PV system/ plant / Other items etc. must conform to the relevant standards and certifications given below:

7.1.1	Solar Photovoltaic (PV) Modules / Panels	IEC 61210/ IS 14286	Design Qualification and Type Approval for Crystalline Silicon Terrestrial PV Modules
		IEC 61701	Salt Mist Corrosion Testing of PV Modules
		IEC 61730-Part – 1 and Part 2	PV Module Safety Qualification – a) Part - 1: Requirements for Construction b) Part - 2: Requirements for testing safety qualification.
7.1.2	Power Conditioning Unit / Solar PV Inverters	IEC 61683 / IS 61683 and must conform to the relevant National / International Electrical Safety Standards.	Efficiency Measurements for Inverter
		IEC 60068-2 (1, 2, 14, 30) /Equivalent BIS Std.	Environmental Testing for Inverter
		IEC 62093 and must conform to the relevant National / International Electrical Safety Standards.	Design Qualification for Charge Controller / MPPT Unit
		IEC 60068-2 (1, 2, 14, 30) /Equivalent BIS Std.	Environmental Testing for Charge Controller / MPPT Unit
		IEC 62109-1, IEC 62109-2	Safety of power converters for use in photovoltaic power system safety compliance (Protection degree IP 65 for outdoor mounting, IP 54 for indoor mounting)
		IEC 60255-27	Measuring relays and protections equipment – Part 27: Product safety requirements
		IEC 61000-2,3,5	Electromagnetic Interference (EMI) and Electromagnetic Compatibility (EMC) testing of PV Inverters (as applicable)
		[NOTE: In case if the Charge controller is in-built in the Inverter, no separate IEC 62093 test is required but must additionally conform to the relevant National/ International Electrical Safety Standards wherever applicable]	
7.1.3	Tubular GEL Battery	IEC 61427 / IS 16270	Secondary Cells and Batteries for Photovoltaic Energy Systems (PVES)– General Requirements and Methods of Test

		IS:10549:2005:	Stationary Valve Regulated Lead Acid Batteries – specifications
		IEC 60896 – 21 & 22: 2004-02:	Stationary lead-acid batteries – Valve regulated types – Methods of test
7.1.4	Fuses	IS/IEC 60947 (Part 1, 2 & 3), EN 50521	General safety requirements for connectors, switches, circuit breakers (AC/DC)
		IEC 60269-6	Low voltage fuses – Part 6: Supplementary requirements for fuse-links for the protection of Solar PV Energy Systems
7.1.5	Surge Arrestors	IEC 61643-11:2011 / IS 10086 (SPD)	Low voltage surge protective devices – Part 11: Surge protective devices connected to low voltage power systems – Requirements and test methods
7.1.6	Cables	IEC 60227/IS 694, IEC 60502/IS1054 (Part 1 & 2)	General test and measuring method for PVC (Polyvinyl Chloride) insulated cables for working voltage upto and include 1100V, and UV resistant for outdoor installation
		BS EN 50618	Electric cables for photovoltaic systems [BT(DE/NOT)258], mainly for DC cables.
7.1.7	Earthing / Lightning	IEC 62561 Series (Part 1,2 & 7) (Chemical earthing)	a) IEC 62561-1: Lightning protection system components (LPSC) – Part 1: Requirements for connection components b) IEC 62561-2: Lightning protection system components (LPSC) – Part 2: Requirements for conductors and earth electrodes. c) IEC 62561: Lightning protection system components (LPSC) – Part 7: Requirements for earthing enhancing compounds.
7.1.8	Junction Boxes	IEC 60529	Junction Boxes and Solar panel terminal boxes shall be of the thermo plastic type with IP 65 protection for outdoor use and IP 54 protection for indoor use.
7.1.9	Solar Mounting Structure PV	IS 2062/IS4759	Materials for the structure mounting.
7.1.10	Switches/Circuit Breakers /Connectors	IEC 60947 part I,II, III / IS 60947 Part I,II,III EN 50521	General Requirements Connectors – safety A.C./D.C.

FORMAT - 1**[To be submitted in the letterhead of the bidder]**
PARTICULARS OF BIDDER

A. GENERAL DETAILS		
1.	Details Bidder	
a)	Name of Bidder	
b)	Postal Address	
c)	Mobile No.	
d)	e-mail address	
e)	Geo coordinates of the office	
2.	Status of bid	a. SPV SYSTEM MANUFACTURER b. SPV SYSTEM INTEGRATOR
3.	a) Legal status of bidder (Please specify)	a) Body incorporated in India under the Companies Act, 1956 or Companies Act, 2013 including any amendment there to b) Government of India / State Government Undertaking/ PSUs.
	b) In case of Manufacturer, (Please specify the component)	a) SPV Module [Enlisted under ALMM of MNRE published by MNRE time to time] b) Power Conditioning Unit (PCU) of SPV Power Plants c) Tubular GEL Battery
4.	Name, designation of the authorized representative of the Bidder to whom all references shall be made	
5.	Mobile No. of authorized representative	
6.	Name and address of the Indian / Foreign Collaboration if any	
7.	Have anything/extra other than price of items (as mentioned in price Schedule) been written in the price schedule.	
8.	Have the Bidder ever been debarred by any Govt. Deptt. / Undertaking for undertaking any work? (Upload undertaking)	
9.	Details of offer (please mention number of pages and number of Drawings uploaded)	
10.	Reference of any other	



	information attached by the bidder (please Mention no. of pages & no. of drawings)	
B. FINANCIAL DETAILS		
1.	Permanente Account Number (PAN)	
2.	Goods & Services Tax (GST) details	
A. GSTIN		
B. Bank Account linked with GSTIN		
C. Name of Bank		
D. Name of Branch		
E. Address		
F. Account Type		
G. IFSC		
H. MICR		

(Signature of the authorized signatory)

Name:

Designation:

Company Seal:

FORMAT 2

[To be submitted in the letterhead of the bidder] (SCHEDULE OF EXPERIENCE / CREDENTIAL OF BIDDER

Name of Work	Design, survey, manufacture, supply, erection, testing, commissioning of 10 kWp & 5 kWp Solar Photovoltaic Power Plant (Hybrid) at Block Offices etc. including Warranty /Guarantee, Annual Maintenance Contract and insurance coverage for 5 (five) years on turn-key basis throughout the state of Tripura
DNIE-T	Insert DNIE-T Number & date

1.	Name of State Nodal Agency for RE / DISCOM (Government / Private) / Govt. Departments / Government Organization		
	Type of System (Off-grid / Hybrid Solar Power Plants AND / OR Grid connected Solar Power Plants in India)		
	LoA / Work Order No. and date		
	Awarded capacity (In kWp)		
	Contract value (In lakh)		
	Schedule date of completion of Work as per Letter of Award (LoA)	Start date: DD/MM/YYYY	
	Actual date of completion of work	End date: DD/MM/YYYY	
	Capacity of completed work (In kWp)		
	Cost of completed work (In lakh)		
	Details of the contact person of Government Department / Govt. PSU/DISCOM (Government / Private) to whom reference may be made	NAME:	
		Designation:	
		Mobile Number:	
		e-mail ID:	
	Copy of the Letter of Award and corresponding Commissioning certificate# - Enclosed or Not	YES / NO	
2.	-----		
3.	-----		
[NOTE: More than three (3) entries are permitted]			

NOTE:

1. Projects commissioned during the last seven (7) years ending last day of the month previous to the month of the floating of the tender will be considered.
2. Upload Letter of Award / Work Order, corresponding Works completion / Commissioning Certificates / Grid Synchronization Report issued by the authorised representative of State Nodal Agency for RE / DISCOM (Government / Private) / Govt. Departments / Government Organization of the concerned State.



3. Copy of the Letter of Award / Work Order, corresponding Works completion / Commissioning Certificates / Grid Synchronization Report should be marked with ANNEXURE NUMBERS before uploading.
4. Letter of Award / Work Order, corresponding Works completion / Commissioning Certificates NOT issued by the authorised representative of State Nodal Agency for RE / DISCOM (Government / Private) / Govt. Departments / Government Organization of the concerned State will not be considered.

(Signature of the authorized signatory)

Name:

Designation:

Company Seal:

FORMAT 3

[SHOULD BE SUBMITTED IN THE LETTERHEAD OF THE BIDDER AND SHOULD BE CERTIFIED BY THE PRACTICING CA OF THE BIDDER]

Name of Work	Design, survey, manufacture, supply, erection, testing, commissioning of 10 kWp & 5 kWp Solar Photovoltaic Power Plant (Hybrid) at Block Offices etc. including Warranty /Guarantee, Annual Maintenance Contract and insurance coverage for 5 (five) years on turn-key basis throughout the state of Tripura
DNIE-T	(Insert DNIE-T Number & date)
NAME OF BIDDER	

PART 1

ANNUAL TURNOVER AND NET WORTH DATA FOR 4 YEARS

Year	Net Worth (In INR)	Annual Turnover (In INR)
ANNUAL AVERAGE		

PART 2

INFORMATION ON PROFITABILITY

PROFIT BEFORE TAX DATA FOR 4 YEARS

Year	Amount (In INR)

(Signature of the authorized signatory of bidder)

Name:

Designation:

Company Seal:

Certified by Chartered Account (CA) in original on their letterhead with Membership No., FRN, UDIN, Contact details (e-mail and Mobile Number)

FORMAT 4

PERFORMANCE SECURITY BOND FORMAT **(MODEL BANK GUARANTEE FORMAT FOR PERFORMANCE SECURITY)**

To

The Director General,

Tripura Renewable Energy Development Agency,
Deptt. Of Power, Govt. of Tripura,
Gorkhabasti, Agartala, West Tripura District, Tripura,
Pin:799006, e-mail: tredatenders25@gmail.com.

WHEREAS (Name and address of the agency) (Hereinafter called as “The Contractor”) has undertaken, in pursuance of **Letter of Intent (LoI) No., dated** for “-----
----- **Name of work** -----” (Hereinafter called “The Contract”).

AND WHEREAS it has been stipulated by you in the said Contract that the Contractor shall have to furnish a Bank Guarantee by a Nationalized Bank having Branches at Agartala, Tripura for the sum specified therein as security for compliance with its obligations in accordance with the Contract.

AND WHEREAS we have agreed to give the Contractor such Bank Guarantee.

NOW THEREFORE, we hereby affirm that we are guarantor and responsible to you, on behalf of the Contractor, up to a total of (amount of guarantee in words and figures), and we undertake to pay you, upon your first written demand declaring the Contractor to be default under the Contract and without cavil or argument, any sum or sums within the limits of (Amount of guarantee) as aforesaid, without your needing to prove or to show grounds or reasons for your demand or the sum specified therein.

We hereby waive the necessity of your demanding the said debt from the Contractor before presenting us with the demand.

We further agree that no change or addition to or other modifications of the terms of the Contract to be performed there under or of any of the Contract document which may be made between you and the Contractor shall in any way release us from any liability under this guarantee and we hereby waive notice of any such changes, addition or modification.

This guarantee shall be valid until the day of 20.....

Name of Bank:

Branch:

Address:

Phone Number:

(Signature of the authorized officer of the Bank)

Name, Designation & Code No.:

Date:

Seal:

FORMAT 5

POWER OF ATTORNEY

(To be on non-judicial stamp paper of appropriate value as per Stamp Act relevant to place of execution)

Power of Attorney to be provided by the Bidding Company in favour of its representative as evidence of authorized signatory's authority.

Know all men by these presents, We

.....
(Name and address of the registered office of the Bidding Company as applicable) do hereby constitute, appoint and authorize Mr./Ms.
(name & residential address) who is presently employed with us and holding the position of..... as our true and lawful attorney, to do in our name and on our behalf, all such acts, deeds and things necessary in connection with or incidental to submission of our Bid for “----- **Name of work** -----” in response to **DNIE-T No.**, **dated** **(Insert DNIE-T Number & date)** issued by Tripura Renewable Energy Development Agency, constituent organization of Deptt. of Power, Gorkhabasti, Agartala, West Tripura District, Tripura, Pin:799006 including signing and submission of the Bid and all other documents related to the Bid, including but not limited to undertakings, letters, certificates, acceptances, clarifications, guarantees or any other document which the TREDA may require us to submit. The aforesaid Attorney is further authorized for making representations to the TREDA, Agartala and providing information / responses to TREDA, Agartala representing us in all matters before TREDA, Agartala and generally dealing with TREDA, Agartala in all matters in connection with this Bid till the completion of the bidding process as per the terms of the above mentioned DNIE-T.

We hereby agree to ratify all acts, deeds and things done by our said attorney pursuant to this Power of Attorney and that all acts, deeds and things done by our aforesaid attorney shall be binding on us and shall always be deemed to have been done by us.

All the terms used herein but not defined shall have the meaning ascribed to such terms under the DNIE-T.

Signed by the within named

..... (Insert the name of the executant company)
through the hand of Mr.duly authorized by the
Board(vide Board resolution No_____) to issue such Power of Attorney
Dated this day of

Accepted

Signature of Attorney

(Name, designation and address of the Attorney)

Attested

.....
(Signature of the executant)

SIGNATURE OF BIDDER WITH SEAL & DATE





(Name, designation and address of the executant)

Signature and stamp of Notary of the place of execution Common seal of has been affixed in my/our presence Pursuant to Board of Director's Resolution dated.....(Board of Director's Resolution is also enclosed)

WITNESS

i)

(Signature)

Name.....

Designation

ii)

(Signature)

Name.....

Designation

Notes:

The mode of execution of the power of attorney should be in accordance with the procedure, if any, laid down by the applicable law and the charter documents of the executant(s) and the same should be under common seal of the executant affixed in accordance with the applicable procedure. Further, the person whose signatures are to be provided on the power of attorney shall be duly authorized by the executant(s) in this regard.

SIGNATURE OF BIDDER WITH SEAL & DATE

do
15/11/2016

FORMAT 6

(Format for sole bidder and to be submitted in the letterhead)

AUTHORIZATION LETTER OF BIDDER

Name of Work	Design, survey, manufacture, supply, erection, testing, commissioning of 10 kWp & 5 kWp Solar Photovoltaic Power Plant (Hybrid) at Block Offices etc. including Warranty /Guarantee, Annual Maintenance Contract and insurance coverage for 5 (five) years on turn-key basis throughout the state of Tripura
---------------------	--

To

The Joint Director,

Tripura Renewable Energy Development Agency,
Deptt. Of Power, Govt. of Tripura,
Gorkhabasti, Agartala, West Tripura District, Tripura,
Pin:799006. e-mail: tredatenders25@gmail.com.

<Bidder's Employee Name> _____,

<Designation> _____ is hereby authorized to sign relevant documents on behalf of the company/ firm in dealing with Bid of reference < Tripura Renewable Energy Development Agency, constituent organization of Deptt. Of Power, Gorkhabasti, Agartala, West Tripura District, Tripura, Pin-799006 vide **DNIE-T No., dated (Insert DNIE-T Number & date).** He is also authorized to attend meetings & submit pre- qualification, technical & commercial information as may be required by you in the course of processing the above said Bid. For the purpose of validation, his/ her verified signatures are as under.

Signature of authorized representative (As per POA):

Verified Signature:

Name of the Bidder: -

Authorized representative executing Power of Attorney (POA): -

Seal of the Organization: -

Date:

Place:



FORMAT 7

(To be on non-judicial stamp paper of appropriate value as per Stamp Act relevant to place of execution and should be attested by NOTARY PUBLIC)

AFFIDAVIT – NO BLACK LISTING

In response to the **DNIE-T No.**, dated **(Insert DNIE-T Number & date)** for “----- **Name of work** -----” as an owner/ partner/ Director of _____, I/ We hereby declare that presently our Company/ firm _____ is having unblemished record and is not declared ineligible for corrupt & fraudulent practices either indefinitely or for a particular period of time by any State/ Central Government/ PSU.

We further declare that presently our Company/ firm _____ is not blacklisted and not declared ineligible for reasons other than corrupt & fraudulent practices by any State/ Central Government/ PSU on the date of Bid Submission.

If this declaration is found to be incorrect then without prejudice to any other action that may be taken, my/ our security may be forfeited in full and the tender if any to the extent accepted may be cancelled.

(Signature of the authorized signatory)

Name:

Designation:

Company Seal:

FORMAT 8

(To be submitted in the official letter head of the company)

DECLARATION BY THE BIDDER

Name of Work	Design, survey, manufacture, supply, erection, testing, commissioning of 10 kWp & 5 kWp Solar Photovoltaic Power Plant (Hybrid) at Block Offices etc. including Warranty /Guarantee, Annual Maintenance Contract and insurance coverage for 5 (five) years on turn-key basis throughout the state of Tripura
DNIE-T	Insert DNIE-T Number & date

I/We _____ (hereinafter referred to as the bidder) being desirous of tendering for the work under the above mentioned tender and having fully understood the nature of the work and having carefully noted all the terms and conditions, specifications etc. as mentioned in the tender document, DO HEREBY DECLARE THAT-

- 1) The Bidder is fully aware of all the requirements of the tender document and agrees with all provisions of the tender document.
- 2) The Bidder is capable of executing and completing the work as required in the tender.
- 3) The Bidder accepts all risks and responsibilities directly or indirectly connected with the performance of the tender.
- 4) The Bidder has no collusion with other contractor, any employee of TREDA / Department of Power, Govt. of Tripura or its autonomous bodies or with any other person or firm in the preparation of the bid.
- 5) The bidder confirm that any genuine changes made by TREDA in the interest of the project with respect to the Technical Specifications, Design & Drawing during the course of performance of the Letter of Award shall be fully acceptable to the bidder without any cost implication whatsoever to TREDA.
- 6) The Bidder has not been influenced by any statement or promises of TREDA / Department of Power, Govt. of Tripura or its autonomous bodies or any of its employees but only by the tender document.
- 7) The Bidder is financially solvent and sound to execute the work.
- 8) The Bidder is sufficiently experienced and competent to perform the contract to the satisfaction of TREDA.
- 9) The information and the statements submitted with the tender are true and correct and also accept that misinterpretation contained in it may lead to our disqualification and rejection of the bid. In case of any false document submitted and found any time in future, we shall be liable to be proceeded as per Applicable Law.
- 10) The bidder declare and confirm that only the bidder is submitting their bid and that our parent, affiliate, the ultimate parent or any group companies with which the bidder have direct or indirect relationships are not separately submitting their bid.
- 11) The bidder confirms that the bidder does not have any conflict of interest in accordance with the provisions of DNIE-T.
- 12) The bidder confirms that in case the bidder is directly or indirectly through an Agency engaged in Corrupt Practice, Fraudulent Practice, Coercive Practice, Collusive Practice, Obstructive Practice or Integrity violation, then our bid shall be considered as informal / non-responsive and shall be liable for rejection.

SIGNATURE OF BIDDER WITH SEAL & DATE



Page **91** of **105**

- 13) Bid Security / EMD The Bidder is familiar with all general and special laws, acts, ordinances, rules and regulations of the Municipal, District, State and Central Government that may affect the work, its performance or personnel employed therein.
- 14) The Bidder has not been debarred or Black Listed from similar type of work by TREDA and or Central / State Government Departments /Undertaking during last three years.
- 15) This offer shall remain valid for acceptance for 180 (one hundred eighty) days from the date of opening of the tender.
- 16) The Bidder gives the assurance to execute the work as per technical specifications, terms and conditions of the DNle-T.
- 17) The Bidder assured to execute the work in accordance to the time schedule as per PERT Network duly approved by TREDA.
- 18) The terms and conditions of DNle-T will be binding upon bidder in the event of acceptance of their tender.
- 19) The Bidder has electronically paid the Tender Fee and BID SECURITY / EMD as required in the tender document.
- 20) The Bidder accepts that the Bid Security / Earnest Money Deposit / Performance Bank Guarantee be partially / absolutely forfeited by TREDA as per the terms & conditions laid down in this DNle-T.
- 21) The bidder confirms that the bidder shall establish a Central Office at Agartala, Tripura and also establish Office cum Service Centre throughout the State of Tripura as per DNle-T requirement as to deliver un-interrupted and sustainable comprehensive maintenance during the AMC period.

Date: ____/____/20____

(Signature of the authorized signatory)

Name:

Designation:

Company Seal:

FORMAT 9

COVERING LETTER

(To be submitted in the official letter head of the company)

To
The Joint Director,
Tripura Renewable Energy Development Agency,
Vigyan Bhawan, P.N. Complex, Gorkhabasti,
Agartala, West Tripura District,
Pin; 799006.

SUB: Offer in response to DNIE-T No.F.6(542)/TREDA/NCES/2025-26/2110, dated 12/06/2026 for “Design, survey, manufacture, supply, erection, testing, commissioning of 10 kWp & 5 kWp Solar Photovoltaic Power Plant (Hybrid) at Block Offices etc. including Warranty /Guarantee, Annual Maintenance Contract and insurance coverage for 5 (five) years on turn-key basis throughout the state of Tripura”.

Sir,

We are submitting our offer in full compliance of the terms & conditions of the above cited DNIE-T. A blank copy of tender duly signed on each page is also uploaded as a proof of our acceptance of all specifications as well as all the terms & conditions.

We confirm that, we have the capability for design, survey, manufacture, supply, erection, testing, commissioning of 10 kWp & 5 kWp Solar Photovoltaic Power Plants and Internal Electrification Works etc on turn-key basis within the time period specified in the DNIE-T with other necessary infrastructures for providing after sales service for 5 (five) years Warranty /Guarantee and Annual Maintenance Contract from the date of commissioning.

We confirm that, we have also submitted all the relevant documents as per tender equipment and have paid the TENDER FEE & BID SECURITY / EMD as per the stipulation of the tender document.

We confirm that our offered rate would be valid for at least 180 (one hundred eighty) days from the date of opening of tender.

The e-tender is uploaded in two separate files named Part-A for technical bid & Part-B for Price Bid.

(Signature of the authorized signatory)

Name:

Designation:

Company Seal:

FORMAT 10

[To be submitted in the letterhead of the bidder including full postal address, telephone, faxes and e-mail address]

FORMAT FOR QUERIES OF BIDDER ON TENDER FOR PRE-BID MEETING

Name of Tender		Design, survey, manufacture, supply, erection, testing, commissioning of 10 kWp & 5 kWp Solar Photovoltaic Power Plant (Hybrid) at Block Offices etc. including Warranty /Guarantee, Annual Maintenance Contract and insurance coverage for 5 (five) years on turn-key basis throughout the state of Tripura		
DNIE-T		Insert DNIE-T Number & date		
Tender ID				
Bid Opening Date				
Name of bidder				
Name of contact person from Bidder with address, e-mail and Contact Number				
Sl. No.	Section No. Page No. Clause No. Para No.	Description as per DNIE-T	Queries / Clarifications of the bidder	Remarks
1	Section No.			
	Page No.			
	Clause No.			
	Sub-clause No.			
2	Section No.			
	Page No.			
	Clause No.			
	Sub-clause No.			

[NOTE:

- a) Clarifications should be sought through e-procurement portal **<https://tripuratenders.gov.in>** in above format within the timeline stipulated in the BID INFORMATION SHEET.
- b) **Queries / Clarifications sought through e-mail of TRED OR in any other form will not be accepted.**

(Signature of the authorized signatory)

Name:

Designation:

Company Seal:

FORMAT 11

[LETTER OF INTENT (LoI) TO THE L1 BIDDER FOR DEPOSITING PERFORMANCE BANK GUARANTEE IN LIEU OF EMD RETURN]

TRIPURA RENEWABLE ENERGY DEVELOPMENT AGENCY

(Power Department, Government of Tripura)

Vigyan Bhawan, Pandit Nehru Complex, Gorkhabasti, Agartala, Tripura

Tele-fax: 0381-2326139, email-tredatenders25@gmail.com

No.F.6(542)/TREDA/NCES/2025-26/_____

Date ____/____/20__

To

M/S _____

SUB: LETTER OF INTENT (LoI) against your bid submitted to this office for <**Name of Work**>.

REF : 1. Tender ID No. ____
2. DNIE-T -----
3. Online bid reference of e-procurement portal:

Sir / Madam,

This is to inform you that, the bid submitted as stated above is under consideration by the appropriate authority subject to your fulfilment of the terms & conditions of the tender and all codal procedures.

You are therefore requested to deposit the "PERFORMANCE BANK GUARANTEE" for an amount of Rs. _____ (Rupees _____) vide Clause No. _____ under SECTION ----- of PART ----- on or before ____/____/202__ by 5:30 PM for completion of Pre-Award codal procedures.

Failure to comply with the same shall result to your Bid being cancelled and the EMD deposited with respect to the stated tender will be forfeited. Further your submission of "Performance Bank Guarantee" does not entail you with right to get the order, unless all the other Tender conditions are satisfied with due verification.

Yours sincerely

Authorized Signatory
Tripura Renewable Energy Development Agency

FORMAT 12

[TO BE SUBMITTED IN THE LETTERHEAD OF THE BIDDER]

[FORMAT OF DECLARATION FOR SUBMITTING PART 6 AND TEST CERTIFICATES / BIS CERTIFICATES]

Name of Work	Design, survey, manufacture, supply, erection, testing, commissioning of 10 kWp & 5 kWp Solar Photovoltaic Power Plant (Hybrid) at Block Offices etc. including Warranty /Guarantee, Annual Maintenance Contract and insurance coverage for 5 (five) years on turn-key basis throughout the state of Tripura
DNIE-T	Insert DNIE-T Number & date

DECLARATION

I/We, on behalf of _____ (Name of Bidder) do hereby declare that, within **30 (thirty) days** from the date of issue of the Letter of Award (LoA), I will submit PART 6: TECHNICAL INFORMATION along with all relevant documents in compliance with the CLAUSE NO. 8 under PART 3 of the DNIE-T.

(Signature of the authorized signatory)

Name:

Designation:

Company Seal:

FORMAT 13

[TO BE SUBMITTED IN THE LETTERHEAD OF THE BIDDER]

SITE VISIT DECLARATION

Name of Work	Design, survey, manufacture, supply, erection, testing, commissioning of 10 kWp & 5 kWp Solar Photovoltaic Power Plant (Hybrid) at Block Offices etc. including Warranty /Guarantee, Annual Maintenance Contract and insurance coverage for 5 (five) years on turn-key basis throughout the state of Tripura
DNIE-T	Insert DNIE-T Number & date

DECLARATION

I/We, _____ (Name of bidder or his representative) on behalf of _____ (Name of Bidder) do hereby declare that I / we visited the sites of work as per provision laid down in the DNIE-T in support of compliance of site visit as provided in the above said DNIE-T for preparation & submission of the bid and thereby to complete the work satisfactorily in all respects as per specification & conditions of the Contract.

(Signature of the authorized signatory)

Name:

Designation:

Company Seal:

Date:

FORMAT 14

(BG SHOULD BE VALID FOR 6 MONTHS + 60 DAYS CLAIM PERIOD)

FORMAT OF BANK GUARANTEE FOR EARNEST MONEY DEPOSIT (EMD)

(To be on non-judicial stamp paper of appropriate value as per Stamp Act relevant to place of execution)

Ref. _____
Date: _____

Bank Guarantee No. _____

In consideration of the ----- [Insert name of the Bidder] (hereinafter referred to as 'Bidder') submitting the response to e-tender for “----- **Name of work** -----” in response to the **DNle-T No., dated** **(Insert DNle-T Number & date)** issued by Tripura Renewable Energy Development Agency, constituent organization of Deptt. of Power, Gorkhabasti, Agartala, West Tripura District, Tripura, Pin: 799006 (hereinafter referred to as TREDA) and TREDA considering such response to the bid of[insert the name of the Bidder] as per the terms of the DNle-T, the _____ [insert name & address of bank] hereby agrees unequivocally, irrevocably and unconditionally to pay to TREDA at [Insert Name of the Place from the address of TREDA] forthwith on demand in writing from TREDA or any Officer authorized by it in this behalf, any amount upto and not exceeding Rupees _____ [Insert amount of EMD] only, on behalf of M/S. _____ [Insert name of the Bidder] .

This guarantee shall be valid and binding on this Bank upto and including _____ [insert date of validity in accordance with DNle-T] and shall not be terminable by notice or any change in the constitution of the Bank or the term of contract or by any other reasons whatsoever and our liability hereunder shall not be impaired or discharged by any extension of time or variations or alternations made, given, or agreed with or without our knowledge or consent, by or between parties to the respective agreement.

Our liability under this Guarantee is restricted to Rs. _____ (Rs. _____ only). Our Guarantee shall remain in force until _____ [insert date of validity in accordance with DNle-T]. TREDA shall be entitled to invoke this Guarantee till _____ [Insert date which is 60 days after the date in the preceding sentence].

The Guarantor Bank hereby agrees and acknowledges that the TREDA shall have a right to invoke this BANK GUARANTEE in part or in full, as it may deem fit.

The Guarantor Bank hereby expressly agrees that it shall not require any proof in addition to the written demand by TREDA, made in any format, raised at the above mentioned address of the Guarantor Bank, in order to make the said payment to TREDA.

The Guarantor Bank shall make payment hereunder on first demand without restriction or conditions and notwithstanding any objection by ----- [Insert name of the Bidder] and/or any other person. The Guarantor Bank shall not require TREDA to justify the invocation of this BANK GUARANTEE, nor shall the Guarantor Bank have any recourse against TREDA in respect of any payment made hereunder.

This BANK GUARANTEE shall be interpreted in accordance with the laws of India and the High Court of Tripura shall have exclusive jurisdiction.

The Guarantor Bank represents that this BANK GUARANTEE has been established in such form and with such content that it is fully enforceable in accordance with its terms as against the Guarantor Bank in the manner provided herein.

This BANK GUARANTEE shall not be affected in any manner by reason of merger, amalgamation, restructuring or any other change in the constitution of the Guarantor Bank.

This BANK GUARANTEE shall be a primary obligation of the Guarantor Bank and accordingly TREDA shall not be obliged before enforcing this BANK GUARANTEE to take any action in any court or arbitral proceedings against the Bidder, to make any claim against or any demand on the Bidder or to give any notice to the Bidder or to enforce any security held by TREDA or to exercise, levy or enforce any distress, diligence or other process against the Bidder.

Not with standing anything contained hereinabove, our liability under this Guarantee is restricted to Rs. _____ (Rs. _____ only) and it shall remain in force until _____ [Date to be inserted on the basis of this DNle-T] with an additional claim period of sixty (60) days thereafter. We are liable to pay the guaranteed amount or any part thereof under this Bank Guarantee only if TREDA serves upon us a written claim or demand.

Signature _____

Name _____

Power of Attorney No. _____

Email ID _____

For _____ [Insert Name of the Bank]

Banker's Stamp and Full Address.

Dated this ____ day of ____, 20__

FORMAT 15 **(PART A)**

[SHOULD BE SUBMITTED IN THE LETTERHEAD OF THE PRACTICING CA AND SHOULD BE CERTIFIED BY THE PRACTICING CA OF THE BIDDER]

Name of Tender	Design, survey, manufacture, supply, erection, testing, commissioning of 10 kWp & 5 kWp Solar Photovoltaic Power Plant (Hybrid) at Block Offices etc. including Warranty /Guarantee, Annual Maintenance Contract and insurance coverage for 5 (five) years on turn-key basis throughout the state of Tripura
DNIE-T	Insert DNIE-T Number & Date
NAME OF BIDDER	

DETAILS OF WORKS FOR CALCULATION OF THE VALUE “A” OF THE BID CAPACITY

1. Value of Engineering works in respect to SOLAR PV projects (Turnkey projects/item rate contract/Construction works) undertaken by the Bidder during the last five (5) years is as follows:

Sl. No.	Financial year	Value of Engineering works undertaken with respect to SOLAR PV projects (Rs in INR)	Updation factor	Value updated to the current costing level (Rs in INR)
(1)	(2)	(3)	(4)	(5) = Col 3 X Col 4
1	2025-26 (1 st year)		1.00	
2	2024-25 (2 nd year)		1.07	
3	2023-24 (3 rd Year)		1.14	
4	2022-23 (4 th Year)		1.21	
5	2021-22 (5 th Year)		1.28	

2. Maximum value of SOLAR PV projects that have been undertaken during the F.Y ----- out of the last five (5) years and value thereof is Rs ----- (Rupees-----)

Certified by Chartered Account (CA) in original on their letterhead with Membership No., FRN, UDIN, Contact details (e-mail and Mobile Number)

(Signature of the authorized signatory of bidder)

Name:

Designation:

Company Seal:

FORMAT 15

(PART B)

[SHOULD BE SUBMITTED IN THE LETTERHEAD OF THE PRACTICING CA AND SHOULD BE CERTIFIED BY THE PRACTICING CA OF THE BIDDER]

Name of Tender	Design, survey, manufacture, supply, erection, testing, commissioning of 10 kWp & 5 kWp Solar Photovoltaic Power Plant (Hybrid) at Block Offices etc. including Warranty /Guarantee, Annual Maintenance Contract and insurance coverage for 5 (five) years on turn-key basis throughout the state of Tripura
DNIE-T	Insert DNIE-T Number & Date
NAME OF BIDDER	

DETAILS OF EXISTING COMMITMENTS FOR CALCULATION OF THE VALUE “B” OF THE BID CAPACITY

1. Value of all the existing commitments and on-going workings to be completed during the next ___ **years** (prescribed time for completion of the works for which bids are invited) is as follows:

Sl. No.	Name of the work/ project	Name of State Nodal Agency for RE / DISCOM / Govt. Departments / Government Organization	Percentage of participation of bidder in the project	Stipulated period of completion as per Agreement/ LoA with the start date	Value of Contract as per Agreement/ LoA (In Rs.)	Value of work completed (In Rs.)	Balance value of work to be completed (In Rs.)	Anticipated date of completion	Financial liability to be incurred for the said work / project during the period of the subject contract (In Rs.)
1	2	3	4	5	6	7	8	9	10

2. The value of total financial liability to be incurred for the above works / projects during the period of the subject contract is Rs ----- (Rupees-----).

Certified by Chartered Account (CA) in original on their letterhead with Membership No., FRN, UDIN, Contact details (e-mail and Mobile Number)

(Signature of the authorized signatory of bidder)

Name:

Designation:

Company Seal:



PART B (PRICE BID)

[NOTE: BOQ FORMAT FOR REFERENCE ONLY, PRICE TO BE FILLED IN SEPARATE BOQ FILE]

Validate Print Help

Item Wise BoQ

Tender Inviting Authority: <Joint Director, Tripura Renewable Energy Development Agency (TREDA), Department of Power, Govt. of Tripura >

Name of Work: <Design, survey, manufacture, supply, erection, testing, commissioning of 10 kWp & 5 kWp Solar Photovoltaic Power Plant (Hybrid) at Block Offices etc. including Warranty / Guarantee, Annual Maintenance Contract and insurance coverage for 5 (five) years on turn-key basis throughout the state of Tripura>

DNle-T: <DNle-T No.F.6(542)/TREDA/NCES/2025-26/2110, dated 12/06/2026>

Name of the Bidder/
Bidding Firm
/ Company :

PRICE SCHEDULE

(DOMESTIC TENDERS - RATES ARE TO GIVEN IN RUPEES (INR) ONLY)

(This BOQ template must not be modified/replaced by the bidder and the same should be uploaded after filling the relevant columns, else the bidder is liable to be rejected for this tender. Bidders are allowed to enter the Bidder Name and Values only)

NUMBER #	TEXT #	TEXT #	NUMBER	TEXT #	NUMBER #	NUMBER	NUMBER	NUMBER #	NUMBER #	TEXT #
Sl. No.	Item Description	Item Code / Make	Quantity	Units	BASIC RATE In Figures To be entered by the Bidder in Rs. P	GST Rate in Percentage %	GST Amount in INR Auto Calculated	TOTAL AMOUNT Without Taxes Rs. P	TOTAL AMOUNT With Taxes Rs. P	TOTAL AMOUNT In Words
1	2	3	4	5	7	9	21	13	14	15
1	Design, manufacture, supply, erection, testing and commissioning of 5 kWp Solar Photovoltaic Power Plant (Hybrid) as per Scope of work and technical specifications mentioned at DNle-T on turn-key basis	ITEM1	29	Nos.			0	0.00	0.00	INR Zero Only
2	Design, manufacture, supply, erection, testing and commissioning of 10 kWp Solar Photovoltaic Power Plant (Hybrid) as per Scope of work and technical specifications mentioned at DNle-T on turn-key basis	ITEM2	58	Nos.			0	0.00	0.00	INR Zero Only
3	Wiring for light / fan / exhaust fan / call bell point with 2 x1.5 sq. mm PVC insulated Copper conductor cable (FR) in pvc casing and capping complete with 6/10 amps modular type switch, cover plate, surface modular switch board including connection etc as required:-									
3.1	SHORT POINT (Upto 3 meters)	ITEM3.1	725	Nos.			0	0.00	0.00	INR Zero Only
3.2	MEDIUM POINT (Above 3 meters to 6 meters)	ITEM3.2	725	Nos.			0	0.00	0.00	INR Zero Only
3.3	LONG POINT (Above 6 meters to 10 meters)	ITEM3.3	812	Nos.			0	0.00	0.00	INR Zero Only

SIGNATURE OF BIDDER WITH SEAL & DATE

Signature



4	Providing and fixing of 3 pin 6/10 amps plug point with modular type 6/10 amps switch and socket including connection with 2x1.5 sq. mm PVC insulated Copper conductor cable (FR) on surface modular switch board, earthing the 3rd pin etc. as required:-	ITEM4	812	Nos.			0	0.00	0.00	INR Zero Only
5	Wiring for Circuit/Sub-main with following sizes PVC insulated Copper conductor cable (FR) in PVC casing and capping etc. as required:-									
5.1	2 X 2.5 sq. mm	ITEM5.1	130500	Meters			0	0.00	0.00	INR Zero Only
5.2	2 X 4 sq. mm	ITEM5.2	43500	Meters			0	0.00	0.00	INR Zero Only
6	Supplying and drawing 2.5 sq. mm Copper conductor wire along with other wire for loop earthing including connection etc. as required.	ITEM6	174000	Meters			0	0.00	0.00	INR Zero Only
7	Supplying and fixing of following surface/recess mounted LED luminaries in powder coated die cast housing complete with high efficiency polycarbonate diffuser and necessary connection etc. as required (with 50000 burning hours and minimum lumen output of 100 lm/watt):-									
7.1	Surface mounted integrated 10 watt LED batten with polycarbonate diffuser:-	ITEM7.1	725	Nos.			0	0.00	0.00	INR Zero Only
7.2	Surface mounted integrated 20 watt LED batten with polycarbonate diffuser:-	ITEM7.2	1450	Nos.			0	0.00	0.00	INR Zero Only
8	Supplying and installation of Bus bar chamber of the following ratings and sizes made of steel sheet enclosure with all accessories including necessary connection etc. as required:-									
8.1	32 amps 415 volts four strip Copper busbar chamber 4 ways of size (300mmx240mmx80mm):-	ITEM8.1	87	Nos.			0	0.00	0.00	INR Zero Only

SIGNATURE OF BIDDER WITH SEAL & DATE



9	Supplying and installation of Switch disconnecter fuse unit (main switch) of the following ratings with all accessories including necessary connection etc. as required:-									
9.1	32 amps 240 volts DP main switch (rewireable):-	ITEM9.1	29	Nos.			0	0.00	0.00	INR Zero Only
9.2	32 amps TPN 415 volts front handle cubical main switch (HRC):-	ITEM9.2	58	Nos.			0	0.00	0.00	INR Zero Only
10	Supplying & installation of 10 KA breaking capacity, B/C curve MCB of following ratings in the existing MCB DB including connection as required:-									
10.1	6 to 32 amps DP MCB:-	ITEM10.1	1160	Nos.			0	0.00	0.00	INR Zero Only
10.2	6 to 32 amps TP MCB:-	ITEM10.2	58	Nos.			0	0.00	0.00	INR Zero Only
11	Supplying and installation of powder coated steel sheet enclosure (MCB DB) suitable for 240/415 volts MCB/Isolator/RCCB (SP/SPN/DP/TP/TPN/FP) complete with necessary bus bar, neutral link, din rail, earth bar etc i/c connection as required.									
11.1	2 way SPN MCB DB (single door):-	ITEM11.1	87	Nos.			0	0.00	0.00	INR Zero Only
11.2	4 way SPN MCB DB (single door):-	ITEM11.2	290	Nos.			0	0.00	0.00	INR Zero Only
12	Supplying and installation of Automatic Transfer Switch (ATS) of following the ratings with enclosure along with UV/OV protection controller including necessary connections etc. as required:-									
12.1	32 amps,240 volt Two Pole ATS : -	ITEM12.1	29	Nos.			0	0.00	0.00	INR Zero Only
12.2	32 amps,440volt Four Pole ATS : -	ITEM12.2	58	Nos.			0	0.00	0.00	INR Zero Only
13	Supplying and fixing of following outdoor LED luminaries in powder coated die cast housing complete with heat resistant tempered glass/diffuser including necessary connection etc. as required (50000 burning hours and minimum lumen output of 100 lm/watt):-									
13.1	25 watt LED street light fixture with integral driver and secondary lens (IP 66):-	ITEM13.1	87	Nos.			0	0.00	0.00	INR Zero Only
14	Drawing of armoured/un-armoured cable with proper saddling etc as required:-	ITEM14	435	Meters			0	0.00	0.00	INR Zero Only

SIGNATURE OF BIDDER WITH SEAL & DATE



15	Supplying and laying of the following sizes XLPE insulated PVC sheathed aluminium conductor armoured cable with necessary connection etc as required								
15.1	2 core 10 Sq mm	ITEM15.1	290	Meters			0	0.00	0.00 INR Zero Only
15.2	4 core 16 Sq mm	ITEM15.2	580	Meters			0	0.00	0.00 INR Zero Only
16	Excavation of earth for cable trench of 800mm depth & 300mm width in all kinds of soil including back filling of cable trench with the excavated soil with 20cm layer of sand, 3.5 nos bricks per metre for protection including ramming, consolidating to the original shape as per direction of Site Engineer :-	ITEM16	725	Meters			0	0.00	0.00 INR Zero Only
17	Supplying and fixing of 1 phase Net Meter: 2 Wire, 5-30 Amp, 240 V with meter box for net metering (Solar Application)	ITEM17	29	Nos.			0	0.00	0.00 INR Zero Only
18	Supplying and fixing of Net Meter: 3 Phase, 4 wire, 3 x 10-60 Amp, 3 x 240 V Meter with meter box for net metering (Solar Applicable)	ITEM18	58	Nos.			0	0.00	0.00 INR Zero Only
Total in Figures								0.00	0.00 INR Zero Only
Quoted Rate in Words		INR Zero Only							